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ARTICLE

Quantifying Partisan Gerrymandering: An Evaluation of the Efficiency Gap Proposal

Benjamin Plener Cover*

Abstract. Electoral districting presents a risk of partisan gerrymandering: the manipulation of electoral boundaries to favor one political party over another. For three decades, the U.S. Supreme Court has failed to settle on a legal test for partisan gerrymandering, and such claims have uniformly failed. Until recently. Plaintiffs prevailed before a three-judge federal panel in Wisconsin by leveraging a new measure called the “efficiency gap,” which quantifies partisan gerrymandering in terms of two parties’ relative efficiency at translating votes for their party into seats in government. The case is now before the Court, which may embrace the efficiency gap approach and thereby remake the law of electoral districting. Through a synthesis of mathematical and legal analysis, this Article examines the efficiency gap measure, focusing particularly on its underlying methodological choices and electoral assumptions as well as its relationship to competitiveness, seats-votes proportionality, and voter turnout.

The efficiency gap is a useful indicative measure of partisan gerrymandering under the circumstances of cases like the one currently before the Court, in which each party earns about half the votes and a large efficiency gap persists under plausible variations in voter behavior. Relying in part on the efficiency gap measure, the Court should rule in favor of the plaintiffs. However, a mapmaker can achieve a below-threshold efficiency gap with a skewed bipartisan gerrymander that carves a state up into uncompetitive districts denying minority parties sufficient representation. For example, a party that earns only 59% of the vote can secure a filibuster- and veto-proof 75% supermajority of the legislature with a below-

* Associate Professor, University of Idaho College of Law. J.D., Yale Law School; M.Sc., London School of Economics. For helpful discussions as I developed these ideas, I extend my thanks to Nicholas Stephanopoulos, Eric McGhee, Heather Gerken, Richard Re, Raymond Dacey, Lisa Carlson, Steven Radil, Michele Wiest, and Kenton Bird, as well as participants in the Fifth Annual State & Local Government Law Works-in-Progress Conference; the Rocky Mountain Junior Scholars Forum; the Malcolm M. Renfrew Interdisciplinary Colloquium; and the Political Science, Economics, and Geography Working Group at the University of Idaho. For useful feedback during the writing process, I am grateful to Justin Levitt, Moon Duchin, Wendy Couture, Kate Evans, David Pimentel, and Sandy Mayson. Kacey Jones, Cody Witko, Justin Bowles, and Kelsey Gooden provided excellent research assistance. A special thank you to Samson J. Schatz and the entire editing team at the *Stanford Law Review* for their patience, diligence, and insights throughout the editing process. As always, I am particularly indebted to Aliza Plener Cover.

threshold efficiency gap. For this and other reasons, the Court should not adopt the efficiency gap as the exclusive definitional measure of partisan gerrymandering, such that a plan would be invalid if and only if it exhibited a large, durable, and unjustified efficiency gap. Instead, the Court should permit some flexibility for scholars, litigants, and courts to refine measurement approaches over time and under varying circumstances. One approach worth future exploration is a variation on the efficiency gap that defines a surplus vote in terms of the full margin of victory and compares wasted vote shares instead of totals. Finally, the Court should be aware that any measure, like the efficiency gap, that compares votes to seats entails the perverse risk that partisan voter suppression may operate to reduce the apparent severity of partisan gerrymanders.

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Introduction

We may be approaching a watershed moment in the U.S. Supreme Court's gerrymandering jurisprudence. In three cases over the last three decades, partisan gerrymandering has eluded the Court's grasp.¹ The Court has recognized that partisan gerrymandering poses a problem of constitutional significance² but has repeatedly fractured on whether and how to intervene. A minority of Justices have insisted that partisan gerrymandering presents a nonjusticiable political question susceptible to no judicially discernible and manageable standard,³ while a majority of Justices have agreed that partisan gerrymandering is justiciable but have disagreed among themselves about the proper legal standard.⁴ Justice Kennedy, the current swing vote, has rejected each proposal for assessing partisan gerrymandering claims while expressing hope that a suitable standard may one day materialize.⁵ In the first thirty years after the Court held partisan

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1. *League of United Latin Am. Citizens v. Perry (LULAC)*, 548 U.S. 399 (2006); *Vieth v. Jubelirer*, 541 U.S. 267 (2004); *Davis v. Bandemer*, 478 U.S. 109 (1986), *abrogated in part by Vieth*, 541 U.S. 267.
 2. *See Ariz. State Legislature v. Ariz. Indep. Redistricting Comm'n*, 135 S. Ct. 2652, 2658 (2015) (“[P]artisan gerrymanders,’ this Court has recognized, [are incompatible] with democratic principles.” (alterations in original) (quoting *Vieth*, 541 U.S. at 292 (plurality opinion))). Even the *Vieth* plurality, while denying courts the ability to adjudicate partisan gerrymandering claims, conceded that “an excessive injection of politics is unlawful” and that “setting out to segregate [voters] by political affiliation is . . . lawful” only “so long as one doesn’t go too far.” *See* 541 U.S. at 293, 305-06 (plurality opinion); *see also* Brief of Amici Curiae Law Professors in Support of Appellees at 4 n.2, *Gill v. Whitford*, No. 16-1161 (U.S. Sept. 5, 2017), 2017 WL 4311104 (“All nine Members of the *Vieth* Court accepted the proposition that excessive partisan gerrymandering violates the Constitution.”).
 3. *See LULAC*, 548 U.S. at 511 (Scalia, J., concurring in the judgment in part and dissenting in part) (joined by Justice Thomas); *Vieth*, 541 U.S. at 281 (plurality opinion) (authored by Justice Scalia and joined by Chief Justice Rehnquist and Justices O’Connor and Thomas); *Bandemer*, 478 U.S. at 144 (O’Connor, J., concurring in the judgment) (joined by Chief Justice Burger and then-Justice Rehnquist).
 4. *See LULAC*, 548 U.S. at 413-14 (majority opinion) (“A plurality of the Court in *Vieth* would have held [partisan gerrymandering] challenges to be nonjusticiable political questions, but a majority declined to do so. We do not revisit the justiciability holding . . .” (citations omitted)); *id.* at 447, 456-57 (Stevens, J., concurring in part and dissenting in part); *id.* at 483 (Souter, J., concurring in part and dissenting in part); *id.* at 491-92 (Breyer, J., concurring in part and dissenting in part); *Vieth*, 541 U.S. at 306 (Kennedy, J., concurring in the judgment); *id.* at 317, 321-23 (Stevens, J., dissenting); *id.* at 344, 346-47 (Souter, J., dissenting); *id.* at 355, 364-67 (Breyer, J., dissenting).
 5. *See Vieth*, 541 U.S. at 306 (Kennedy, J., concurring in the judgment) (“I would not foreclose all possibility of judicial relief if some limited and precise rationale were found . . .”); *see also LULAC*, 548 U.S. at 414 (affirming *Vieth*’s five-Justice vote against declining to hear all partisan gerrymandering cases on justiciability grounds).

gerrymandering justiciable,⁶ dozens of plaintiffs raised claims of partisan gerrymandering, but not one was granted relief⁷—until recently.

In *Whitford v. Gill*,⁸ plaintiffs challenged the 2012 Wisconsin State Assembly district map as a partisan gerrymander, relying in part on a newly proposed numeric measure and associated legal test called the “efficiency gap.”⁹ In 2014, political scientist Eric McGhee proposed the measure.¹⁰ In 2015, McGhee and leading election law scholar Nicholas Stephanopoulos developed the measure into a legal test specifically designed to address concerns with prior proposals for assessing partisan gerrymandering.¹¹ In brief, the efficiency gap measure counts the relative number of votes “wasted” by each of two competing political parties; it thereby quantifies the relative efficiency with which each party is able to convert popular support (votes) into governmental power (seats).¹² The legal test classifies as an invalid partisan gerrymander any plan that produces a large, durable, and unjustified efficiency gap.¹³

6. See *Bandemer*, 478 U.S. at 125-27.

7. See *Vieth*, 541 U.S. at 279-80 (plurality opinion) (“[I]n all of the cases we are aware of involving [the] most common form of political gerrymandering, [that involving the drawing of district lines,] relief was denied.”); Nicholas O. Stephanopoulos & Eric M. McGhee, *Partisan Gerrymandering and the Efficiency Gap*, 82 U. CHI. L. REV. 831, 832-33 (2015) (“By our count, claimants’ record over this generation-long period [from 1986 to 2015] is roughly zero wins and fifty losses.”); Easha Anand, Comment, *Finding a Path Through the Political Thicket: In Defense of Partisan Gerrymandering’s Justiciability*, 102 CALIF. L. REV. 917, 933 (2014) (“[O]f the thirty-nine decisions surveyed . . . , only one found a gerrymander unconstitutional, and that one decision was subsequently dismissed as moot.”).

8. 218 F. Supp. 3d 837, 854, 910 (W.D. Wis. 2016), stay granted, 137 S. Ct. 2289, and jurisdiction postponed, 137 S. Ct. 2268 (2017).

9. See Complaint ¶¶ 1, 5, *Whitford*, 218 F. Supp. 3d 837 (No. 3:15-cv-00421-bbc), 2015 WL 4651084.

10. See Eric McGhee, *Measuring Partisan Bias in Single-Member District Electoral Systems*, 39 LEGIS. STUD. Q. 55, 68, 77 (2014). In his 2014 article, McGhee called the metric “relative wasted votes.” See *id.* at 77.

11. See Stephanopoulos & McGhee, *supra* note 7, at 833-34. Throughout this Article, I refer to Eric McGhee and Nicholas Stephanopoulos as the “academic proponents” or simply the “proponents” of the proposed efficiency gap measure and legal test.

12. See *id.* at 851 (defining the “efficiency gap” as “the difference between the parties’ respective wasted votes, divided by the total number of votes cast in the election” (emphasis omitted)); *id.* at 852 (“A gap in a party’s favor enables the party to claim more seats, relative to a zero-gap plan, without claiming more votes.”).

13. See *id.* at 864-65, 885. Specifically, “large” means that the gap exceeds a set numeric threshold, see *id.* at 886-89 (recommending two seats for congressional plans and an 8% efficiency gap for state house plans); “durable” means that the gap is robust to sensitivity testing that models plausible shifts in voting patterns, see *id.* at 889-90; and “unjustified” means that the gap cannot be explained as the product of consistently applying legitimate districting criteria to the jurisdiction’s “underlying political geography,” see *id.* at 891.

Armed with this new measure and associated legal test, the *Whitford* plaintiffs not only survived the motions stage but also won at trial before a panel of three federal judges.¹⁴ The majority opinion does not endorse wholesale the plaintiffs' proposal, but it extensively discusses the efficiency gap as strong evidence in support of its conclusion that the map was a partisan gerrymander.¹⁵ Wisconsin appealed directly to the U.S. Supreme Court,¹⁶ which stayed the panel's remedial order, ordered full briefing, and heard oral argument on October 3, 2017.¹⁷ *Whitford* offers the Court the opportunity to decide whether the efficiency gap provides the legal test it has been waiting for. Were the Court to affirm the panel's finding of partisan gerrymandering—based on the efficiency gap analysis, other evidence, or some combination thereof—it would remake the law of electoral districting in advance of the 2020 redistricting cycle.¹⁸

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14. *Whitford*, 218 F. Supp. 3d at 843, 856-57, 930. When a plaintiff sues in federal court to challenge the constitutionality of an electoral districting plan for "any statewide legislative body" or congressional delegation, Congress has required the convening of a three-judge panel, 28 U.S.C. § 2284(a) (2016), with direct appeal to the Supreme Court after injunctive relief is granted or denied, *id.* § 1253. See *Cooper v. Harris*, 137 S. Ct. 1455, 1464 n.2 (2017); see also *Shapiro v. McManus*, 136 S. Ct. 450, 454-55 (2015) (clarifying that § 2284's requirement of convening a three-judge panel is mandatory). See generally Joshua A. Douglas, *The Procedure of Election Law in Federal Courts*, 2011 UTAH L. REV. 433 (discussing the operation of relevant statutory provisions in election law cases).
 15. See *Whitford*, 218 F. Supp. 3d at 903 ("[The] evidence is further bolstered by the plaintiffs' use of the 'efficiency gap' . . . to demonstrate that . . . their representational rights have been burdened."); see also *id.* at 933 (Griesbach, J., dissenting) ("Despite the central role the efficiency gap has played in the case from the beginning, . . . the majority has declined the Plaintiffs' invitation to adopt their standard and uses it only as confirming evidence. . . .").
 16. See Defendants' Notice of Appeal at 1, *Whitford v. Gill*, No. 3:15-cv-00421-bbc (W.D. Wis. Feb. 24, 2017).
 17. See *Gill v. Whitford*, 137 S. Ct. 2289, 2289 (2017) (mem.) (granting a stay of the three-judge court's judgment pending appeal to the Court); *Gill v. Whitford*, 137 S. Ct. 2268, 2268 (2017) (mem.) (postponing consideration of the jurisdictional question pending hearing the case on the merits); Transcript of Oral Argument, *Gill v. Whitford*, No. 16-1161 (U.S. Oct. 3, 2017), 2017 WL 4517131.
 18. Justice Kennedy was the swing vote in *Vieth*, see *Vieth v. Jubelirer*, 541 U.S. 267, 306 (2004) (Kennedy, J., concurring in the judgment), and may remain the swing vote in *Whitford* if each of the five Justices who joined the court since *Vieth* (Chief Justice Roberts and Justices Alito, Sotomayor, Kagan, and Gorsuch) votes like his or her predecessor on the question of partisan gerrymandering. Cf., e.g., *Kerr v. Hickenlooper*, 759 F.3d 1186, 1193, 1196 (10th Cir. 2014) (Gorsuch, J., dissenting from the denial of rehearing en banc) (citing Justice Scalia's plurality opinion in *Vieth*). But this possibility is no foregone conclusion. And even if the Court embraces a partisan gerrymandering claim this Term in a predicted 5-4 decision, see, e.g., Robert Barnes, *Supreme Court Takes Up Wisconsin as Test in Partisan Gerrymandering Claims*, WASH. POST (Oct. 3, 2017), <https://perma.cc/HN7V-5GKS>, the evolution and refinement of partisan gerrymandering doctrine over time will be determined by the Court as a whole rather than by any single Justice.

As the Court considers *Whitford*, the efficiency gap measure and associated legal test warrant careful and comprehensive examination. Thus far, the reactions in popular media,¹⁹ scholarship,²⁰ and litigation²¹ have been strong and conflicting. This Article contributes to this evaluative effort by offering a new analysis of the proposed efficiency gap measure, focusing particularly on its underlying methodological choices and electoral assumptions, as well as its relationship to competitiveness, seats-votes proportionality, and voter turnout.

19. For a small sampling of media coverage of the efficiency gap proposal, see David Daley, *Will Justice Kennedy Be the Supreme Court's Hero on Gerrymandering? Don't Count On It*, SALON (June 5, 2017, 1:59 AM), <https://perma.cc/GRT5-G7VS>; and Adam Liptak, *When Does Political Gerrymandering Cross a Constitutional Line?*, N.Y. TIMES: SIDEBAR (May 15, 2017), <https://perma.cc/S8RJ-MSLZ>.
20. For scholarly examination of the efficiency gap proposal, see Theodore S. Arrington, *A Practical Procedure for Detecting a Partisan Gerrymander*, 15 ELECTION L.J. 385 (2016); Jowei Chen, *The Impact of Political Geography on Wisconsin Redistricting: An Analysis of Wisconsin's Act 43 Assembly Districting Plan*, 16 ELECTION L.J. 443 (2017); Edward B. Foley, *Due Process, Fair Play, and Excessive Partisanship: A New Principle for Judicial Review of Election Laws*, 84 U. CHI. L. REV. 655 (2017); Anthony J. McGann et al., *A Discernable and Manageable Standard for Partisan Gerrymandering*, 14 ELECTION L.J. 295 (2015); John F. Nagle, *How Competitive Should a Fair Single Member Districting Plan Be?*, 16 ELECTION L.J. 196 (2017); Samuel S.-H. Wang, *Three Practical Tests for Gerrymandering: Application to Maryland and Wisconsin*, 15 ELECTION L.J. 367 (2016); and Jonathan Krasno et al., *Can Gerrymanders Be Measured?: An Examination of Wisconsin's State Assembly* (May 28, 2016) (unpublished manuscript), <https://perma.cc/7TFY-JQFB>.
21. In the *Whitford* three-judge panel's decision on the merits, both the majority and dissent discussed the efficiency gap proposal in depth. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 854-57 (W.D. Wis. 2016), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017); *id.* at 933-34, 937-38 (Griesbach, J., dissenting). The panel also discussed the proposal when denying Wisconsin's motions to dismiss and for summary judgment. See *Whitford v. Nichol*, 151 F. Supp. 3d 918, 920-22 (W.D. Wis. 2015) (denying motion to dismiss); *Whitford v. Nichol*, 180 F. Supp. 3d 583, 585, 588-93 (W.D. Wis. 2016) (denying motion for summary judgment).

Meanwhile, a federal lawsuit challenging North Carolina's congressional redistricting plan and relying on the efficiency gap measure also survived a motion to dismiss. See *Common Cause v. Rucho*, 240 F. Supp. 3d 376, 377-78, 380 (M.D.N.C. 2017) (*per curiam*). The three-judge panel convened in the Middle District of North Carolina consolidated two cases, one brought by a group of plaintiffs led by Common Cause, the other by a group of plaintiffs led by the League of Women Voters. See *id.* at 377 & n.1. The two groups of plaintiffs make similar legal arguments, but only the League of Women Voters plaintiffs have used the efficiency gap in the discriminatory effect element of the proffered legal test. See *id.* at 380. The *Rucho* panel briefly discussed the efficiency gap proposal when denying the state's motion to dismiss. See *id.* at 380-81. It discussed the efficiency gap in much greater detail when, in January 2018, it held that the state's congressional districting plan was an unconstitutional partisan gerrymander. See *Common Cause v. Rucho*, 279 F. Supp. 3d 587, 597, 658-64, 668-69 (M.D.N.C. 2018) (discussing evidence related to the efficiency gap and concluding that the gap, along with other statistical measures of partisan asymmetry, "provided 'strong proof' of the plan's discriminatory effects (quoting *Sylvester v. SOS Children's Vill. Ill., Inc.*, 453 F.3d 900, 903 (7th Cir. 2006))), *stay granted*, No. 17A745, 2018 WL 472142 (U.S. Jan. 18, 2018), and *appeal docketed*, No. 17-1295 (U.S. Mar. 14, 2018).

This analysis bears on the questions before the Court in *Whitford*. Is partisan gerrymandering justiciable? If so, what is the governing legal standard? Under that standard, is the Wisconsin State Assembly plan a partisan gerrymander? And what role, if any, should the efficiency gap measure play in that standard? I would suggest the following answers to the first three questions: Yes, partisan gerrymandering is justiciable; the principle of partisan symmetry is an appropriate legal standard; and the Wisconsin plan is a partisan gerrymander. The fourth question is the subject of this Article.

The efficiency gap is one of multiple useful indicative measures of partisan asymmetry under circumstances like those in *Whitford*, where each party earns about half the votes and a large efficiency gap persists under plausible variations in voter behavior. However, the Court should not adopt the efficiency gap as the exclusive definitional measure of partisan gerrymandering, such that a plan would be invalid if and only if it exhibited a large, durable, and unjustified efficiency gap. Instead, the Court should permit some flexibility for scholars, litigants, and courts to refine measurement approaches over time and under varying circumstances. Note that this is precisely the approach suggested by leading academics in an amicus brief filed in *Whitford*.²² Furthermore, the Court should acknowledge that partisan gerrymandering is not the only form of political gerrymandering that subverts democratic values and should signal its receptiveness to efforts to define and proscribe other forms of political gerrymandering. Just as excessive departures from partisan symmetry can trigger a partisan gerrymandering claim, perhaps excessive departures from competitiveness should trigger a bipartisan gerrymandering claim or excessive departures from seats-votes proportionality should trigger a minority protection claim.

Were the Court to embrace an approach of measurement refinement over time, this Article would prove relevant to the process through which “lower

22. See Brief of Heather K. Gerken et al. as Amici Curiae in Support of Appellees at 4, *Whitford*, No. 16-1161 (U.S. Aug. 30, 2017), 2017 WL 3774485 (urging the Court to announce “[p]artisan symmetry” as a “workable principle” for adjudicating partisan gerrymandering cases, “one that lends itself to a manageable test, while allowing the lower courts to work out the precise contours of that test with time and experience”); *id.* at 25 (“Partisan-symmetry tests all answer the same, simple question and rely on a shared standard. But they are flexible enough to accommodate contextual differences, thus allowing courts to choose the test best suited for assessing a particular plan. More importantly, a court can assess a districting plan using more than one symmetry test. Extreme gerrymanders will certainly perform poorly along more than one symmetry measure. Judges can thereby use multiple symmetry tests to assure themselves of the robustness of their assessment and identify extreme outliers.” (citations omitted)); see also Brief of Robin Best et al. as Amici Curiae in Support of Appellees at 3, *Whitford*, No. 16-1161 (U.S. Sept. 5, 2017), 2017 WL 4311099 (classifying the Wisconsin State Assembly plan as a partisan gerrymander based on an analysis employing an alternative measure and 10,000 simulated alternative maps).

courts . . . work out the precise contours of [partisan gerrymandering claim analysis] with time and experience.”²³ Additionally, the efficiency gap measure represents a contribution to the election law and political science literatures independent from the role it may play in *Whitford*. Political scientists are exploring the relationship between partisan gerrymandering and other variables of interest, using the efficiency gap measure as the operational definition of partisan gerrymandering.²⁴ An evaluation of the efficiency gap measure is thus relevant not only to whether and how courts proscribe partisan gerrymandering but also to how political scientists study it. Finally, in developing the efficiency gap measure, McGhee has discovered significant, surprising relationships between seats-votes curves and properties of wasted vote measures, such as the fact that under traditional definitions parties waste an equal number of votes when a party translates a 1% increase in votes into a 2% increase in seats.²⁵ This Article identifies other relationships of interest between wasted vote measures, seats-votes proportionality, competitiveness, and voter turnout.

This Article proceeds in five Parts. Part I relates the necessary background in a way that frames the subsequent analysis, suggesting the utility and limits of the efficiency gap measure. Political gerrymandering is a multinormative structural problem the Court has struggled to regulate. The efficiency gap is designed to better measure partisan asymmetry using the ideal of equal wasted votes. But a legal standard for partisan gerrymandering must cohere with both an individual rights framework and a structural account of electoral democracy attentive to the multiple norms at stake. This suggests an inquiry into the efficiency gap’s conceptual design and its relationship to competitiveness and seats-votes proportionality.

Part II explores the efficiency gap’s conceptual design, examining five choices underlying the measure: the “efficiency principle” McGhee developed as a guide to the measure’s design; the equal voter turnout assumption used to reduce the long-form equation to the simplified formula; the method of aggregating wasted votes to produce a single number; the definition and weight of surplus votes; and the two-party assumption.

Part III examines the efficiency gap’s relationship to seats-votes proportionality and competitiveness. It shows that the efficiency gap can be understood as

23. Brief of Heather K. Gerken et al. as Amici Curiae in Support of Appellees, *supra* note 22, at 4.

24. See, e.g., Devin Caughey et al., *Partisan Gerrymandering and the Political Process: Effects on Roll-Call Voting and State Policies*, 16 ELECTION L.J. 453, 454-55 (2017).

25. See McGhee, *supra* note 10, at 68-69 (presenting equation 5, which shows that the efficiency gap reduces to a simplified formula when the same number of ballots are cast in each district, and determining that the balance of wasted votes under that simplified formula “will be equal whenever the majority’s margin in seats is twice its margin in votes”); *id.* app. B at 79-82 (deriving equation 5).

a competitiveness gap expressed in terms of turnout and margin of victory rather than wasted votes or undeserved seats. The efficiency gap measure may allow or even encourage mapmakers to draw plans that undermine electoral competitiveness and proportionality between votes earned and seats won. This creates a false positive problem, where the measure disfavors normatively desirable plans, and a false negative problem, where the measure favors normatively undesirable plans. The doctrinal analyses of intent and justification, as well as sensitivity analysis, only partially address the false positive problem because mapmakers may fear not just invalidation but also litigation. And these tools fail to address the false negative problem because the efficiency gap proposal offers no mechanism to overcome the presumption of validity triggered by a below-threshold gap. This analysis also suggests unacknowledged measure convergence, in which scholars or jurists invoke the competitiveness gap without realizing that it is mathematically equivalent to the efficiency gap. Finally, the definition and weight of surplus votes determines the efficiency gap's relationship to the norms of electoral competitiveness and seats-votes proportionality. With a voter-centric definition of surplus votes (using the full margin of victory rather than half the margin) and a party-centric scale (comparing wasted vote totals rather than shares), the efficiency gap would idealize triple proportionality, exacerbating the extreme vote share problem.

Part IV presents a new wasted vote measure designed to exhibit greater discernibility and structural coherence. This measure defines a surplus vote as the entire (rather than half of the) margin of victory and then compares the parties' wasted vote shares (rather than totals). This conceptual design is more voter-centric in terms of how wasted votes are measured and compared. And the measure bears a relationship to competitiveness and proportionality that better aligns with structural values and electoral reality.

Part V concludes, drawing doctrinal implications from the mathematical and legal analysis preceding it. First, questions of robustness and scope must be addressed when setting the numeric threshold and computing a challenged plan's efficiency gap. Second, given the measure's normatively fraught relationship with competing democratic norms, courts should use it only as an indicative measure and not as the exclusive definition of partisan gerrymandering.

I. The Efficiency Gap's Power and Limits

This Part contextualizes the efficiency gap within the broader effort to define and curb political gerrymandering. It describes the challenges posed by gerrymandering, explains the appeal of the efficiency gap measure, and identifies the questions that motivate this Article's analysis.

A. Gerrymandering as a Multinormative Structural Problem

Jurists, scholars, politicians, media, reformers, and ordinary citizens agree that gerrymandering²⁶ poses a profound threat to democratic values.²⁷ And for good reason. Electoral districting confers on the mapmaker the power to shape electoral destiny—a power too easily abused.²⁸ To favor one party over another,

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26. This Article employs the following terminology throughout: *Gerrymandering* refers broadly to any manipulation of electoral boundaries; *malapportionment* refers to distortion of the population sizes of electoral districts; *racial gerrymandering* refers to manipulation of electoral boundaries on the basis of race; *political gerrymandering* refers to manipulation with political intent and effect; *partisan gerrymandering* refers to manipulation intended to benefit one party over another; and *bipartisan gerrymandering* refers to manipulation intended to preserve safe seats for incumbents from both parties. Thus, partisan gerrymandering is one type of political gerrymandering.
27. See, e.g., *Ariz. State Legislature v. Ariz. Indep. Redistricting Comm'n*, 135 S. Ct. 2652, 2658 (2015) (“[P]artisan gerrymanders . . . [are incompatible] with democratic principles.” (first and third alterations in original) (quoting *Vieth v. Jubelirer*, 541 U.S. 267, 292 (2004) (plurality opinion))); Jack M. Balkin, Closing Keynote Address, *The Last Days of Disco: Why the American Political System Is Dysfunctional*, 94 B.U. L. REV. 1159, 1165 (2014) (including “exclusively single-member districts,” “first-past-the-post election rules,” and “[p]olitical gerrymandering” in a list of causes of “features of our current system that make it dysfunctional”); Robert Draper, *The League of Dangerous Mapmakers*, ATLANTIC (Oct. 2012), <https://perma.cc/M8W5-YXAV> (“[R]edistricting today has become the most insidious practice in American politics . . .”).
28. See Samuel Issacharoff, *Gerrymandering and Political Cartels*, 116 HARV. L. REV. 593, 595 (2002) (“There is a core understanding in American politics, going back to the evocative imagery of the gerrymander, that geographically districted elections are subject to ends-oriented manipulation.”). The risk of gerrymandering is an inherent feature of the practice of *geographic electoral districting*, by which I mean the system in which individual representatives for a multimember body are selected through separate elections conducted in geographic subunits (called electoral districts) of the jurisdiction. Many countries eschew districting entirely, opting instead for some system of proportional representation, whereby representation of the entire body is distributed according to the support each party earns in a single election conducted over the entire jurisdiction. See *Electoral Systems Around the World*, FAIRVOTE, <https://perma.cc/VX93-UFQR> (archived Feb. 20, 2018) (surveying 35 major democracies as of 2012 and finding that 29 of them use some form of proportional or mixed proportional system). But from its inception to the present, the U.S. electoral system has relied heavily on geographic electoral districting. See Paul L. McKaskle, Essay, *Of Wasted Votes and No Influence: An Essay on Voting Systems in the United States*, 35 HOUS. L. REV. 1119, 1124, 1136 (1998). Most states have adopted a *single-member simple plurality system*, under which each electoral district is assigned one seat in the multimember body and each district awards its seat to the candidate who earns the most votes in that district’s race. See Bruce E. Cain, Commentary, *Garrett’s Temptation*, 85 VA. L. REV. 1589, 1601 (1999); Justin Levitt, *What Is Redistricting?*, ALL ABOUT REDISTRICTING, <https://perma.cc/5REF-W3LT> (archived Feb. 20, 2018) (“Most of our federal legislators, all of our state legislators, and many of our local legislators in towns and counties are elected from districts. These districts divide states and the people who live there into geographical territories.”). Electoral districting may in fact offer some advantages over proportional representation systems. See Nathaniel Persily, Reply, *In Defense of Foxes Guarding Henhouses: The Case for Judicial Acquiescence to Incumbent-Protecting Gerrymanders*, 116 HARV. L. REV. 649, 650 (2002); Peter H. Schuck, *The Thickest Thicket: Partisan Gerrymandering and Judicial Regulation of Politics*, 87 COLUM. L. REV. 1325, 1350–51 (1987). But it has one profound disadvantage: It is vulnerable to manipulation by political cartographers.

the mapmaker can simply dilute the influence of the disfavored party's supporters by assigning them to districts where their votes have less impact: either by *packing* them into a few districts where their preferred candidates win by overwhelming margins or by *cracking* them into many districts so that their preferred candidates lose each one. Aided by powerful computers—and prevailing patterns of residence and voting—the modern mapmaker can pack and crack with exquisite precision, thereby distorting the way political parties translate popular support (votes) into governmental power (seats).²⁹ With the stroke of a pen (or a few taps on a keyboard), the mapmaker can confer a legislative majority on a party supported by a minority of voters or a legislative supermajority on a party supported by a slim majority of voters. As one state legislator put it, the practice of gerrymandering turns the process of electoral districting into “the business of rigging elections.”³⁰ This is why legislatures guard their districting power so jealously,³¹ why the districting process is often so partisan and secretive,³² and why parties expend so many resources drawing and litigating electoral districting plans.³³

Electoral districting entails districting power; such power invites abuse; we call such abuse gerrymandering. The term—a portmanteau of the surname

29. See *Vieth*, 541 U.S. at 345–46 (Souter, J., dissenting) (citing Issacharoff, *supra* note 28, at 624; Pamela S. Karlan, *The Fire Next Time: Reapportionment After the 2000 Census*, 50 STAN. L. REV. 731, 736 (1998); and Richard H. Pildes, *Principled Limitations on Racial and Partisan Redistricting*, 106 YALE L.J. 2505, 2553–54 (1997)).

30. See *id.* at 317 (Kennedy, J., concurring in the judgment) (quoting John Hoeffel, *Six Incumbents Are a Week Away from Easy Election*, WINSTON-SALEM J., Jan. 27, 1998, at B1, B1).

31. For example, when Arizona voters, acting through initiative, transferred districting power from the state legislature to an independent commission, the legislature (unsuccessfully) challenged the constitutionality of this initiative all the way up to the U.S. Supreme Court. See *Ariz. State Legislature*, 135 S. Ct. at 2658–59.

32. For example, the Wisconsin State Assembly plan challenged in *Whitford* was produced with the use of nondisclosure agreements, expedited legislative procedures, a war room with limited access, and consultation exclusively with members of one party. See Emily Bazelon, *The New Front in the Gerrymandering Wars: Democracy vs. Math*, N.Y. TIMES MAG. (Aug. 29, 2017), <https://perma.cc/E7EM-XG94> (noting that “[n]early all of the 79 Republicans in the Wisconsin Senate and Assembly [visited] the map room, signing the same secrecy pledge to see the new shape of their districts,” whereas “[n]o Democrat was invited,” and adding that “[t]he Legislature passed the plan a week later, with the support of every Republican . . . and no Democrats”).

33. See, e.g., Lisa Marshall Manheim, *Redistricting Litigation and the Delegation of Democratic Design*, 93 B.U. L. REV. 563, 612–13 (2013); David Daley, *The House the GOP Built: How Republicans Used Soft Money, Big Data, and High-Tech Mapping to Take Control of Congress and Increase Partisanship*, NEW YORK (Apr. 24, 2016, 9:02 PM), <https://perma.cc/22ST-G8QV>. Former President Obama and former Attorney General Eric Holder are focusing on re-districting reform through a newly formed organization called the National Democratic Redistricting Committee. See Edward-Isaac Dove, *Obama, Holder to Lead Post-Trump Redistricting Campaign*, POLITICO (Oct. 17, 2016, 5:06 AM EDT), <https://perma.cc/C84D-VTWD>.

“Gerry” and the word “salamander”—was coined in 1812 by a critic of the districting plan for the Massachusetts Senate, who likened its serpentine appearance to a salamander and suggested that Governor Elbridge Gerry was behind it.³⁴ The term colorfully captures our intuitive sense—and visceral disgust—that manipulation of electoral districts subverts fundamental democratic norms.

But political gerrymandering, like its amphibian namesake, is slippery, repeatedly eluding efforts to curb it, in part because gerrymandering is a slippery concept resistant to precise, consensus-garnering definition and quantification. This is so because gerrymandering is an inherently structural phenomenon concerning the functioning of a healthy electoral system,³⁵ and the relevant structural analysis is irreducibly multinormative. Electoral districting implicates, and gerrymandering threatens, multiple democratic norms—including electoral competition, voter participation, majoritarianism, minority protection, and partisan fairness. There is a high-level consensus that districting power may be abused but dissensus on the right way to draw electoral districts and thus disagreement on precisely how to define and measure gerrymandering.³⁶ Just as different doctors may disagree on the most salient components of health at stake in any given treatment decision, legal scholars “are divided as to what [is] the most important structural consideration” that “capture[s] what is truly at stake” in electoral districting.³⁷ As Stephanopoulos puts it, “Two approaches to redistricting have dominated the academic debate over the last generation: the partisan fairness approach, advocating that district plans treat the major parties

34. See *Ariz. State Legislature*, 135 S. Ct. at 2658 n.1 (citing ELMER C. GRIFFITH, *THE RISE AND DEVELOPMENT OF THE GERRYMANDER* 16-19 (Arno Press 1974) (1907)); *Vieth*, 541 U.S. at 274 (plurality opinion) (citing *Gerrymander*, WEBSTER'S NEW INTERNATIONAL DICTIONARY (2d ed. 1945)).

35. See Heather K. Gerken, Essay, *Lost in the Political Thicket: The Court, Election Law, and the Doctrinal Interregnum*, 153 U. PA. L. REV. 503, 508 (2004) (“It is hard to figure out what is ‘fair’ or ‘equal’ in districting without speaking in structural terms. Any such conclusion would require a theory of representation, an idea about how a healthy democracy is supposed to function.”).

36. See *Vieth*, 541 U.S. at 307 (Kennedy, J., concurring in the judgment) (“No substantive definition of fairness in districting seems to command general assent.”); Krasno et al., *supra* note 20, at 1 (“Partisan gerrymandering shares both of the characteristics of pornography that Potter Stewart famously wrestled with in his concurring opinion in *Jacobellis*: it is difficult to measure objectively and (therefore) a matter of subjective opinion.”); see also *Jacobellis v. Ohio*, 378 U.S. 184, 197 (1964) (Stewart, J., concurring) (“[U]nder the First and Fourteenth Amendments criminal laws in this area are constitutionally limited to hardcore pornography. I shall not today attempt further to define the kinds of material I understand to be embraced within that shorthand description But I know it when I see it” (footnote omitted)).

37. Nicholas O. Stephanopoulos, *Elections and Alignment*, 114 COLUM. L. REV. 283, 295-96 (2014).

symmetrically, and the competitiveness approach, advising that districts be made as competitive as is feasible.”³⁸

Some of the key democratic norms at stake involve the relationship between the votes a party earns and the seats it wins. I will now introduce some notation and basic properties that help to analyze this relationship; these will feature prominently throughout this Article. Let V and S respectively denote the total number of ballots cast and the total number of seats awarded in the election. Assume that there is a set of parties (P), and for each party $p \in P$ let V_p , $\bar{V}_p$, and V_p^* respectively denote that party’s vote total, vote share, and vote margin, and let S_p , $\bar{S}_p$, and S_p^* respectively denote that party’s seat total, seat share, and seat margin.³⁹

A party p ’s vote total (V_p) is simply the number of ballots cast for that party (across all districts); its vote share ($\bar{V}_p$) is its vote total divided by the total number of ballots cast ($\bar{V}_p = \frac{V_p}{V}$); and its vote margin (V_p^*) is the difference between its vote share and 50% ($V_p^* = \bar{V}_p - \frac{1}{2}$). For example, if 100 ballots are cast ($V = 100$) and a party earns sixty of them ($V_p = 60$), its vote share is 60% ($\bar{V}_p = \frac{V_p}{V} = \frac{60}{100} = 0.6$), and its vote margin is 10% ($V_p^* = \bar{V}_p - \frac{1}{2} = 0.6 - 0.5 = 0.1$).

The seat variables are defined similarly: A party p ’s seat total (S_p) is the number of seats won by candidates of that party; its seat share ($\bar{S}_p$) is its seat total divided by the total number of seats ($\bar{S}_p = \frac{S_p}{S}$); and its seat margin (S_p^*) is the difference between its seat share and 50% ($S_p^* = \bar{S}_p - \frac{1}{2}$). For example, if a congressional plan consists of ten seats ($S = 10$) and a party earns four of them ($S_p = 4$), its seat share is 40% ($\bar{S}_p = \frac{S_p}{S} = \frac{4}{10} = 0.4$), and its seat margin is -10% ($S_p^* = \bar{S}_p - \frac{1}{2} = 0.4 - 0.5 = -0.1$). Note that under a two-party assumption, a positive vote (seat) margin connotes majority status while a negative vote (seat) margin connotes minority status.

These variables relate in simple ways in the special case where there are only two parties, an assumption generally adopted in the efficiency gap approach⁴⁰ and in much of this Article’s analysis. In this case, I refer to the two parties as party x and party y ; that is, $P = \{x, y\}$. Because every ballot is cast for, and every seat won by, one party or the other—ignoring the possibility of a tie—simple

38. Nicholas O. Stephanopoulos, *The Consequences of Consequentialist Criteria*, 3 U.C. IRVINE L. REV. 669, 673 (2013).

39. Under the notation used throughout this Article, the bar accent indicates a share, while the star superscript indicates a margin.

40. See Stephanopoulos & McGhee, *supra* note 7, at 853.

relationships apply. The overall vote (seat) total is the sum of the two parties' vote (seat) totals.

$$V_x + V_y = V$$

$$S_x + S_y = S$$

Respective vote (seat) shares sum to one.⁴¹

$$\bar{V}_x + \bar{V}_y = 1$$

$$\bar{S}_x + \bar{S}_y = 1$$

Respective vote (seat) margins sum to zero.⁴²

$$V_x^* + V_y^* = 0$$

$$S_x^* + S_y^* = 0$$

This means that the parties' vote (seat) margins have equal magnitudes but opposite signs: $V_x^* = -V_y^*$ and $S_x^* = -S_y^*$. For this reason, I will sometimes assume, without loss of generality, that party x enjoys a positive vote (seat) margin and refer to the vote margin simply as V^* instead of V_x^* and the seat margin simply as S^* instead of S_x^* . Each party's vote (seat) share can then be expressed in terms of the vote (seat) margin.⁴³

$$\bar{V}_x = \frac{1}{2} + V^*; \bar{V}_y = \frac{1}{2} - V^*$$

$$\bar{S}_x = \frac{1}{2} + S^*; \bar{S}_y = \frac{1}{2} - S^*$$

And the difference between vote (seat) shares is twice the vote (seat) margin.⁴⁴

$$\bar{V}_x - \bar{V}_y = 2V^*$$

$$\bar{S}_x - \bar{S}_y = 2S^*$$

41. Proof: $\bar{V}_x + \bar{V}_y = \frac{V_x}{V} + \frac{V_y}{V} = \frac{V_x + V_y}{V_x + V_y} = 1$; $\bar{S}_x + \bar{S}_y = \frac{S_x}{S} + \frac{S_y}{S} = \frac{S_x + S_y}{S_x + S_y} = 1$.

42. Proof: $V_x^* + V_y^* = \left(\bar{V}_x - \frac{1}{2}\right) + \left(\bar{V}_y - \frac{1}{2}\right) = (\bar{V}_x + \bar{V}_y) - \left(\frac{1}{2} + \frac{1}{2}\right) = 1 - 1 = 0$;

$S_x^* + S_y^* = \left(\bar{S}_x - \frac{1}{2}\right) + \left(\bar{S}_y - \frac{1}{2}\right) = (\bar{S}_x + \bar{S}_y) - \left(\frac{1}{2} + \frac{1}{2}\right) = 1 - 1 = 0$.

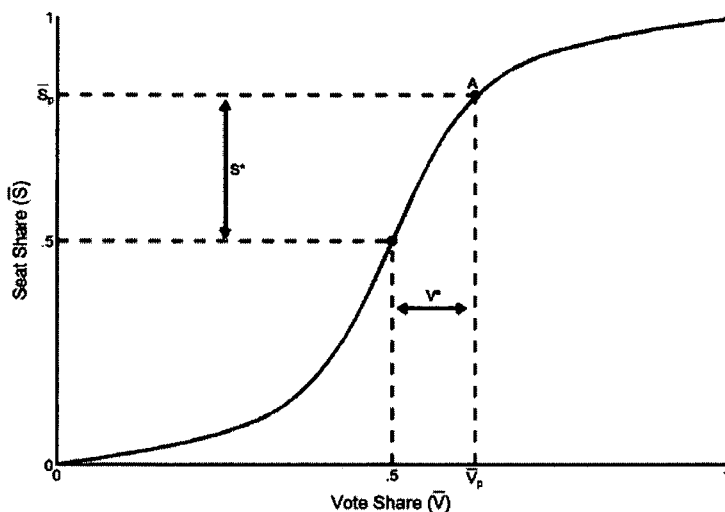
43. Proof: $V^* = V_x^* = \bar{V}_x - \frac{1}{2}$. Thus: $\bar{V}_x = \frac{1}{2} + V^*$; $\bar{V}_y = 1 - \bar{V}_x = 1 - \left(\frac{1}{2} + V^*\right) = \frac{1}{2} - V^*$.

$S^* = S_x^* = \bar{S}_x - \frac{1}{2}$. Thus: $\bar{S}_x = \frac{1}{2} + S^*$; $\bar{S}_y = 1 - \bar{S}_x = 1 - \left(\frac{1}{2} + S^*\right) = \frac{1}{2} - S^*$.

44. Proof: $\bar{V}_x - \bar{V}_y = \left(\frac{1}{2} + V^*\right) - \left(\frac{1}{2} - V^*\right) = 2V^*$; $\bar{S}_x - \bar{S}_y = \left(\frac{1}{2} + S^*\right) - \left(\frac{1}{2} - S^*\right) = 2S^*$.

Political scientists analyze how a party's seat share ($\bar{S}_p$) does and should vary with its vote share ($\bar{V}_p$) by conceptualizing seat share as a function of vote share.⁴⁵ They illustrate this relationship graphically by drawing a *seats-votes curve*.⁴⁶ Figure 1 below provides an example of a seats-votes curve. A single point on the curve represents an electoral outcome, that is, the seat share a party earns at a given vote share; the curve itself represents a range of outcomes corresponding to different values of the party's vote share.

Figure 1



Point A represents a single election outcome at which the party earns seat share $\bar{S}_p$ with vote share $\bar{V}_p$. The curve represents a range of outcomes corresponding to different values of vote share.

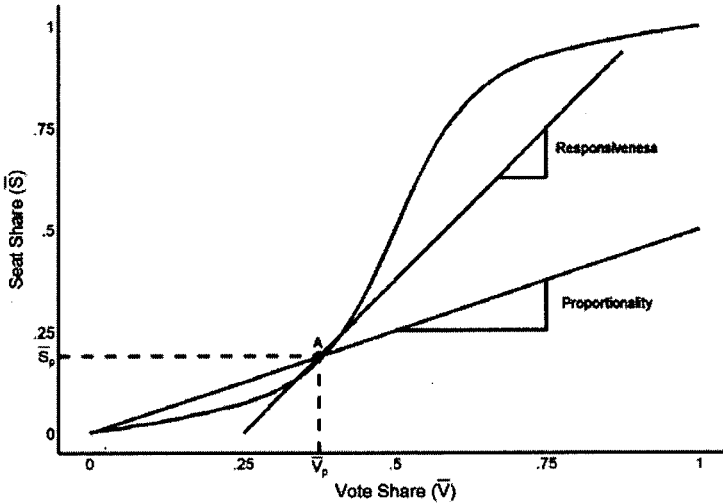
Figure 2 below illustrates the concepts of *seats-votes proportionality* and *seats-votes responsiveness*. Seats-votes proportionality captures the absolute relationship between vote share and seat share. Graphically, it is the slope of the line connecting the origin (0,0) to the point of the observed electoral outcome. Seats-votes responsiveness captures the marginal relationship: the ratio between an incremental change in vote share and the corresponding incremental change in

45. See, e.g., Edward R. Tufte, *The Relationship Between Seats and Votes in Two-Party Systems*, 67 AM. POL. SCI. REV. 540 (1973).

46. A *descriptive* seats-votes curve estimates the relationship that actually exists in the real world. A *prescriptive* seats-votes curve indicates the ideal relationship that ought to exist in a healthy, well-functioning democracy.

seat share. Graphically, it is the slope of the tangent at the point of the observed electoral outcome. In the language of differential calculus, it is the derivative of the seats-votes function depicted in Figure 1 above at that point. Seats-votes responsiveness is one measure of electoral competitiveness: The more districts with close races won by small margins of victory, the more seat flips (and therefore seat share changes) for a given shift in vote share.⁴⁷

Figure 2



Seats-votes proportionality at point A is the slope of the line connecting point A to the origin (0,0), which is the ratio of $\bar{S}_p$ to $\bar{V}_p$. Seats-votes responsiveness at point A is the slope of the tangent line passing through point A, which is the derivative of $\bar{S}(\bar{V})$ at $\bar{V}_p$.

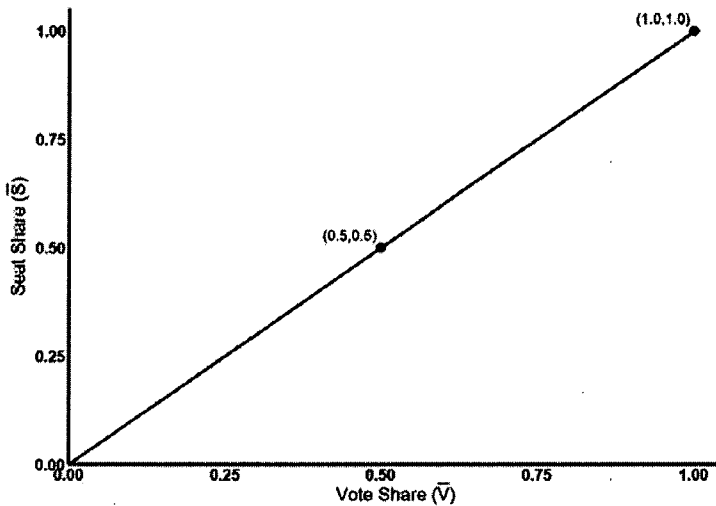
Only when a seats-votes curve is truly curved can proportionality and responsiveness diverge. If a seats-votes curve is a straight line, proportionality and responsiveness are equal. For example, strict proportionality, where a party's seat share is identical to its vote share, corresponds to a straight line from the point (0,0) (where a party receives no votes and no seats) to the point (1,1) (where a party receives all the votes and all the seats), passing through the point (0.5,0.5) (where a party receives half the votes and half the seats). Some argue for an ideal of strict proportionality between a party's vote share and its seat share and, on

47. See, e.g., Stephanopoulos, *supra* note 38, at 678 ("[E]lectoral responsiveness indicate[s] both how competitive individual districts are and how responsive a jurisdiction's electoral system is as a whole.").

this basis, propose that we replace our districting-based electoral system with one explicitly based on proportional representation.⁴⁸

In the real world, the seats-votes curves estimated by political scientists are not straight lines, but rather S-shaped curves that exhibit lower responsiveness (flatter slopes) when one party enjoys a large majority and higher responsiveness (steeper slopes) when the electorate is more evenly split between the two parties. Figures 3 and 4 below provide examples of linear (strictly proportional) and nonlinear (real-world) seats-votes curves.

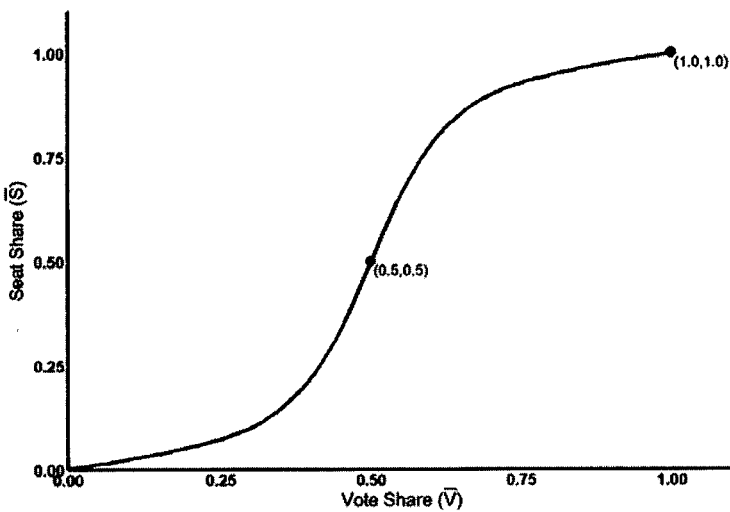
Figure 3



A linear seats-votes curve satisfying strict seats-votes proportionality.

48. See generally, e.g., DOUGLAS J. AMY, *REAL CHOICES/NEW VOICES: HOW PROPORTIONAL REPRESENTATION ELECTIONS COULD REVITALIZE AMERICAN DEMOCRACY* (2d ed. 2002).

Figure 4



A nonlinear seats-votes curve exhibiting higher responsiveness (steeper slopes) when the electorate is more evenly split between the two parties and lower responsiveness (flatter slopes) when one party enjoys a large majority.

Political scientists have found that average seats-votes responsiveness is usually not 1, as strict proportionality would require, but generally closer to 2.⁴⁹ The result is a “seat bonus”: the majority translates a positive vote margin into an even larger seat margin.⁵⁰ For example, with a 51% vote share (1% vote margin), a party may earn a 52% seat share (2% seat margin) instead of the 51% seat share that would be required under strict proportionality. Some argue that seat

49. See, e.g., Nicholas M. Goedert, *Redistricting, Risk, and Representation: How Five State Gerrymanders Weathered the Tides of the 2000s*, 13 ELECTION L.J. 406, 413 (2014) (“[T]he slope in U.S. congressional elections is often found to be around 2.” (citations omitted)); Edward R. Tufte, *The Relationship Between Seats and Votes in Two-Party Systems*, 67 AM. POL. SCI. REV. 540, 542 (1973) (finding a “swing ratio” of 1.9 over a period with twelve congressional elections). The *Whitford* majority emphasized that both parties stipulated that the simplified formula’s “implied 2-to-1 votes-to-seats relationship reflects the ‘observed average seat/votes curve in historical U.S. congressional and legislative elections.’” *Whitford v. Gill*, 218 F. Supp. 3d 837, 907 (W.D. Wis. 2016) (quoting Joint Final Pretrial Report ¶ 105, *Whitford*, 218 F. Supp. 3d 837 (No. 3:15-cv-00421-bbc)), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

50. See Mitchell N. Berman, *Managing Gerrymandering*, 83 TEX. L. REV. 781, 806 n.165 (2005) (discussing the seat bonus and noting that it is sometimes termed the “winner’s bonus”); Adam Cox, Commentary, *Partisan Fairness and Redistricting Politics*, 79 N.Y.U. L. REV. 751, 765 (2004) (same).

bonuses are normatively undesirable because of the departure from strict proportionality;⁵¹ others argue that they are normatively desirable because the bonus incentivizes robust campaigning and promotes a stable, functioning legislative majority.⁵²

The S-shaped nature of actual seats-votes curves reflects an accommodation between competing norms, ensuring representation for minority parties while rewarding majority parties. As Robert Browning and Gary King aptly put it, this approach “reflects an important principle of the United States two-party, democratic system. It helps majorities form, yet protects the minority party.”⁵³ Specifically, it helps the majority party by exhibiting high seats-votes responsiveness when the two parties earn similar vote shares but protects the minority party by exhibiting low seats-votes responsiveness when one party earns most of the vote share.⁵⁴

The seats-votes framework helps illustrate how electoral districting implicates tradeoffs between multiple democratic values. This in turn helps explain why there is no universal agreement on the right way to draw electoral districts. Of course, the task with which the Supreme Court is currently faced is not to prescribe ideal districting practices as a matter of policy or abstract democratic theory, but rather to distinguish valid from invalid districting practices as a matter of federal constitutional law. And the question fracturing the Court is less whether partisan gerrymandering presents a constitutional problem than whether the Court can address it. In the language of federal courts jurisprudence, the question is whether gerrymandering constitutes a justiciable legal claim the courts can adjudicate or a nonjusticiable political question the courts cannot address, which turns on whether the Court can identify a judicially discernible and manageable standard to channel and limit judicial intervention.⁵⁵

51. See, e.g., AMY, *supra* note 48, at 34–41 (“This system unfairly diminishes the power of minority parties and artificially enhances the power of the largest party.”).

52. See, e.g., Schuck, *supra* note 28, at 1350–51 (“By raising the returns to victory . . . , [this system] may attract more talented, entrepreneurial individuals into public life. . . . Citizens and parties may prefer the stability, power aggregation, and accountability to voters that a victory bonus encourages.”).

53. Robert X. Browning & Gary King, *Seats, Votes, and Gerrymandering: Estimating Representation and Bias in State Legislative Redistricting*, 9 LAW & POL’Y 305, 313 (1987).

54. See *supra* Figure 4.

55. See, e.g., Jurisdictional Statement at i, *Gill v. Whitford*, No. 16–1161 (U.S. Mar. 24, 2017), 2017 WL 1131500. The presence or absence of a judicially discernible and manageable standard is just one of six factors under the political question doctrine. The first is whether the constitutional text provides for resolution of the issue by a coordinate branch; another is whether there is “an unusual need for unquestioning adherence to a political decision already made”; and the remaining factors concern whether adjudication would require courts to make policy determinations they are ill equipped to make, express disrespect to a coordinate branch, decide issues without “judicially discoverable and manageable standards,” or risk embarrassment from interbranch dissensus. See *Baker v. Carr*, 369 U.S. 186, 217 (1962).

Thus, gerrymandering presents two distinct but related questions: justiciability (is there an adequate standard to guide judicial intervention?) and identification (what is that standard?). These questions implicate both the relationship between electoral districting practices and constitutionally significant representational norms as well as the proper role of the federal judiciary in regulating electoral districting practices pursuant to these norms. Both are democratic problems of profound constitutional significance on which the Constitution provides limited explicit guidance. Although political gerrymanders undoubtedly implicate constitutional values,⁵⁶ the Constitution's text offers limited procedural guidance on congressional and state legislative elections.⁵⁷ And the precise scope of the federal judicial power to adjudicate federal constitutional claims is also a question without an explicit textual answer.⁵⁸ In this sense, both action and inaction by the Court on political gerrymandering claims present real but ineffable constitutional risks.

56. See sources cited *supra* note 2. The Constitution provides that members of the House of Representatives are to be "chosen . . . by the People," U.S. CONST. art. I, § 2, cl. 1; "guarantee[s] to every State . . . a Republican Form of Government," *id.* art. IV, § 4; protects freedoms of expression and association, *id.* amend. I; enshrines due process and equal protection, *id.* amend. V; *id.* amend. XIV, § 1; and prohibits race-based electoral discrimination, *id.* amend. XIV, § 1; amend. XV, § 1. Each of these provisions codifies values threatened by gerrymandering.

57. Article I vests the federal legislative power in a Congress composed of two multimember legislative bodies: the House of Representatives, apportioned on a population basis and popularly elected, see U.S. CONST. art. I, § 2, cls. 1, 3, and a Senate, apportioned on the basis of equal state suffrage, see *id.* art. I, § 3, cl. 1; *id.* art. V, with each state's senators originally "chosen by the Legislature thereof," *id.* art. I, § 3, cl. 1 (amended 1913), and now "elected by the people thereof," *id.* amend. XVII. But the Elections Clause does not mandate how to conduct congressional elections; instead, it gives the choice to individual states and Congress. See *id.* art. I, § 4, cl. 1. And while the Constitution assumes that each state has at least one popularly elected legislative body, see, e.g., *id.* art. I, § 2, cl. 1; *id.* amend. XVII, it says nothing about how to conduct state legislative elections, implicitly leaving that choice to each state as well.

58. Article III, Section 2 provides that "[t]he judicial Power shall extend" to an enumerated set of "Cases" and "Controversies," the first of which is "all Cases, in Law and Equity, arising under" federal law. U.S. CONST. art. III, § 2, cl. 1. While "[i]t is emphatically the province and duty of the judicial department to say what the law is," *Marbury v. Madison*, 5 U.S. (1 Cranch) 137, 177 (1803), some "subjects are political" and so "can never be examinable by the courts," *id.* at 166. The political question doctrine, like all justiciability doctrines partially discerned from the text of Article III, relates in part to "an idea, which is more than an intuition but less than a rigorous and explicit theory, about the constitutional and prudential limits to the powers of an unelected, unrepresentative judiciary in our kind of government." See *Elk Grove Unified Sch. Dist. v. Newdow*, 542 U.S. 1, 11 (2004) (quoting *Allen v. Wright*, 468 U.S. 737, 750 (1984)), *abrogated in other part by Lexmark Int'l, Inc. v. Static Control Components, Inc.*, 134 S. Ct. 1377 (2014). One may question whether the Court, in developing the political question doctrine, has succeeded in divining, from Article III and its animating structural principles, a discernible and manageable test for justiciability.

B. The Efficiency Gap as an Improved Measure of Partisan Symmetry

Because political gerrymandering is so slippery, it has repeatedly eluded the Court's grasp. The Court has constrained race-based manipulation of district shape⁵⁹ and political manipulation of district population size⁶⁰—meaningfully limiting mapmakers' packing and cracking abilities. But the Court has repeatedly fractured on whether and how to intervene when political cartographers manipulate district shape based on party support.

In *Davis v. Bandemer*, three Justices concluded that “partisan gerrymandering claims . . . raise a nonjusticiable political question,”⁶¹ but six Justices insisted that

59. When electoral districting implicates race, the Court and Congress have imposed two principal legal constraints. First, pursuant to the Reconstruction Amendments, the Court evaluates a districting plan under strict scrutiny whenever considerations of race predominate, thereby curtailing intentional race-based cracking or packing. See *Cooper v. Harris*, 137 S. Ct. 1455, 1464 (2017); *Bethune-Hill v. Va. State Bd. of Elections*, 137 S. Ct. 788, 798 (2017); *Shaw v. Reno*, 509 U.S. 630, 642 (1993). Second, the prohibition on racial vote dilution codified in section 2 of the Voting Rights Act as amended in 1982, see Voting Rights Act Amendments of 1982, Pub. L. No. 97-205, 96 Stat. 131 (codified as amended in scattered sections of 52 U.S.C.), constrains the ability of a mapmaker to intentionally or in effect dilute the voting strength of racial minorities. See 52 U.S.C. § 10301 (2016); *Bartlett v. Strickland*, 556 U.S. 1, 11-12 (2009) (opinion of Kennedy, J.); *Johnson v. De Grandy*, 512 U.S. 997, 1011 (1994) (citing *Grove v. Emison*, 507 U.S. 25, 39-40 (1993)); *Thornburg v. Gingles*, 478 U.S. 30, 48-51 (1986). Section 5 of the Voting Rights Act of 1965, see Pub. L. No. 89-110, § 5, 79 Stat. 437, 439 (codified as amended at 52 U.S.C. § 10304), imposed one additional constraint: A covered jurisdiction may only implement a new districting plan, or any other electoral “standard, practice, or procedure,” after persuading the Attorney General or the U.S. District Court for the District of Columbia that the plan would not have a racially retrogressive effect. See 52 U.S.C. § 10304. But because the Court struck down the coverage formula contained in 52 U.S.C. § 10303, see *Shelby County v. Holder*, 133 S. Ct. 2612, 2631 (2013), and Congress has yet to adopt a new one, this third constraint on racial gerrymandering is presently inoperative.

60. For decades, the Court dismissed malapportionment challenges as nonjusticiable political questions, heeding Justice Frankfurter's admonition not to enter the “political thicket,” see *Colegrove v. Green*, 328 U.S. 549, 556 (1946) (opinion of Frankfurter, J.), before reversing course, adopting the one person, one vote principle, and thereby launching the reapportionment revolution, see *Reynolds v. Sims*, 377 U.S. 533, 556-58 (1964); *Baker*, 369 U.S. at 197-98. For examples of scholarship recognizing and exploring the significance of the reapportionment revolution, see GORDON E. BAKER, *THE REAPPORTIONMENT REVOLUTION: REPRESENTATION, POLITICAL POWER, AND THE SUPREME COURT* 3-6 (1966); GARY W. COX & JONATHAN N. KATZ, *ELBRIDGE GERRY'S SALAMANDER: THE ELECTORAL CONSEQUENCES OF THE REAPPORTIONMENT REVOLUTION* 12-13 (2002); Gordon E. Baker, *The Unfinished Reapportionment Revolution*, in *POLITICAL GERRYMANDERING AND THE COURTS* 11, 11-16 (Bernard Grofman ed., 1990).

61. See 478 U.S. 109, 144 (1986) (O'Connor, J., concurring in the judgment) (joined by Chief Justice Burger and then-Justice Rehnquist), *abrogated in part by* *Vieth v. Jubelirer*, 541 U.S. 267 (2004).

such claims were justiciable under the Equal Protection Clause⁶² and agreed that plaintiffs must demonstrate both discriminatory intent and discriminatory effect.⁶³ Yet those six Justices disagreed among themselves on the correct legal test for discriminatory effect.⁶⁴ For the next eighteen years, the lower courts applied the plurality's standard—and rejected at the motions stage every partisan gerrymandering claim they considered.⁶⁵

In *Vieth v. Jubelirer*, the four conservative Justices then on the Court concluded that partisan gerrymandering is a nonjusticiable political question.⁶⁶ The four liberal Justices insisted that it is justiciable but offered three different legal tests.⁶⁷ Justice Kennedy rejected each standard proposed but suggested that partisan gerrymandering may be justiciable if a suitable standard could be

62. See *id.* at 113 (plurality opinion) (authored by Justice White and joined by Justices Brennan, Marshall, and Blackmun); *id.* at 161 (Powell, J., concurring in part and dissenting in part) (joined by Justice Stevens).

63. See *id.* at 127 (plurality opinion); *id.* at 161 (Powell, J., concurring in part and dissenting in part).

64. Justice White, writing for a four-Justice plurality, proposed a stringent but vague “consistent degradation” test under which a departure from seats-votes proportionality would be insufficient to establish discrimination. See *id.* at 132 (plurality opinion) (“[T]he mere lack of proportional representation will not be sufficient to prove unconstitutional discrimination. . . . Rather, unconstitutional discrimination occurs only when the [challenged plan] . . . will consistently degrade a voter’s or a group of voters’ influence on the political process as a whole.”). Justice Powell, joined by Justice Stevens, proposed a standard that would have been easier for courts to apply and plaintiffs to meet. See *id.* at 173-74, 173 n.13 (Powell, J., concurring in part and dissenting in part) (proposing consideration of multiple factors, including “the shapes of voting districts,” “adherence to established political subdivision boundaries,” “the legislative procedures by which the apportionment law was adopted,” “population disparities,” and “disproportionate election results”).

65. See *Vieth*, 541 U.S. at 279-80 (plurality opinion) (“[I]n all of the cases we are aware of involving that most common form of political gerrymandering, [that involving the drawing of district lines,] relief was denied.”); *id.* at 280 n.6 (collecting cases).

66. See *id.* at 281 (authored by Justice Scalia and joined by Chief Justice Rehnquist and Justices O’Connor and Thomas).

67. Justice Stevens suggested a “predominant motivation” standard based on the *Shaw v. Reno* cause of action for racial gerrymandering. See *id.* at 339, 341 (Stevens, J., dissenting). Justice Souter, joined by Justice Ginsburg, proposed a burden-shifting framework modeled on Title VII doctrine with a five-factor *prima facie* case. See *id.* at 346-51 (Souter, J., dissenting). Justice Breyer proposed a test based on “unjustified [partisan] entrenchment,” in which a party with a minority of vote share achieves a majority of seat share through “partisan manipulation.” See *id.* at 360 (Breyer, J., dissenting) (emphasis omitted). The *Vieth* plaintiffs proposed to demonstrate discriminatory effects by showing that a challenged plan “systematically ‘pack[s]’ and ‘crack[s]’ the rival party’s voters” and thereby threatens to “thwart the plaintiffs’ ability to translate a majority of votes into a majority of seats.” See *id.* at 286-87 (plurality opinion) (quoting Brief for Appellants at 19-20, *Vieth*, 541 U.S. 267 (No. 02-1580), 2003 WL 22070244).

identified.⁶⁸ Justice Kennedy emphasized that judicial intervention required “clear, manageable, and politically neutral standards for measuring the particular burden a given partisan classification imposes on representational rights,” and he suggested that the First Amendment may offer a better textual basis than the Equal Protection Clause for such standards.⁶⁹ Specifically, Justice Kennedy wrote that partisan gerrymandering may infringe “the First Amendment interest of not burdening or penalizing citizens because of their participation in the electoral process, their voting history, their association with a political party, or their expression of political views.”⁷⁰

Two years later, in *League of United Latin American Citizens v. Perry* (*LULAC*), the Court fractured along similar lines.⁷¹ Justice Scalia, joined by Justice Thomas, continued to insist that partisan gerrymandering is a political question.⁷² The four liberal Justices continued to favor justiciability and suggest alternative legal tests.⁷³ Justice Kennedy again rejected each proffered standard but left open the possibility that an adequate standard may yet materialize.⁷⁴

68. See *id.* at 306, 308 (Kennedy, J., concurring in the judgment).

69. See *id.* at 307-08; *id.* at 314 (“The First Amendment may be the more relevant constitutional provision in future cases that allege unconstitutional partisan gerrymandering.”); *id.* at 315 (“Where it is alleged that a gerrymander had the purpose and effect of imposing burdens on a disfavored party and its voters, the First Amendment may offer a sounder and more prudential basis for intervention than does the Equal Protection Clause.”).

70. *Id.* at 314.

71. See 548 U.S. 399, 406-07 (2006); *supra* note 4 and accompanying text. The case fractured the Court in a particularly severe and complex fashion because it presented both an unsuccessful claim of partisan gerrymandering and a successful claim of racial vote dilution under section 2 of the Voting Rights Act. Compare *LULAC*, 548 U.S. at 423 (“[A]ppellants have established no legally impermissible use of political classifications.”), with *id.* at 442 (“[T]he totality of the circumstances demonstrates a § 2 violation.”).

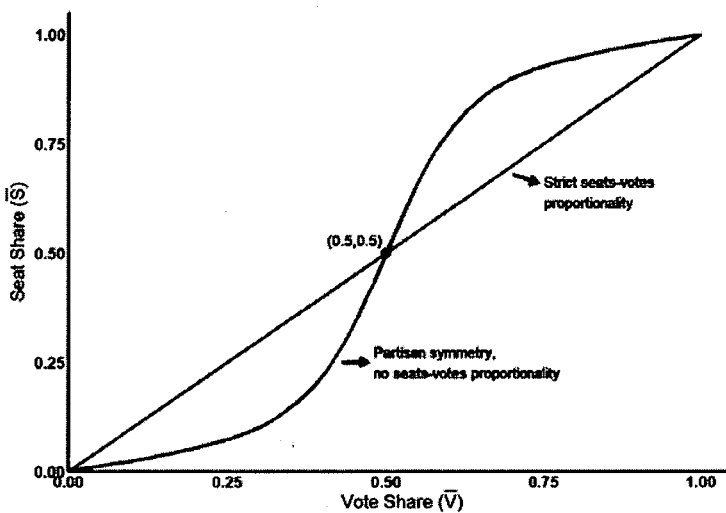
72. See *LULAC*, 548 U.S. at 511 (Scalia, J., concurring in the judgment in part and dissenting in part).

73. See *id.* at 447 (Stevens, J., concurring in part and dissenting in part); *id.* at 483 (Souter, J., concurring in part and dissenting in part) (joined by Justice Ginsburg); *id.* at 491-92 (Breyer, J., concurring in part and dissenting in part).

74. See *id.* at 414 (majority opinion) (recounting the *Vieth* majority’s refusal to hold partisan gerrymanders nonjusticiable, declining to “revisit the justiciability holding,” and “proceed[ing] to examine whether appellants’ claims offer the Court a manageable, reliable measure of fairness for determining whether a partisan gerrymander violates the Constitution”); *id.* at 446 (opinion of Kennedy, J.) (“[A]ppellants . . . lack any reliable measure of partisan fairness.”); *id.* at 418 (“[A] successful claim attempting to identify unconstitutional acts of partisan gerrymandering must . . . show a burden, as measured by a reliable standard, on the complainants’ representational rights.”). Chief Justice Roberts, joined by Justice Alito, agreed with Justice Kennedy’s conclusion that the plaintiffs failed to provide an adequate standard but declined to weigh in on the question of justiciability. See *id.* at 492-93 (Roberts, C.J., concurring in part, concurring in the judgment in part, and dissenting in part).

Additionally, the Justices in *LULAC* considered a new partisan “symmetry” standard proposed by political scientists Gary King and colleagues.⁷⁵ That standard “require[d] that the electoral system treat similarly-situated political parties equally, so that each receives the same fraction of legislative seats for a particular vote percentage as the other party would receive if it had received the same percentage” of the vote.⁷⁶ This principle of partisan symmetry does not require strict seats-votes proportionality.⁷⁷ Rather, in the language of the seats-votes framework, this principle requires that the seats-votes curve be symmetric, which entails in particular that it passes through the point (0.5,0.5) so that each of the two major parties gets half the seats when it earns half the votes. Figure 5 below clarifies the difference between partisan symmetry and strict seats-votes proportionality.

Figure 5



The nonlinear curve satisfies partisan symmetry but not strict seats-votes proportionality. The linear curve satisfies both partisan symmetry and strict seats-votes proportionality.

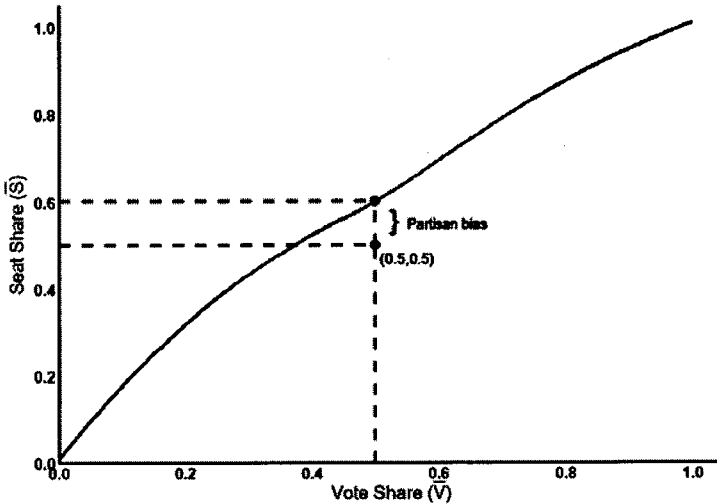
75. See Brief of Amici Curiae Professors Gary King et al. in Support of Neither Party at 4-5, *LULAC*, 548 U.S. 399 (Nos. 05-204 et al.), 2006 WL 53994.

76. *Id.*

77. See *id.* at 7-8 (“Measuring symmetry and partisan bias does *not* require ‘proportional representation’ (where each party receives the same proportion of seats as it receives in votes).”).

If the seats-votes curve accords one party more than half the seats when it earns half the votes, the additional seats constitute a measure of partisan asymmetry that political scientists often refer to as “bias.” Figure 6 below provides an example of a curve that violates partisan symmetry and thereby exhibits bias.

Figure 6



An example of a curve that violates partisan symmetry: With 50% of the votes, one party earns 60% of the seats while the other earns 40%.

This notion of partisan symmetry is normatively appealing, but it necessarily entails some comparison to a counterfactual hypothetical where the parties’ respective vote shares are switched or each party earns 50% of the votes. To evaluate this hypothetical, an analyst must make some assumptions about the geographic distribution of vote-switchers across electoral districts.

The liberal Justices in *LULAC* expressed interest in the partisan symmetry concept.⁷⁸ But Justice Kennedy identified three concerns with this proposal:

78. See *LULAC*, 548 U.S. at 466 (Stevens, J., concurring in part and dissenting in part) (“[T]he symmetry standard, a measure social scientists use to assess partisan bias, . . . is undoubtedly ‘a reliable standard’ . . .” (quoting *id.* at 418 (opinion of Kennedy, J.))); see also *id.* at 483–84 (Souter, J., concurring in part and dissenting in part) (declining to “rule out the utility of a criterion of symmetry as a test,” noting that “[i]nterest in exploring this notion is evident,” and suggesting that “further attention could be devoted to [its] administrability”).

(1) it involved “conjecture”;⁷⁹ (2) it relied on a counterfactual “hypothetical” rather than a directly observed election;⁸⁰ and (3) it provided no guidance on how much departure from the ideal is “too much.”⁸¹

Since *LULAC*, partisan gerrymandering has remained in doctrinal limbo. Without a clear legal test for partisan gerrymandering, plaintiffs have an incentive to attack political gerrymanders as racial gerrymanders, and defendants have an incentive to justify partisan plans as efforts to comply with the Voting Rights Act.⁸² In the context of *conjoined polarization*—“[t]he more consistent alignment of race, party, and ideology since 1965,” when the original Voting Rights Act was enacted⁸³—race and party are easy for litigants to conflate and hard for courts to distinguish.⁸⁴ And perversely, by requiring perennial redistricting, the one person, one vote standard gave mapmakers new opportunities to manipulate electoral district boundaries, making electoral districting a moving target resistant to judicial oversight.⁸⁵

It was against this backdrop that McGhee and Stephanopoulos entered the scene.⁸⁶ They designed the efficiency gap to operationalize the same principle of

79. *Id.* at 420 (opinion of Kennedy, J.) (“The existence or degree of asymmetry may in large part depend on conjecture about where possible vote-switchers will reside.”).

80. *Id.* (“[W]e are wary of adopting a constitutional standard that invalidates a map based on unfair results that would occur in a hypothetical state of affairs.”).

81. *Id.* (“[T]he counterfactual plaintiff would face the same problem as the present, actual appellants: providing a standard for deciding how much partisan dominance is too much.”).

82. See Brief of Amici Curiae Law Professors in Support of Appellees, *supra* note 2, at 27–31.

83. Bruce E. Cain & Emily R. Zhang, *Blurred Lines: Conjoined Polarization and Voting Rights*, 77 OHIO ST. L.J. 867, 869 (2016); see also Voting Rights Act of 1965, Pub. L. No. 89-110, 79 Stat. 437 (codified as amended in scattered sections of 52 U.S.C.).

84. See *Cooper v. Harris*, 137 S. Ct. 1455, 1472–81 (2017); Richard L. Hasen, Essay, *Race or Party, Race as Party, or Party All the Time: Three Uneasy Approaches to Conjoined Polarization in Redistricting and Voting Cases*, 59 WM. & MARY L. REV. (forthcoming 2018) (manuscript at 1–3) (on file with author).

85. See Pamela S. Karlan, Essay, *John Hart Ely and the Problem of Gerrymandering: The Lion in Winter*, 114 YALE L.J. 1329, 1339 (2005) (“[W]e have moved from entrenchment through inaction to a perhaps even more pathological phenomenon of entrenchment through nonstop action.”).

86. The “efficiency gap” proposal was presented in the academic literature through two related articles: a political science article published in 2014 by McGhee, see McGhee, *supra* note 10, and a 2015 law review article coauthored by Stephanopoulos and McGhee, see Stephanopoulos & McGhee, *supra* note 7. The 2014 article introduced the numeric measure and demonstrated its key technical properties, see McGhee, *supra* note 10, at 68–70; *id.* app. B, while the 2015 article developed the measure into a proposed legal standard, see Stephanopoulos & McGhee, *supra* note 7, at 884–95.

Plaintiffs then adopted an efficiency gap approach in litigation challenging Wisconsin’s state assembly plan (the *Whitford* litigation) and North Carolina’s congressional plan (the *Rucho* litigation). See *supra* note 21. In 2017, McGhee published an article responding to critiques articulated here and elsewhere, addressing previously unexamined technical and conceptual aspects of the efficiency gap measure, and offering a more refined and generalized conceptualization. See Eric McGhee, *Measuring Efficiency in Redistricting*, 16 ELECTION L.J. 417, 426–31 (2017) [hereinafter McGhee, *Measuring Efficiency*].

partisan symmetry five Justices found appealing in *LULAC* while addressing each of Justice Kennedy's concerns. The proponents designed the efficiency gap to "aggregate[] all of a district plan's cracking and packing choices into a single, tidy number," thereby distilling "the essence of what critics have in mind when they refer to partisan gerrymandering."⁸⁷ The proponents define a partisan gerrymander as "a district plan that results in one party wasting many more votes than its adversary."⁸⁸ "Wasted votes include both 'lost' votes (those cast for a losing candidate) and 'surplus' votes (those cast for a winning candidate but in excess of what she needed to prevail)."⁸⁹

To calculate the efficiency gap (ΔW), "[e]ach party's wasted votes are totaled, one sum is subtracted from the other, and then . . . this difference is divided by the total number of votes cast."⁹⁰ The proponents then deploy an assumption to reduce the long-form equation for the efficiency gap into a simplified formula: the difference between the seat margin and twice the vote margin.⁹¹

$$\Delta W = S^* - 2V^*$$

According to this formula, in an ideal world without partisan gerrymandering (a world in which each party wastes equal votes), each party's seat margin (S^*) would be twice its vote margin (V^*).⁹²

$$\Delta W = 0 \text{ if and only if } S^* = 2V^*$$

According to this simplified formula, the efficiency gap has a simple interpretation within the seats-votes framework, as illustrated in Figure 7 below.

87. Stephanopoulos & McGhee, *supra* note 7, at 834, 852.

88. *See id.* at 849-50.

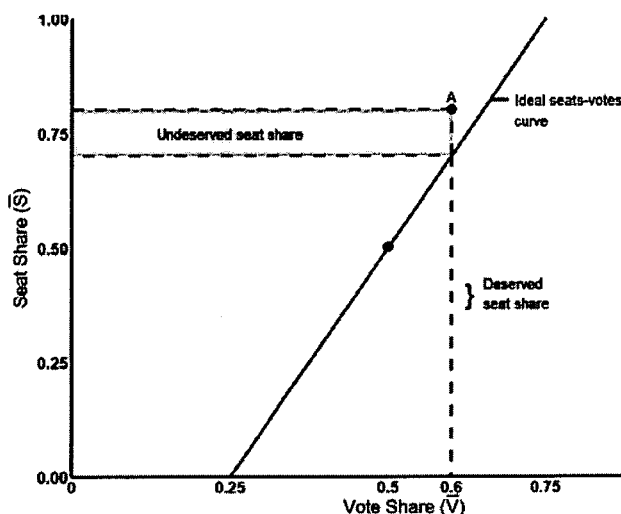
89. *Id.* at 851.

90. *Id.* at 851-52.

91. *See* McGhee, *supra* note 10, at 68 (presenting equation 5); *id.* app. B at 79-82 (deriving equation 5); *see also* Stephanopoulos & McGhee, *supra* note 7, at 853.

92. *See* Stephanopoulos & McGhee, *supra* note 7, at 854 ("[E]ach additional percentage point of vote share for a party should result in an extra two percentage points of seat share. This relationship is implied by the efficiency gap formula noted above. If the gap is zero, it can remain at this level only if any shift in seat share is twice the size of any shift in vote share.").

Figure 7



The simplified formula corresponds to an ideal seats-votes curve—a straight line passing through the point (0.5,0.5) with a slope of 2. The efficiency gap can be understood as a measure of undeserved seat share relative to the baseline of this ideal seats-votes curve. At point A, a party earns 80% seat share with 60% vote share. According to the simplified formula, the party deserves to earn 70% seat share. The difference between the seat share actually earned and the deserved seat share is the undeserved seat share. The efficiency gap measures this undeserved seat share, here 10%.

The efficiency gap represents the vertical distance between the observed seats-votes combination and the “ideal” seats-votes curve corresponding to outcomes where the parties waste equal votes. On this basis, the proponents present the gap as a measure of the “undeserved seat share” attributable to partisan gerrymandering rather than the party’s popularity.⁹³ For a given vote share, the ideal curve tells us what seat share a party would achieve under the ideal of partisan symmetry where each party wastes equal votes. Any seat share above that ideal is undeserved in that it is attributable to partisan asymmetry in which the favored party wastes fewer votes than the disfavored party.

Like the seats-votes curve for strict proportionality, this ideal seats-votes curve is a straight line, running through the point where each party equally splits votes and seats (0.5,0.5). At this one point, both strict proportionality and equal wasted votes are achieved. But this new seats-votes curve has a slope of 2 instead of 1. This means that according to the efficiency gap, the ideal seats-votes

93. See *id.* (“The efficiency gap[] . . . is a measure of undeserved seat share: the proportion of seats a party receives that it would *not* have received under a plan with equal wasted votes.”).

relationship is one of double responsiveness and double proportionality. As the proponents explain:

[T]he gap offers what scholars to date have been unable to supply: a normative guide as to how large [the seat] bonus should be. To produce partisan fairness—in the sense of equal wasted votes for each party—the bonus should be a precisely two-fold increase in seat share for a given increase in vote share.⁹⁴

For example, if a party earns 52% vote share, it deserves 54% seat share; if it actually obtains 56% seat share, there is an efficiency gap of 2%. Note that the proponents defined partisan symmetry as equal wasted votes and then demonstrated that this requirement (under definitions and assumptions I analyze below) is mathematically equivalent to a requirement that the ideal seats-votes curve exhibit double proportionality and responsiveness.⁹⁵ Thus, the proponents claim that the ratio of seat share to vote share should be 2:1 because that corresponds to the normative ideal of equal wasted votes.⁹⁶

The 2015 article proposed a legal test for political gerrymandering based on the efficiency gap measure.⁹⁷ If the plan's efficiency gap exceeds a numeric threshold and sensitivity analysis suggests that the plan will continue to produce an above-threshold gap in future elections, it is presumptively invalid.⁹⁸ This presumption can be overcome only if the plan's partisan effect can be justified or explained as the product of legitimate redistricting criteria consistently applied to the jurisdiction's underlying political geography.⁹⁹ Such criteria include contiguity, compactness, preservation of local political boundaries, preservation of communities of interest, and compliance with the Voting Rights Act.¹⁰⁰

The proponents originally proposed a numeric threshold of 8% based on their analysis of historical practice—gaps above 8% represent outliers relative to the distribution of gaps produced by modern electoral maps.¹⁰¹ The *Whitford*

94. *Id.* (footnote omitted).

95. McGhee first made this move in the 2014 article. *Compare* McGhee, *supra* note 10, at 68 (defining the measure algebraically in equation 2), *with id.* (presenting the simplified formula in equation 5), *and id.* app. B at 79-80 (deriving the simplified formula). Stephanopoulos and McGhee followed suit a year later. *Compare* Stephanopoulos & McGhee, *supra* note 7, at 851 & n.110 (defining the measure linguistically and citing the algebraic definition in the 2014 article), *with id.* at 853 & n.114 (presenting the simplified formula and citing its derivation in the 2014 article).

96. *See* Stephanopoulos & McGhee, *supra* note 7, at 854.

97. *See id.* at 884-95.

98. *See id.* at 884-85, 891.

99. *See id.* at 884-85.

100. *See id.* at 892; *see also, e.g.,* *Vieth v. Jubelirer*, 541 U.S. 267, 284 (2004) (plurality opinion).

101. *See* Stephanopoulos & McGhee, *supra* note 7, at 884, 888-89.

plaintiffs' expert, Simon Jackman, proposed a numeric threshold of 7% based on durability—gaps above 7% tend to persist for the life of an electoral map.¹⁰²

The proposed efficiency gap measure and associated legal test were explicitly framed as an effort to improve upon the partisan bias proposal offered by amici and considered by the Court in *LULAC*.¹⁰³ It relies on an intuitive and constitutionally discernible concept of symmetric partisan treatment viewed favorably by five Justices in *LULAC* while addressing the inadequacies Justice Kennedy identified with the symmetry measure proposed by the *LULAC* amici.¹⁰⁴ The measure of partisan symmetry proposed by the *LULAC* amici necessarily relied on assumptions to compare party performance in counterfactual scenarios.¹⁰⁵ The efficiency gap compares party performance directly observed in actual election results without necessarily relying on inferential techniques.¹⁰⁶ And the proposed efficiency gap test answers the question how much advantage is too much¹⁰⁷ with a numeric threshold of presumptive validity based on historical practice and durability.¹⁰⁸ An above-threshold efficiency gap is a concrete indication that the electoral map favors one party in a way that is likely to persist for the life of the map and that departs from historical practice. Finally, the measure is distinct from a requirement of strict proportionality, a standard the Court has already rejected.¹⁰⁹ Unlike strict proportionality, this double proportionality measure permits some seat bonus but limits the size of

102. See Assessing the Current Wisconsin State Legislative Districting Plan, Simon Jackman, at 66-69, *Whitford v. Gill*, 218 F. Supp. 3d 837 (W.D. Wis. 2016) (No. 3:15-cv-00421-bbc), 2015 WL 10091020 [hereinafter Jackman Report].

103. See Stephanopoulos & McGhee, *supra* note 7, at 895.

104. See *id.* at 895-99.

105. See *League of United Latin Am. Citizens v. Perry (LULAC)*, 548 U.S. 399, 420 (2006) (opinion of Kennedy, J.) ("Amici's proposed standard does not compensate for appellants' failure to provide a reliable measure of fairness. The existence or degree of asymmetry may in large part depend on conjecture about where possible vote-switchers will reside. . . . [W]e are wary of adopting a constitutional standard that invalidates a map based on unfair results that would occur in a hypothetical state of affairs.").

106. See Stephanopoulos & McGhee, *supra* note 7, at 857. Inferential techniques are, however, used for sensitivity testing and in the case of uncontested districts. See *id.* at 866-67, 889-90.

107. Cf. *LULAC*, 548 U.S. at 420 (opinion of Kennedy, J.) (faulting partisan gerrymandering plaintiffs for failing to "provid[e] a standard for deciding how much partisan dominance is too much").

108. See Jackman Report, *supra* note 102, at 66-69; Stephanopoulos & McGhee, *supra* note 7, at 884, 888-89.

109. See, e.g., *Vieth v. Jubelirer*, 541 U.S. 267, 288 (2004) (plurality opinion) ("[The appellants' proposed] standard rests upon the principle that groups . . . have a right to proportional representation. But the Constitution contains no such principle."); see also *LULAC*, 548 U.S. at 419 (opinion of Kennedy, J.) ("To be sure, there is no constitutional requirement of proportional representation . . .").

this bonus. The limit aligns with electoral reality: Political scientists have consistently found approximately 2:1 seat share to vote share ratios in state legislative and congressional elections.¹¹⁰ And the limit is simple: Seats-votes responsiveness cannot depart significantly from 2.

C. Two Potential Concerns with the Efficiency Gap

Faced with the competing constitutional risks presented by claims of partisan gerrymandering, the Justices have understandably sought “clear, manageable, and politically neutral standards” and “rules to limit and confine judicial intervention.”¹¹¹ In an effort to supply such a standard, the *LULAC* amici focused on the principle of partisan symmetry, defined in terms of a counterfactual comparison.¹¹² The efficiency gap similarly focuses on partisan symmetry but instead defines it as symmetric efficiency in the sense of equal wasted votes.¹¹³ To help the Court measure and thereby proscribe partisan gerrymandering, the efficiency gap must simultaneously cohere with an individual rights framework based on a principle of nondiscrimination¹¹⁴ and with a structural account of electoral democracy attentive to the multiple values implicated by electoral districting. To satisfy this double coherence, the efficiency gap must sensibly define and compare parties’ wasted votes in a way that distinguishes normatively desirable from undesirable plans. Notably, there may be tension between the individual rights framework and the structural account.

This need for double coherence suggests two potential concerns with the efficiency gap that motivate this Article’s analysis: one regarding its coherence with the individual rights framework, the other regarding its structural implications. First, the efficiency gap addresses concerns with the *LULAC* amici’s proposed measure by offering an alternative definition of partisan symmetry based on observed election results rather than counterfactual results. Instead of requiring that the actual seats-votes curve exhibit (or approximate) symmetry, the efficiency gap requires that the observed election result fall on (or close to) a prescriptive seats-votes curve corresponding to an ideal of equal wasted votes. Thus, the efficiency gap replaces one equality norm (symmetric hypothetical outcomes) with a new equality norm (equal wasted votes). How intuitive and compelling this new equality norm is depends on how wasted votes are defined

110. See *supra* note 49 and accompanying text.

111. See, e.g., *Vieth*, 541 U.S. at 307-08 (Kennedy, J., concurring in the judgment).

112. See *supra* notes 75-78 and accompanying text.

113. See *supra* text accompanying notes 88-89.

114. This principle of nondiscrimination may be grounded in the Equal Protection Clause right to be free from unjustified differential treatment or the First Amendment right to be free from discrimination or punishment on the basis of political affiliation or belief. Cf. *Vieth*, 541 U.S. at 313-14 (Kennedy, J., concurring in the judgment) (discussing both bases and suggesting that the First Amendment may be preferable).

and compared. This suggests the importance of examining the efficiency gap's conceptual design. Part II below undertakes this examination.

Second, the principle of partisan symmetry the efficiency gap is designed to capture may be a necessary but insufficient condition for a well-functioning democracy. Thus, it is worth considering how the efficiency gap relates to other democratic norms, like seats-votes proportionality and competitiveness, that are relevant to electoral districting but that the efficiency gap was not designed to capture. Part III explores these relationships.

II. The Efficiency Gap's Conceptual Design

As it has been defined by its proponents, the efficiency gap reflects a series of interrelated electoral assumptions and methodological choices that warrant careful examination. First, the proponents assume that each district's general election is a two-candidate contest between two candidates, one from each of the two major parties (to which I refer throughout as party x and party y). This ensures that every ballot is cast for, and every district race is won by, either party x or party y . Thus, the set of districts (D) can be split into the set of x -won districts (D_x) and the set of y -won districts (D_y), and voter turnout (T_i) in district i is simply the sum of the parties' respective vote totals (V_{xi} and V_{yi}).

$$D = D_x \cup D_y$$

$$T_i = V_{xi} + V_{yi}$$

When a district has an uncontested general election, the proponents use imputation techniques to estimate what the vote totals would have been had the election been contested.¹¹⁵

Second, the proponents deploy a cluster of definitions culminating in the concept of a wasted vote. A lost vote (L_{pi}) is one cast for the losing party: $L_{pi} = V_{pi}$. A surplus, or excess,¹¹⁶ vote (E_{pi}) is one cast for the winning party beyond the threshold needed to win (V_{ni}): $E_{pi} = V_{pi} - V_{ni}$. The threshold needed to win (in an assumed two-party race) is half of actual district turnout: $V_{ni} = \frac{T_i}{2}$.¹¹⁷ And a wasted vote is a lost or a surplus vote.¹¹⁸ Note that the possibility of a tie is ignored, so every district race is won by one party or the other.

115. See Stephanopoulos & McGhee, *supra* note 7, at 865-67.

116. I use E for "excess" to denote the concept rather than S for "surplus," which could be confused with "seat" or "share."

117. See Stephanopoulos & McGhee, *supra* note 7, at 851.

118. *Id.* ("Wasted votes include both 'lost' votes (those cast for a losing candidate) and 'surplus' votes (those cast for a winning candidate but in excess of what she needed to prevail)."); *id.* ("[A]ny vote for a losing candidate is wasted by definition, but so too is any vote beyond the 50 percent threshold needed (in a two-candidate race) to win a seat.")

The total wasted votes (W_{pi}) for a party p in district i is the sum of lost votes and surplus votes cast for that party: $W_{pi} = L_{pi} + E_{pi}$. A party incurs surplus votes when it wins a district ($i \in D_p$) and incurs lost votes when it loses a district ($i \notin D_p$).

$$W_{pi} = \begin{cases} E_{pi} = V_{pi} - V_{ni} = V_{pi} - \frac{T_i}{2} & i \in D_p \\ L_{pi} = V_{pi} & i \notin D_p \end{cases}$$

In short, the proponents define wasted votes by equally weighting lost and surplus votes and defining surplus votes relative to a threshold of half of turnout. Judge Griesbach, the dissenting voice on the *Whitford* panel, described this definition as “opaque” and “absurd.”¹¹⁹ I will argue that the proponents’ definition is reasonable but not the only plausible way to define surplus votes. The proponents’ threshold of half of turnout represents a party-centric approach. A more voter-centric approach would use a threshold based on the votes cast for the runner-up candidate.

Third, the proponents aggregate values by party and district to produce a single number for the entire plan.¹²⁰ For each party, the number of wasted votes over the entire plan (W_p) is simply the sum of its wasted votes in each district.

$$W_p = \sum_{i \in D} W_{pi}$$

The plan’s efficiency gap (ΔW) is “the difference between the parties’ respective wasted votes, divided by the total number of votes cast in the election.”¹²¹

$$\Delta W = \frac{\sum_{i \in D} W_{yi} - \sum_{i \in D} W_{xi}}{\sum_{i \in D} V_{yi} + \sum_{i \in D} V_{xi}} = \frac{W_y - W_x}{V_y + V_x}$$

This approach compares the parties’ relative wasted vote totals: It is zero when each party wastes the same raw number of votes.

Fourth, the proponents adopt an electoral assumption to simplify the long-form equation. Specifically, they assume that “each district has exactly the same

119. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 958 (W.D. Wis. 2016) (Griesbach, J., dissenting), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

120. See Stephanopoulos & McGhee, *supra* note 7, at 851-52; *id.* at 852 fig.1 (illustrating the computation of the efficiency gap for a hypothetical plan by aggregating wasted votes by party and district).

121. *Id.* at 851 (emphasis omitted).

number of voters.”¹²² Under this equal voter turnout assumption, district-level turnout (T_i) is equal to average turnout (T^*) for every district (i) in the plan (D).

$$T_i = T^* = \frac{\sum_{i \in D} T_i}{S} \quad \text{for each } i \in D$$

The proponents use this assumption to reduce the long-form equation (ΔW) to a simple function of seat margin (S^*) and vote margin (V^*).¹²³

$$\Delta W = S^* - 2V^*$$

This simplified formula is much easier to compute, so the proponents use it when analyzing historical election data,¹²⁴ as did *Whitford* expert Simon Jackman.¹²⁵ Moreover, using this simplified formula, the proponents recast the efficiency gap as a measure of the undeserved seat share attributable to partisan gerrymandering rather than to the party’s popularity.¹²⁶

In sum, the efficiency gap measure relies on four conceptual moves: (1) the two-party assumption; (2) the definition and weight of surplus votes; (3) the aggregation from district to plan to compare the parties’ wasted vote totals; and (4) the equal voter turnout assumption, which allows them to derive the simplified formula and exploit the associated double proportionality seats-votes interpretation. In this Part, I examine each move to assess whether the efficiency gap defines and compares wasted votes in a sufficiently discernible and manageable manner. It will prove useful to examine them in reverse order, considering first the equal voter turnout assumption, then the aggregation method from district to plan, then the definition and weight of surplus votes, and finally the assumption that every district’s general election is a two-party contest. But first, I briefly examine the principle McGhee has identified as informing these various methodological choices.

122. McGhee, *supra* note 10, app. B at 79.

123. *See id.* app. B at 79-82.

124. The 2015 article does not specify which computation method (long-form equation or simplified formula) was used to analyze historical election data. But in a phone conversation with Eric McGhee, I confirmed that this analysis was performed using the simplified formula. Telephone Interview with Eric McGhee, Research Fellow, Pub. Policy Inst. of Cal. (Oct. 21, 2016).

125. *See* Jackman Report, *supra* note 102, at 16 (“The assumption of equally-sized districts is especially helpful for the analysis reported below, since the calculation of [the efficiency gap] in a given election then reduces to using the jurisdiction-level quantities [seat share] and [vote share] as in [the simplified formula]. For the analysis of historical election results reported below, it isn’t possible to obtain measures of district populations, meaning that we really have no option other than to rely on the jurisdiction-level . . . [seat share] and [vote share] when estimating the [efficiency gap].”).

126. *See* Stephanopoulos & McGhee, *supra* note 7, at 854 (“The efficiency gap’s second interesting property follows from [the simplified formula]. Simply put, it is a measure of undeserved seat share: the proportion of seats a party receives that it would *not* have received under a plan with equal wasted votes.”).

A. McGhee's Efficiency Principle

McGhee's design of the efficiency gap is guided by an overarching criterion he calls the *efficiency principle*.

Any measure of efficiency must indicate a greater advantage for (against) a party when the seat share for that party increases (decreases) without any corresponding increase (decrease) in its vote share.¹²⁷

This principle reflects McGhee's insight that what makes partisan gerrymandering so attractive to partisan mapmakers and so troubling for democracy is that it permits mapmakers to increase their party's power (seat share) at a constant or even decreasing level of popularity (vote share). According to McGhee, this principle "is a bedrock condition for a measure of efficiency"; if a measure violates it, "it might be an adequate measure of something else, but it is missing the very essence of an efficient gerrymander."¹²⁸ This Subpart briefly explores and critiques this principle.

1. Implications of the efficiency principle

In one sense, the efficiency principle is quite strict. Mathematically, it requires that a measure be a function of seat share, specifically one that has a positive partial derivative with respect to seat share.¹²⁹ This means that holding all else (including vote share) constant, an increase in seat share increases the measure, and a decrease in seat share decreases the measure. Assuming that seat share is a function of vote share with a nonnegative derivative,¹³⁰ then any measure satisfying the efficiency principle must itself be a function of vote share, specifically one that has a nonpositive derivative with respect to vote share. Thus, any measure satisfying the efficiency principle must be a function of both seat share and vote share with a positive partial derivative with respect to seat share and a nonpositive partial derivative with respect to vote share. But the measure cannot depend on any variable independent of seat share and vote share, such as overall competitiveness or voter turnout. If it did, one could hold

127. McGhee, *Measuring Efficiency*, *supra* note 86, at 418.

128. *Id.*

129. *See id.* at 427 ("This can also be phrased more formally: to satisfy the [efficiency principle], a measure's partial derivative with respect to seat share must be positive."). The partial derivative of a function with respect to a given variable captures the incremental rate of change of the function with respect to that variable when all other variables are held constant.

130. This is a modest assumption, one violated only in the odd event that a party's vote share goes up but its seat share goes down (or vice versa). McGann and his coauthors call this the assumption of "nonnegative responsiveness" and describe it as "a minimal requirement that is met by every reasonable single vote electoral system, including the first-past-the-post system used in the United States." *See McGann et al.*, *supra* note 20, at 302-03 (emphasis omitted).

vote share constant, increase seat share and thereby increase the measure, but then modify the independent variable so as to return the measure to its original value. The result would be an increase in seat share at constant vote share without any corresponding increase in the measure—a violation of the efficiency principle.

Thus, any measure satisfying the efficiency principle must be a function *only* of seat share and vote share, with a positive partial derivative with respect to seat share, a nonpositive partial derivative with respect to vote share, and no dependence whatsoever on any variable independent of vote share and seat share. This means that we can set the measure equal to zero and then solve for seat share as a function of vote share. The result is a seats-votes curve corresponding to symmetric partisan efficiency as defined by this measure.¹³¹ The measure must then represent the extent to which a given electoral outcome departs from this ideal symmetric seats-votes curve. In sum, the efficiency principle implicitly requires that the analyst specify a single ideal seats-votes curve and then measure the extent to which an electoral outcome departs from that ideal curve. Each measure that complies with the efficiency principle corresponds to exactly one such ideal curve.

In another sense, though, the efficiency principle is quite flexible because it imposes minimal constraints on what the ideal seats-votes curve may be. A function satisfies the efficiency principle so long as it is a function of seat share ($\bar{S}$) and vote share ($\bar{V}$) with properly signed partial derivatives. If we impose the further requirement that the measure is linear in both $\bar{S}$ and $\bar{V}$, the unsurprising result is a linear seats-votes curve, and the only remaining task is to determine its slope—that is, its degree of proportionality and responsiveness. As we shall see, when we define the efficiency gap in terms of parties' relative wasted vote totals, the result is a linear seats-votes curve, with the degree of proportionality and responsiveness determined by the definition and weight of surplus votes.

Any measure that satisfies the efficiency principle will encounter several challenges. First, the measure may be better suited to plans involving a large number of districts, like state legislative plans and congressional plans in populous states, than to plans involving fewer districts, like congressional plans in less populous states or plans for local bodies like city councils and school boards. The reason is that when plans have a small number of districts, seat share ($\bar{S}$) is constrained to a few values, so the observed electoral outcome will necessarily depart significantly from the ideal seats-votes curve unless the system happens

131. For example, the simplified formula satisfies the efficiency principle because it depends only on seat margin and vote margin. If one sets the simplified formula to zero and then solves for seat margin as a function of vote margin, the result is the seats-votes curve illustrated in Figure 7 above.

to exhibit the right vote share ($\bar{V}$). For example, if a state (like Idaho¹³²) has only two congressional districts, there are only two possible values for seat margin: zero (each party wins one district) or fifty percent (one party wins both districts). Unless this state has the right vote margin, it will necessarily have a large efficiency gap.¹³³ This suggests that the efficiency gap measure may prove more useful when analyzing state legislative plans like the one at issue in *Whitford* than congressional plans like the one at issue in *Rucho*.¹³⁴

Second, any measure that satisfies the efficiency principle is vulnerable to the perverse risk that partisan voter suppression will reduce the apparent severity of a partisan gerrymander as quantified by the measure. If a party's supporter is unable to cast a ballot, assuming that this ballot does not determine the outcome of the district election, the result will be that the party achieved the same seat share with less vote share, indicating greater efficiency for that party. This suggests that partisan voter suppression could reduce the apparent severity of a partisan gerrymander. I explore this risk in greater detail in Part II.E.2 below.

2. A modified efficiency principle

The proponents assume that a partisan mapmaker's goal is to maximize her party's seat share at a fixed level of vote share.¹³⁵ The efficiency principle aligns the measure with this objective. But this may be a simplification of the strategic calculus motivating real-world partisan mapmakers. It may be more accurate to assume that the mapmaker seeks to maximize her party's expected seat share over a plausible range of vote shares.¹³⁶ If the mapmaker is too confident in her

132. See *Directory of Representatives*, U.S. HOUSE REPRESENTATIVES, <https://perma.cc/JB87-SC8K> (archived Feb. 24, 2018).

133. For a detailed treatment of the efficiency gap's difficulty handling plans with low numbers of districts, see Wendy K. Tam Cho, *Measuring Partisan Fairness: How Well Does the Efficiency Gap Guard Against Sophisticated as Well as Simple-Minded Modes of Partisan Discrimination?*, 166 U. PA. L. REV. ONLINE 17, 21-27 (2017).

134. *Whitford* involved a challenge to the Wisconsin State Assembly, a body currently subdivided into ninety-nine districts. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 843 (W.D. Wis. 2016), stay granted, 137 S. Ct. 2289, and jurisdiction postponed, 137 S. Ct. 2268 (2017); 2017 *Wisconsin State Representatives*, WIS. ST. LEGISLATURE, <https://perma.cc/N2QJ-GS94> (archived Feb. 24, 2018). *Rucho*, conversely, involved a challenge to North Carolina's thirteen congressional districts. See *Common Cause v. Rucho*, 279 F. Supp. 3d 587, 597 (M.D.N.C. 2018), stay granted, No. 17A745, 2018 WL 472142 (U.S. Jan. 18, 2018), and appeal docketed, No. 17-1295 (U.S. Mar. 14, 2018); *Directory of Representatives*, *supra* note 132.

135. See Stephanopoulos & McGhee, *supra* note 7, at 850 ("Our analysis begins with the premise that the goal of a partisan gerrymander is to win as many seats as possible given a certain number of votes." (emphasis omitted)).

136. See Guillermo Owen & Bernard Grofman, *Optimal Partisan Gerrymandering*, 7 POL. GEOGRAPHY Q. 5, 7 (1988) (arguing that efforts to maximize expected seat share "lead to strategies for sophisticated optimal partisan gerrymandering which differ from the classic 'recipe' of seeking to control as many districts as possible by paper-thin margins").

prediction of vote share and too greedy in her desire to maximize seat share at predicted vote share, the result may be a “dummymander”¹³⁷ that inures to the benefit of the other party when actual vote share departs from the mapmaker’s prediction.¹³⁸ To account for this strategic calculus under conditions of electoral uncertainty, I propose the following *modified efficiency principle*.

Any measure of efficiency must indicate a greater advantage for (against) a party when the seat share for that party increases (decreases) without any corresponding increase (decrease) in its vote share *unless its expected seat share decreases (increases) under plausible variation in that vote share*.

To see the difference between the original principle and this modified one, consider a mapmaker who alters a plan so that the majority party wins more seats by smaller margins with the same number of votes. This change allows the majority party to increase its seat share at constant vote share, but only by making the plan more competitive and thereby increasing the risk that seats will flip to the other party under plausible variation in vote share. In this scenario, a measure that satisfies the original efficiency principle would indicate greater advantage for the majority party because it won more seat share with the same vote share. In contrast, a measure that satisfies the modified efficiency principle might indicate less advantage for the majority party because its expected seat share over plausible variation in vote share could decrease. Unlike the original efficiency principle, the modified one permits a measure to take into account the plan’s increased competitiveness.

B. The Equal Voter Turnout Assumption

Recall that McGhee derives the simplified formula ($\Delta W = S^* - 2V^*$) by explicitly assuming that “each district has exactly the same number of voters.”¹³⁹

137. Bernard Grofman & Thomas L. Brunell, *The Art of the Dummymander: The Impact of Recent Redistrictings on the Partisan Makeup of Southern House Seats*, in REDISTRICTING IN THE NEW MILLENNIUM 183, 184 (Peter F. Galderisi ed., 2005) (“A dummymander is a gerrymander by one party that, over the course of the decade, benefits the other party . . .”).

138. This is why Justice O’Connor, arguing for nonjusticiability in *Bandemer*, suggested that “political gerrymandering is a self-limiting enterprise.” See *Davis v. Bandemer*, 478 U.S. 109, 152 (1986) (O’Connor, J., concurring in the judgment), *abrogated in part by* *Vieth v. Jubelirer*, 541 U.S. 267 (2004).

139. McGhee, *supra* note 10, app. B at 79. The first line of McGhee’s derivation explicitly assumes that “each district has exactly the same number of voters” so that “proportions can be substituted for raw votes in all of the formulas . . . [such that] the total vote in each district becomes equal to 1.0” and the sum of district vote totals “is simply the total number of districts in the electoral system.” See *id.*; see also Stephanopoulos & McGhee, *supra* note 7, at 853 & n.114 (incorporating this derivation).

Under this equal voter turnout assumption, for every district (i) in the plan (D), district-level turnout (T_i) is equal to average turnout (T^*).

$$T_i = T^* \text{ for each } i \in D$$

Thus, the simplified formula, by design, does not account for interdistrict variation in voter turnout.¹⁴⁰ If a real-world election were to satisfy this assumption, the long-form formula and the simplified seats-votes formula would compute the same number.

$$\text{If } T_i = T^* \text{ for each } i \in D \text{ then } \frac{w_y - w_x}{v_y + v_x} = S^* - 2V^*$$

However, questions of plausibility and robustness arise whenever an assumption (like equal voter turnout) underlies a shorthand equation (like the simplified formula) for a measure.¹⁴¹ How plausible is the assumption of equal voter turnout in each district? How sensitive is the efficiency gap to departures from this condition? To address these questions, I relax the equal voter turnout assumption and derive a more generalized simplified formula that expresses the

140. See McGhee, *supra* note 10, at 68 & 83 n.6 (“This necessarily assumes away differences in efficiency due to turnout. . . . [B]ut turnout variation is still a worthy topic of study. In fact, future research could use [the long-form equation] instead of [the simplified formula] to explore the subject.”).

141. Two Terms ago, the Court clarified that a state may—but declined to address whether it must—comply with the one person, one vote principle by equalizing the number of people (as opposed to the number of eligible voters, for instance) in each district. See *Evenwel v. Abbott*, 136 S. Ct. 1120, 1126-27 (2016). Thus, equal total population might be constitutionally required, but neither equal *voter* population nor equal voter *turnout* is constitutionally required. For example, “it is plainly permissible for jurisdictions to measure equalization by the *total* population of state and local legislative districts,” *id.* at 1126-27 (emphasis added), even if this produces an electoral map that “measured by a *voter*-population baseline . . . [exhibits a] maximum population deviation exceed[ing] 40%,” *id.* at 1125 (emphasis added).

The scholarship presenting the efficiency gap proposal, published before *Evenwel*, characterized the relevant equality conditions in a way that may generate confusion. See McGhee, *supra* note 10, at 68 (“In the special case where . . . districts are equal in *population*[,] . . . [the long-form equation] reduces to [the simplified formula].” (emphasis added)); *id.* at 83 n.6 (“Ignoring *turnout* differences in this way is legally *mandated* for redistricting in the United States . . .” (emphasis added)); Stephanopoulos & McGhee, *supra* note 7, at 853 (stating that the simplified formula “assume[s] that all districts are equal in *population* (which is constitutionally *required*)” (emphasis added)). If the assumption used to derive the simplified formula were constitutionally required, it would necessarily be satisfied in real elections, and so questions of plausibility and robustness would be moot. Yet the relevant assumption is not equal total population but rather equal voter turnout. This assumption is not constitutionally required; a constitutionally valid electoral map may exhibit small interdistrict variation in total population but large variation in the population of eligible voters and even larger variation in actual voter turnout. Thus, there is no guarantee that real elections will exhibit (or even approximate) equal voter turnout, and so questions of plausibility and robustness warrant attention.

efficiency gap in terms of statewide seat and vote margin but makes no ex ante assumption about interdistrict variation in voter turnout.

1. The turnout gap

To explain the effect of turnout variation on the efficiency gap, I first introduce the concept of a turnout gap. When we relax the equal voter turnout assumption, we can still denote by T^* the average turnout across all districts: $T^* = \frac{\sum_{i \in D} T_i}{S}$. But now each district may have a turnout above or below (or equal to) average turnout. Let ΔT_i denote the proportional difference between actual turnout in district i and average turnout over all districts: $\Delta T_i = \frac{T_i - T^*}{T^*}$.¹⁴² Let ΔT_p denote the average value of ΔT_i over districts won by party p : $\Delta T_p = \frac{\sum_{i \in D_p} \Delta T_i}{S_p}$.¹⁴³ Define the turnout gap (ΔT)¹⁴⁴ as the product of party x seat share ($\bar{S}_x$) and the average proportional difference in x -won districts (ΔT_x).

$$\Delta T = \bar{S}_x \Delta T_x$$

142. ΔT_i is therefore positive when district i has higher-than-average turnout, negative when district i has lower-than-average turnout, and zero when district i 's turnout is identical to the average. For example, $\Delta T_i = 0.05$ when that district's turnout is 5% higher than average turnout.

143. By design, with only two parties, the seat-share-weighted sum of ΔT_x and ΔT_y is zero. $0 = \sum_{i \in D} \Delta T_i = \sum_{i \in D_x} \Delta T_i + \sum_{i \in D_y} \Delta T_i = S_x \Delta T_x + S_y \Delta T_y = \bar{S}_x \Delta T_x + \bar{S}_y \Delta T_y$.

Equivalently, $\bar{S}_x \Delta T_x = -\bar{S}_y \Delta T_y$. This makes intuitive sense because x -won districts will have above-average turnout only if y -won districts have correspondingly below-average turnout (and vice versa).

144. Relying on the fact that $\bar{S}_x \Delta T_x = -\bar{S}_y \Delta T_y$ (as derived in note 143 above), the turnout gap can also be expressed as half the seat-share-weighted difference of ΔT_x and ΔT_y .

$$\Delta T = \bar{S}_x \Delta T_x = \frac{\bar{S}_x \Delta T_x + \bar{S}_x \Delta T_x}{2} = \frac{\bar{S}_x \Delta T_x - \bar{S}_y \Delta T_y}{2}$$

Alternatively, the turnout gap can be expressed as the sum of proportional turnout differences over x -won districts divided by the number of districts.

$$\Delta T = \bar{S}_x \Delta T_x = \frac{S_x}{S} \frac{\sum_{i \in D_x} \Delta T_i}{S_x} = \frac{\sum_{i \in D_x} \Delta T_i}{S}$$

This is essentially how McGhee defines the turnout gap. See McGhee, *Measuring Efficiency*, *supra* note 86, at 427 (presenting equations 4 and 5); *id.* app. at 438-39 (deriving equation 4). Thus, McGhee and I derive equivalent turnout-generalized seats-votes formulas. Note that I discussed turnout effects with McGhee while preparing this Article, a draft of which McGhee cites in his 2017 piece. See *id.* at 428, 437.

The turnout-generalized seats-votes formula is:¹⁴⁵

$$\Delta W = S^* - 2V^* + \Delta T.$$

This more generalized simplified formula is similar to the proponents', except that it now contains an additional term, ΔT , that precisely quantifies the effect of voter turnout on the efficiency gap. The long-form equation reduces to the original simplified formula if and only if the turnout gap is zero. Note that $\bar{S}_x$ is never negative, so the sign of the turnout gap depends on the sign of ΔT_x , which captures whether turnout in x -won districts is above or below average turnout. When x -won districts exhibit above-average turnout, ΔT_x —and thus the turnout gap—is positive and the long-form equation produces a higher number than does the simplified formula, registering greater advantage for party x . When x -won districts exhibit below-average turnout, ΔT_x —and thus the turnout gap—is negative and the long-form equation produces a lower number than does the simplified formula, registering greater advantage for party y .

2. The size of the turnout gap

In some cases, the turnout gap may be so small that it has no meaningful effect on the efficiency gap analysis and can thus be safely ignored. The 2012 Wisconsin State Assembly election under the plan challenged in *Whitford* appears to be one such case. Consider the analysis of Kenneth Mayer, one of the plaintiffs' experts in the *Whitford* litigation. Of the 2,844,676 votes cast in the 2012 election, Democrats wasted 877,445 votes while Republicans wasted only 544,893, leading to an efficiency gap of 11.69% according to the long-form calculation.¹⁴⁶ In this plan, Republicans received only 1,389,958 (48.86%) of the total votes cast but won 57 of 99 (57.58%) districts, producing an efficiency gap of 9.85% according to the simplified formula.¹⁴⁷ The long-form value exceeds the

145. For a proof of this result, see Benjamin Plener Cover, Mathematical Supplement 1 (2018), <https://perma.cc/M3FD-SRMG> [hereinafter Mathematical Supplement].

146. See Analysis of the Efficiency Gaps of Wisconsin's Current Legislative District Plan and Plaintiffs' Demonstration Plan, Kenneth R. Mayer, at 46 tbl.10, *Whitford v. Gill*, 218 F. Supp. 3d 837 (W.D. Wis. 2016) (No. 3:15-cv-00421-bbc), 2015 WL 10091018 [hereinafter Mayer Report]. The long-form calculation is:

$$\frac{W_y - W_x}{V_y + V_x} = \frac{877,445 - 544,893}{2,844,676} = 0.1169.$$

See *id.* at 45-46, 46 tbl.10.

147. See *id.* at 46 tbl.10. The simplified calculation is:

$$S^* - 2V^* = \left(\frac{57}{99} - 0.5\right) - 2\left(\frac{1,389,958}{2,844,676} - 0.5\right) = 0.0985.$$

short-form value because there is a turnout gap of 1.84% in favor of Republicans.¹⁴⁸ But both values indicate above-threshold gaps in favor of Republicans.¹⁴⁹ For this reason, even though the *Whitford* majority regarded the long-form equation as “preferable” to the simplified formula, it was “not troubled” by the choice of computational technique given that “both methods yield an historically large, pro-Republican” efficiency gap.¹⁵⁰

But the turnout gap is not always insignificant. Consider Indiana’s 2014 congressional election, the results of which are provided in Table 1 below.¹⁵¹

148. $\Delta T = \left(\frac{w_y - w_x}{v_y + v_x} \right) - (S^* - 2V^*) = 0.1169 - 0.0985 = 0.0184$.

149. Interestingly, Simon Jackman, the *Whitford* plaintiffs’ other expert, computed a gap of 13% for the 2012 election using the simplified formula. See Jackman Report, *supra* note 102, at 16, 36. However, in addition to employing different computation methods (long-form versus simplified formula), Jackman and Mayer may have also employed different imputation methods to account for uncontested assembly races. See *id.* at 24-30; Mayer Report, *supra* note 146, at 44-45. Of 99 assembly races in the 2012 election, 27 were uncontested. See Mayer Report, *supra* note 146, at 44. For a discussion of the efficiency gap’s sensitivity to alternative imputation strategies, see Part II.E.1 below. Note that the *Whitford* panel majority, in assessing the efficiency gap evidence, compared the value Mayer computed using the long-form equation to the value Jackman computed using the simplified formula—not the values Mayer computed with each formula. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 904-05, 907 (W.D. Wis. 2016), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

150. See *Whitford*, 218 F. Supp. 3d at 907-08. This argument neglects the fact that both the academic proponents and Jackman analyzed historical election data using only the simplified formula and not the long-form equation. See *supra* notes 124-25 and accompanying text. In other words, the majority relied on the fact that the 11.69% gap Mayer calculated using the long-form equation is larger than gaps calculated for historical election data with the simplified formula. Essentially, the majority assumed a high correlation between the results produced by the two computation methods. Such a high correlation may very well exist, but it would ideally be computed rather than assumed. The *Whitford* majority also emphasized that the defendants’ expert Nicholas Goedert “described the simplified method as ‘an appropriate and useful summary measure’” and that both parties stipulated that the simplified formula’s “implied 2-to-1 votes-to-seats relationship reflects the ‘observed average seat/votes curve in historical U.S. congressional and legislative elections.’” See 218 F. Supp. 3d at 907 (quoting Use of Efficiency Gap in Analyzing Partisan Gerrymandering, Report for State of Wisconsin, Nicholas Goedert, at 5-6, *Whitford*, 218 F. Supp. 3d 837 (No. 3:15-cv-00421-bbc), 2015 WL 10091017). Finally, the *Whitford* majority cautioned that “[w]here there record evidence indicating that [the simplified formula] did not correlate highly with both the [long-form equation] and electoral reality, we would have reason to doubt [the simplified formula’s] validity.” *Id.* at 907-08.

151. Table 1 relies on election results as reported by the Indiana Secretary of State. See *Indiana General Election, November 4, 2014*, IN.GOV, <https://perma.cc/D8CD-LNPG> (last updated Mar. 11, 2015, 10:01 AM). It includes results for Republican and Democratic candidates but omits votes for third-party candidates from the “Total Turnout” column.

Table 1
Results of Indiana's 2014 Congressional Election

District	Total Votes by Party		Total Turnout
	R	D	
1	51,000	86,579	137,579
2	85,583	55,590	141,173
3	102,889	39,771	142,660
4	94,998	47,056	142,054
5	105,277	49,756	155,033
6	102,187	45,509	147,696
7	46,887	61,443	108,330
8	103,344	61,384	164,728
9	101,594	55,016	156,610
Totals	793,759	502,104	1,295,863

Out of 1,295,863 ballots cast, the Democrats wasted 379,150 votes, but the Republicans wasted only 268,782 votes, for an efficiency gap of 8.5% in Republicans' favor according to the long-form equation.¹⁵² With only 61.3% of the statewide vote share, the Republicans won 7 of 9 (77.8%) districts, resulting in an efficiency gap of 5.3% according to the simplified formula.¹⁵³ The discrepancy between the efficiency gap under the long-form equation (8.5%) and that under the simplified formula (5.3%) is a function of the turnout gap. District turnout is 143,985 on average, but it ranges from a low of 108,330 in Democrat-won District 7 to a high of 164,728 in Republican-won District 8, and turnout in Republican-won districts is 4.2% above average, for a turnout gap of 3.2%.¹⁵⁴ The turnout gap is the difference between the long-form and simplified computations (3.2% = 8.5% – 5.3%). Note that the long-form value (the departure from equal wasted votes) is greater than the short-form value (the departure from double proportionality) because Republican-won districts exhibit above-average turnout, producing a positive turnout gap. If the “real” efficiency gap is defined by the long-form equation, the simplified formula underestimates it. If an 8% threshold were used for congressional plans—as the proponents originally

152. $\Delta W = \frac{w_y - w_x}{v_y + v_x} = \frac{379,150 - 268,782}{1,295,863} = 0.085$. For a demonstration of this calculation, see

Mathematical Supplement, *supra* note 145, at 2.

153. $S^* - 2V^* = \left(\frac{7}{9} - 0.5\right) - 2(0.613 - 0.5) = 0.053$.

154. $\Delta T = \bar{s}_x \Delta T_x = \frac{7}{9}(0.042) = 0.032$.

recommended¹⁵⁵—whether this election produced an above-threshold or below-threshold gap would depend on the choice of computation method.

Data analysis beyond the scope of this Article can compute the turnout gaps (and thus the efficiency gaps under the long-form equation) for historical elections, but historical patterns do not guarantee future trends. Any factor that disproportionately decreases Democratic turnout will generally tend to generate a turnout gap in Republicans' favor, and vice versa: Democrat-won districts have more Democrats, so a uniform decrease in Democratic turnout will have a larger impact on the turnout in Democrat-won districts than in Republican-won districts. This suggests that the turnout gap has a tendency to systematically increase whenever electoral rules have this differential partisan impact on turnout. Many believe, with good reason, that this is precisely the impact and intent of many recent electoral reforms.¹⁵⁶

However common or rare the occurrence, when the turnout gap is large enough, the choice of computation method matters, presenting questions of robustness and correspondence.¹⁵⁷

Partly in response to a draft of this Article, McGhee recently published a new piece addressing several features of the efficiency gap measure, including its relationship to turnout, that were previously unexamined.¹⁵⁸ In that piece, McGhee derives the same turnout gap as presented above.¹⁵⁹ McGhee views the efficiency gap's dependence on turnout as a violation of his efficiency principle.¹⁶⁰ To see why, consider a district where party *x* wins but wastes fewer votes. Keep constant the parties' respective vote shares in that district, but increase district turnout. This change will increase the wasted vote disparity in that district and thus the efficiency gap as originally defined, indicating greater advantage for party *x*. But party *x* has increased its vote share with no change in seat share, suggesting less advantage for party *x*.

To solve this problem, McGhee essentially proposes a new definition of wasted votes in which the threshold needed to win (V_{ni}) is not half of district

155. See *supra* text accompanying note 101.

156. See *infra* Part II.E.2.

157. In Part V below, I discuss in detail the questions of robustness and correspondence raised by this and other issues.

158. See McGhee, *Measuring Efficiency*, *supra* note 86, at 426-32.

159. See *id.* at 427; *id.* app. at 438-39.

160. See *id.* at 427-28 (calling the turnout problem "the opposite of what would be expected" and a "clear violation" of the efficiency principle).

turnout ($\frac{T_i}{2}$) but half of average turnout ($\frac{T^*}{2}$).¹⁶¹ This definition eliminates the turnout gap from the equation so that the simplified formula expresses the efficiency gap not approximately based on electoral assumptions, but rather exactly based on the new definition.¹⁶² Full consideration of this new approach lies beyond the scope of this Article, which focuses on the efficiency gap as originally presented in scholarship and litigation. But note that this new definition reflects the view of a mapmaker assessing an entire plan rather than of a voter participating in an individual district.¹⁶³

C. The Aggregation Method from District to Plan

That the efficiency gap, traditionally defined, compares the total numbers of wasted votes is noteworthy for two reasons. First, it facilitates an analytical technique that defines a plan's gap as a turnout-weighted average of district-level wasted vote disparities.¹⁶⁴ Second, there is an alternative aggregation method that compares wasted vote shares rather than wasted vote totals.¹⁶⁵

1. The analytical technique: from district-level disparity to plan-level gap

The proponents' approach involves a two-step aggregation process: first, sum wasted votes over districts; second, compute the difference using the total

161. McGhee does not claim to be changing the definition of a surplus vote from $E_{pi} = V_{pi} - \frac{T_i}{2}$ to $E_{pi}^{new} = V_{pi} - \frac{T^*}{2}$. Instead, he writes that a party's total wasted votes (W_p) must be adjusted by adding an "effective vote deviation," defined as $\sum_{i \in D_p} (\frac{T_i}{2} - \frac{T^*}{2})$. See *id.* at 427 (equation 6). But adding this effective vote deviation term is equivalent to changing the definition of a surplus vote.

To see this, let $W_p = \sum_{i \in D_p} E_{pi} + \sum_{i \notin D_p} L_{pi}$, and let $W_p^{new} = \sum_{i \in D_p} E_{pi}^{new} + \sum_{i \notin D_p} L_{pi}$.

Then: $W_p + \sum_{i \in D_p} (\frac{T_i}{2} - \frac{T^*}{2}) = \sum_{i \in D_p} E_{pi} + \sum_{i \notin D_p} L_{pi} + \sum_{i \in D_p} (\frac{T_i}{2} - \frac{T^*}{2})$
 $= \sum_{i \in D_p} (V_{pi} - \frac{T_i}{2} + \frac{T_i}{2} - \frac{T^*}{2}) + \sum_{i \notin D_p} L_{pi} = \sum_{i \in D_p} (V_{pi} - \frac{T^*}{2}) + \sum_{i \notin D_p} L_{pi}$
 $= \sum_{i \in D_p} E_{pi}^{new} + \sum_{i \notin D_p} L_{pi} = W_p^{new}$.

162. See McGhee, *Measuring Efficiency*, *supra* note 86, at 427-28.

163. Consider a plan with average turnout of 100, a high-turnout district that party *x* wins 105-95, and a low-turnout district that party *x* wins 40-10. If surplus votes are defined relative to a baseline of half of average turnout ($100 / 2 = 50$), then party *x* incurs 55 surplus votes in the high-turnout district ($105 - 50 = 55$) and *negative* 10 surplus votes in the low-turnout district ($40 - 50 = -10$).

164. See *infra* Part II.C.1.

165. See *infra* Part II.C.2.

number of votes cast.¹⁶⁶ One can switch the order of aggregation by defining district-level wasted vote disparity and then expressing a plan's efficiency gap as the weighted average of district-level disparities. Specifically, define the district-level wasted vote disparity (Δw_i) between party x and party y in district i as the parties' relative wasted votes in that district, expressed as a proportion of district-level voter turnout (T_i).

$$\Delta w_i = \frac{W_{yi} - W_{xi}}{T_i}$$

A plan's efficiency gap is the weighted average of district-level wasted vote disparities, where each district's wasted vote disparity is weighted by its turnout.¹⁶⁷

$$\Delta W = \frac{\sum_{i \in D} T_i \Delta w_i}{\sum_{i \in D} T_i}$$

Under the equal voter turnout assumption, the plan's efficiency gap is simply the unweighted average district disparity.¹⁶⁸

$$\Delta W = \frac{\sum_{i \in D} \Delta w_i}{S}$$

Formally, define the set of zero-disparity districts (Π_i^0) as the set of districts that produce a wasted vote disparity of zero, and define the set of zero-gap plans (Π^0) as the set of plans that produce an efficiency gap of zero.

$$\begin{aligned}\Pi_i^0 &= \{D_i \mid \Delta w_i = 0\} \\ \Pi^0 &= \{D \mid \Delta W = 0\}\end{aligned}$$

Thus, because a plan's efficiency gap is an average of its districts' disparities, one simple way to achieve a zero-gap plan is to maintain zero disparity in each district.

$$D^0 = \cup D_i \text{ where for each } i \ D_i \in \Pi_i^0$$

166. See Stephanopoulos & McGhee, *supra* note 7, at 851-52.

167. By definition: $\Delta W = \frac{w_y - w_x}{v_x + v_y}$, $\Delta w_i = \frac{W_{yi} - W_{xi}}{T_i}$, $W_p = \sum_{i \in D} W_{pi}$, $V_p = \sum_{i \in D} V_{pi}$, and

$$V_{xi} + V_{yi} = T_i. \text{ Thus: } \Delta W = \frac{w_y - w_x}{v_x + v_y} = \frac{\sum_{i \in D} W_{yi} - W_{xi}}{\sum_{i \in D} V_{xi} + V_{yi}} = \frac{\sum_{i \in D} T_i \left(\frac{W_{yi} - W_{xi}}{T_i} \right)}{\sum_{i \in D} T_i} = \frac{\sum_{i \in D} T_i \Delta w_i}{\sum_{i \in D} T_i}.$$

168. $S = \sum_{i \in D} 1$, and under the equal turnout assumption, $T_i = T^*$ for each $i \in D$. Thus:

$$\Delta W = \frac{\sum_{i \in D} T_i \Delta w_i}{\sum_{i \in D} T_i} = \frac{\sum_{i \in D} T^* \Delta w_i}{\sum_{i \in D} T^*} = \frac{\sum_{i \in D} \Delta w_i}{\sum_{i \in D} 1} = \frac{\sum_{i \in D} \Delta w_i}{S}.$$

I call such a plan a *simple zero-gap plan*. By design, it maintains an efficiency gap of zero.¹⁶⁹ Note that the simple zero-gap plan is not the only zero-gap plan. More generally, a plan can achieve an efficiency gap of zero even if it exhibits nonzero district-level wasted vote disparities, provided those disparities average out such that both parties waste the same number of votes overall. But so long as we restrict our attention to plans with equal voter turnout in each district, each zero-gap plan can be converted to (and from) a simple zero-gap plan by performing the appropriate series of voter swaps, whereby two districts swap two voters—one party *x* supporter for one party *y* supporter—without altering any district election outcome. A voter swap corresponds to a marginal change in the electoral boundary between two adjacent districts. A significant change to a district boundary can be understood as a series of incremental voter swaps. By design, a swap changes the wasted vote disparities in the participating districts but maintains the same number of wasted votes for each party overall. Starting with a simple zero-gap plan, such a swap produces a plan that is still zero-gap but is no longer simple.

This provides a useful way to investigate the efficiency gap: Construct a simple zero-gap plan, examine its properties, and determine which properties vary under gap-preserving voter swaps. I use this technique in Part III below.

2. The alternative approach: comparing wasted vote shares

The proponents compare the parties' total number of wasted votes:

$$\Delta W = \frac{W_y - W_x}{V_y + V_x}.$$

Anthony McGann and his coauthors have criticized this approach, arguing that “it is not obvious that each party having an equal absolute number of wasted votes is uniquely fair” and noting that one could alternatively require that each party should waste the same *share* of votes rather than the same *number* of votes.¹⁷⁰ Following this suggestion, John Nagle has developed what he calls a “voter-centric” measure (ΔW^V) that compares relative wasted vote shares rather than relative wasted vote totals as a proportion of all ballots cast.¹⁷¹

$$\Delta W^V = \frac{W_y}{V_y} - \frac{W_x}{V_x}$$

169. Because both parties waste the same number of votes in each district, the parties must waste the same number of votes overall.

170. McGann et al., *supra* note 20, at 296.

171. See Nagle, *supra* note 20, at 201-02.

The two aggregation methods superficially appear nearly identical, but they can produce markedly different results. Suppose 6 million votes are cast, with 4 million for party *x* and 2 million for party *y*, and suppose each party wastes 1.5 million votes. This means that 75% of ballots cast by party *y* supporters are wasted while only 37.5% of ballots cast by party *x* supporters are wasted.¹⁷² Each party wastes the same number of raw votes, but a party *y* supporter is twice as likely as a party *x* supporter to waste her vote. While a party may care more about the total number of votes it wastes, an individual voter may care more about the likelihood her vote will be wasted, which depends on the share rather than the total of votes wasted by her preferred party. This is why Nagle describes relative wasted vote totals as “party-centric” and relative wasted vote shares as “voter-centric.”¹⁷³ One could argue that this voter-centric approach better coheres with an individual rights framework based on a particular voter’s equal protection or First Amendment interest in participating free from discrimination based on political affiliation, belief, or expressive conduct.¹⁷⁴

Note that the choice between a party-centric (wasted vote totals) and voter-centric (wasted vote shares) scale for the asymmetry comparison may be particularly consequential for minor parties that earn a small number of total votes and usually waste all of them (because they win no seats). Such a minor party will have a low wasted vote total but a high wasted vote share—possibly 100%. Thus, a voter-centric scale may facilitate greater receptivity to a claim that an electoral map is gerrymandered to disadvantage a minor party.

Nagle explored the mathematical properties of a wasted vote measure using a voter-centric scale of wasted vote shares.¹⁷⁵ But he defined a surplus vote as half the victory margin rather than the full victory margin, noting that the alternative was mathematically equivalent to unequal weighting of lost and surplus votes.¹⁷⁶ Nagle then rejected unequal weighting of lost and surplus votes on the ground that it would violate McGhee’s efficiency principle.¹⁷⁷ Part IV below proposes a new measure that compares the shares of wasted votes, with surplus votes defined as the full victory margin. This new measure is voter-centric in terms of both the scale of comparison and the definition of wasted votes.

172. That is, $1,500,000 / 2,000,000 = 0.75$ and $1,500,000 / 4,000,000 = 0.375$.

173. See Nagle, *supra* note 20, at 201.

174. See *supra* note 114 and accompanying text.

175. See Nagle, *supra* note 20, at 201-03.

176. See *id.* at 199 & n.16, 203 & n.24.

177. See *id.* at 203 (“However, as Eric McGhee has kindly pointed out, the possibility that different values of *S* for the same vote *V* may give the same value of bias violates a fundamental principle for bias measures, namely, gerrymandering might be able to increase *S* for the same *V* and not be detected by the measure of bias.” (citation omitted)).

D. The Definition and Weight of Surplus Votes

Both the definition and the weight of surplus votes are methodological choices, not self-defining concepts. They are susceptible to competing interpretations, and selecting among them requires deliberation and transparency. The proponents and the *Whitford* litigants define surplus votes using a threshold of half of total votes, and they equally weight lost and surplus votes.¹⁷⁸ But these two distinct yet related methodological choices have not been adequately explained.

The proponents simply define a wasted vote as a lost *or* surplus vote, assuming implicitly and without explanation that lost and surplus votes should be equally weighted.¹⁷⁹ When considered from the perspective of the party, equal weighting makes sense: Whether lost or surplus, a vote is equally wasted in the sense that it could be more effective if cast in another district. But when considered from the perspective of the individual voter, lost and surplus votes may not be equivalent. True enough, both the voter who casts the lost vote and the voter who casts the surplus vote may regret that her vote could have been more effective in another district. But the voter who casts a surplus vote gets to be represented by the candidate of her choice. Not so for the voter who casts a lost vote. Faced with a choice between casting a lost vote and casting a surplus vote, I would prefer the latter option, and I suspect most other voters would, too.

The definition of surplus votes is similarly unexplained and even less intuitive. The proponents define a surplus vote as one “cast . . . for a winning candidate but *in excess of what [the candidate] needed to prevail*.”¹⁸⁰ Thus, $E_{pi} = V_{pi} - V_{ni}$, where V_{ni} denotes the number of votes the “winning candidate . . . needed to prevail.”¹⁸¹ But what is V_{ni} ? Let V_{1i} and V_{2i} denote the number of ballots cast for the first- and second-place candidates, respectively. Recall that under the two-party assumption, the sum of the parties’ respective vote totals is the district’s total voter turnout: $T_i = V_{yi} + V_{xi}$.¹⁸² The difference

178. See *supra* notes 116-19 and accompanying text.

179. See Stephanopoulos & McGhee, *supra* note 7, at 850-51 (“[A]ny vote for a losing candidate is wasted by definition, but so too is any vote beyond the 50 percent threshold needed (in a two-candidate race) to win a seat.”).

180. *Id.* at 834 (emphasis added); see also *id.* at 851.

181. See *id.* at 851.

182. See *supra* text accompanying note 115.

between V_{1i} and V_{2i} is the victory margin (M_i), which is itself a measure of electoral competitiveness.¹⁸³

$$M_i = V_{1i} - V_{2i} = |V_{xi} - V_{yi}| = \begin{cases} V_{xi} - V_{yi} & i \in D_x \\ V_{yi} - V_{xi} & i \in D_y \end{cases}$$

Under a plurality voting system, the candidate with the most votes wins.¹⁸⁴ This suggests that the number of votes the winning candidate “needed to prevail” is the number of votes earned by her most popular opponent—that is, $V'_{ni} = V_{2i}$.¹⁸⁵ Under this definition, the number of surplus votes (E'_{pi}) is simply the victory margin.

$$E'_{pi} = V_{pi} - V'_{ni} = V_{1i} - V_{2i} = M_i$$

For example, if 100 ballots are cast and the victor prevails with a vote tally of 65 to 35, then $V_{ni} = V_{2i} = 35$ and $E'_{pi} = V_{pi} - V'_{ni} = 65 - 35 = 30$.

But this is not the proponents’ definition. Instead, they define necessary votes (V_{ni}) as half of actual voter turnout ($\frac{T_i}{2}$).¹⁸⁶ Under this definition, the number of surplus votes is half the victory margin.

$$E_{pi} = V_{pi} - V_{ni} = V_{1i} - \frac{T_i}{2} = V_{1i} - \frac{V_{1i} + V_{2i}}{2} = \frac{V_{1i} - V_{2i}}{2} = \frac{M_i}{2}$$

For example, if the victor prevails 65 to 35, $V_{ni} = \frac{T_i}{2} = 50$ and $E_{pi} = V_{pi} - V_{ni} = 65 - 50 = 15$.¹⁸⁷

The proponents justify this approach by invoking “the 50 percent threshold needed (in a two-candidate race) to win a seat.”¹⁸⁸ It is true that a candidate must earn more than half the votes to win a two-way race. But the proponents’ invocation of the 50% threshold fails to clarify the basis for their definition because

183. See Stephanopoulos, *supra* note 38, at 678 (“Metrics such as average margin of victory . . . indicate both how competitive individual districts are and how responsive a jurisdiction’s electoral system is as a whole.”).

184. See *Plurality-Majority Systems*, FAIRVOTE, <https://perma.cc/VBY8-JS7A> (archived Feb. 25, 2018).

185. Technically, the number of necessary votes is one more than the runner-up’s vote total. Like others engaged in efficiency gap analysis, see, e.g., Stephanopoulos & McGhee, *supra* note 7, at 834 n.14, I ignore this “plus one” technicality.

186. See *supra* text accompanying notes 117-18.

187. See Stephanopoulos & McGhee, *supra* note 7, at 851 n.107 (“Assume, for example, that Candidate A receives 65 percent of the vote and Candidate B receives 35 percent. Then 15 percent of Candidate A’s votes . . . are wasted . . .”).

188. See *id.* at 851; see also *id.* at 834 n.14 (“For the sake of simplicity, we also assume that 50 [out of 100] votes are needed to win a district . . .”); *id.* at 851 n.107 (“[V]ictory in a two-candidate race is achieved with 50 percent of the vote . . .”).

both definitional approaches—theirs and the one defined in terms of the victory margin—can be explained with reference to a 50% threshold. The distinction between surplus and necessary votes logically entails a counterfactual in which a candidate earns fewer votes but still prevails. If the victor earned the same vote total as the runner-up ($V_{1i} = V_{2i}$), half of turnout would be the runner-up's vote total: $\frac{T_i}{2} = \frac{V_{1i} + V_{2i}}{2} = \frac{2V_{2i}}{2} = V_{2i}$. But the proponents, without adequate explanation, apply the 50% threshold to the actual vote total rather than the vote total in the counterfactual.

This is the definition of surplus votes Judge Griesbach, the dissenting judge on the *Whitford* panel, attacked as “opaque” and “absurd.”¹⁸⁹ Judge Griesbach suggested that surplus votes must be defined as the entire vote margin rather than half the vote margin:

Just as a baseball game is not decided by reference to total runs, an election is not decided by a fraction of total votes. Instead, the number of votes needed to win is simply the number one more than the losing candidate won, and therefore anything beyond that should be counted as a “wasted” vote¹⁹⁰

The proponents' definition of surplus votes could be explained more clearly. It simply defines surplus votes as the number of voter swaps possible without altering the outcome. Each swap exchanges one party *x* supporter for one party *y* supporter and accordingly decreases the victory margin by two votes, so the total number of possible outcome-preserving swaps is half the victory margin. This reflects the perspective of the mapmaker under the equal voter turnout assumption. Each voter swap represents a marginal adjustment to the district boundaries—one that changes the district of only the two homes where the respective swapped voters reside.

This definition of surplus votes is not absurd, but it does privilege the mapmaker's party-centric perspective under the assumption of equal voter turnout. From the perspective of an individual voter who supports a particular party, her choice is either to cast her ballot or to stay home. Her vote is wasted if she could have stayed home without altering the outcome. From the perspective of the mapmaker focused on that partisan voter, his choice is to which district she should be assigned; he could leave the voter in her current district or swap her with a voter from another district supporting the other party. From the perspective of the mapmaker, then, the partisan voter's vote is wasted if he could swap her without altering the outcome in the original district.

Judge Griesbach's baseball analogy implicitly adopts a voter-centric approach. Judge Griesbach considers a single baseball game just as a voter considers only the district to which she is assigned. But in a real election, there are multiple

189. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 958 (W.D. Wis. 2016) (Griesbach, J., dissenting), stay granted, 137 S. Ct. 2289, and jurisdiction postponed, 137 S. Ct. 2268 (2017).

190. *Id.* (emphasis omitted).

districts, and a partisan mapmaker is concerned with the total number of districts won. To put this in baseball terms: Suppose multiple baseball games are played simultaneously, each game between one team from each of two leagues. Further suppose that a run for team *x* in one game could be swapped for a run for team *y* in another. Each league's ultimate goal is to win as many games as possible with a fixed number of runs. This two-league competition has the salient features of partisan districting. Under these conditions, it is not absurd to define a team's surplus runs as more than half of a game's total runs, as that is simply the number of run-swaps possible without altering a game's outcome.

In short, the proponents' party-centric approach is one plausible way to define and weight surplus votes. But it is neither the only way nor necessarily the most intuitive way. Just as Nagle has recently considered McGann's suggestion of comparing wasted vote shares,¹⁹¹ he has also noted that surplus votes can alternatively be defined as the entire vote margin and that wasted votes can be generalized as a weighted sum of lost and surplus votes.¹⁹² Part III.F below similarly alters the definition and weight of surplus votes, and it derives a more generalized formula that quantifies the precise impact of these methodological choices on the efficiency gap measure. My results accord with Nagle's, but my approach demonstrates the critical role these methodological choices play in calibrating the measure's relationship between the competing norms of electoral competitiveness and seats-votes proportionality.

E. The Two-Party Assumption

The proponents make the assumption that "there are only two parties" in any election.¹⁹³ The efficiency gap measure is by definition a bilateral comparison—it takes two parties and compares their relative efficiency by calculating the difference in their respective wasted vote totals. I use the term *focal parties* to refer to the two parties that are the focus of the measure's bilateral comparison and the term *peripheral candidates* to refer to candidates unaffiliated with either of the two focal parties. I call the two focal parties party *x* and party *y*.¹⁹⁴ When the proponents assume that there are only two parties, they necessarily ignore

191. See *supra* text accompanying notes 171-77.

192. See Nagle, *supra* note 20, at 199 & n.16, 200, 203 & n.24.

193. See Stephanopoulos & McGhee, *supra* note 7, at 853.

194. The efficiency gap is a signed measure: Its absolute value indicates the extent of the gerrymander, while its sign indicates which party the gerrymander favors. I define all relevant concepts so that a positive gap favors party *x* and a negative gap favors party *y*. Obviously, the two primary parties of interest are the major political parties. Whenever I discuss the efficiency gap between Republicans and Democrats, I treat Republicans as party *x* and Democrats as party *y*.

any ballot cast for a peripheral candidate.¹⁹⁵ Because the proponents emphasize partisan fairness between the two major political parties,¹⁹⁶ they make the simplifying assumption that every district race is a contest between one party *x* candidate and one party *y* candidate. This two-party assumption actually consists of three related assumptions: In each district's general election (1) no ballots are cast for peripheral candidates; (2) no more than one candidate runs from each focal party;¹⁹⁷ and (3) no race is uncontested.¹⁹⁸

Subpart E.1 below examines the consequences when the third assumption fails and analysts must impute results for uncontested races. Implicit in the efficiency gap approach is a final assumption: that the vote shares earned by the two parties reflect their relative popular support among the electorate. Subpart E.2 examines the consequences when this final assumption fails because electoral administration differentially prevents or discourages supporters of one party from casting ballots.

1. Uncontested races

When a district race is uncontested, the proponents suggest a strategy of imputation to estimate what would have occurred had the race been contested:

Going forward, we encourage other scholars to explore a range of imputation techniques to ensure that the direction of a gerrymander (if not its size) is robust to any particular strategy. But this catholic philosophy has its limits. We strongly discourage analysts from either dropping uncontested races from the computation or treating them as if they produced unanimous support for a party. The former approach eliminates important information about a plan, while the latter assumes that coerced votes accurately reflect political support. Neither correctly represents how the gerrymandering party itself would view its plan.¹⁹⁹

Uncontested district races present problems of *normative correspondence*, *robustness*, and *scope* for the efficiency gap measure. If imputation were impermissible or impossible, we could either omit uncontested elections from

195. See McGhee, *supra* note 10, at 68 (defining the system as one “with two parties”). The proponents’ 2015 article offers no definition of surplus votes outside the two-party context. See Stephanopoulos & McGhee, *supra* note 7, at 851.

196. See, e.g., Stephanopoulos & McGhee, *supra* note 7, at 838-39 (characterizing the efficiency gap as a “new measure of *partisan* symmetry” designed to capture “the idea that a plan should treat the *major parties* symmetrically” (emphasis added)).

197. The 2015 article defines the measure in terms of the parties’ wasted votes and defines wasted votes in terms of ballots cast for party-affiliated candidates, implicitly assuming that each party has only one candidate. See Stephanopoulos & McGhee, *supra* note 7, at 851.

198. The proponents explicitly recognize this assumption and apply imputation techniques when races are uncontested to estimate what would have happened if those races had been contested. See *id.* at 866-67.

199. Stephanopoulos & McGhee, *supra* note 7, at 867.

the dataset (presenting a problem of scope) or count uncontested districts as registering unanimous support for the winning party (presenting a problem of normative correspondence). If imputation is permitted, we must choose a fair and accurate method of imputation (presenting a problem of robustness).

The need to impute election results in uncontested districts prompts two analytical points related to our overall assessment of the efficiency gap measure. First, the imputation method chosen may, under the right circumstances, have a significant, even outcome-determinative, effect on a plan's efficiency gap. Second, uncontested races are a sign of uncompetitive districts, so there may be an unfortunate association between a plan's uncompetitiveness and the sensitivity of the efficiency gap calculation to imputation method.

My first point is practical. The proponents recognize that employing different imputation methods may present robustness problems²⁰⁰ but do not quantify how significantly the choice of imputation method might affect the resulting efficiency gap. It is possible, however, to mathematically estimate how sensitive the gap is to different imputation approaches. Consider the simplified efficiency gap formula: $\Delta W = S^* - 2V^*$. When an analyst imputes vote share for an uncontested district, she changes only the vote totals, not the winning party. The seat margin (S^*) stays the same, but the vote margin (V^*) changes. Imagine two analysts employing different imputation methods that produce different imputed vote margins in uncontested races. Consider the notation in Table 2 below.

Table 2
Notation for the Effect of Imputation

Notation	Description
$\bar{S}_U$	the proportion of districts that are uncontested
$\bar{V}_C$	the average district vote share in contested districts
$\bar{V}_{1U}$	the average district vote share in uncontested districts as estimated by Analyst 1
$\bar{V}_{2U}$	the average district vote share in uncontested districts as estimated by Analyst 2
$\bar{V}_1$	the overall vote share as estimated by Analyst 1
$\bar{V}_2$	the overall vote share as estimated by Analyst 2
$\Delta W_{1,2}$	the difference between the efficiency gap estimated by Analyst 1 and the efficiency gap estimated by Analyst 2

200. See, e.g., *id.* at 866 (“[I]mputation approaches can be more or less sophisticated, and can bring varying amounts of information to bear on the problem.”).

We can then quantify the impact of imputation under the equal voter turnout assumption:²⁰¹

$$\Delta W_{1,2} = 2\bar{S}_U(\bar{V}_{2U} - \bar{V}_{1U}).$$

The greater the difference between imputed average vote shares, the greater the difference in computed gaps. Note that the analyst who imputes the higher average vote share will compute the lower gap. By increasing the estimate of party *x* support, the analyst increases the estimate of deserved seat share, thereby decreasing the estimate of undeserved seat share and the estimate of the extent to which the plan favors party *x*.

Moreover, the more uncontested races there are, the more the imputation technique matters. For example, assume that one-third of districts in a state hold uncontested elections. Assume also that Analyst 1 concludes that party *x* would have, on average, earned 70% of the vote in uncontested races, but Analyst 2 concludes that party *x* would have, on average, won 73% of the vote in uncontested races. In that scenario, Analyst 1 would compute an efficiency gap 2% higher than that computed by Analyst 2.²⁰²

This effect of the imputation method upon the statewide efficiency gap is not merely a theoretical concern; the imputation method could have significant practical consequences for the overall validity of a given districting plan. In the 2012 Wisconsin State Assembly election, for instance, 27 of 99 assembly races were uncontested and therefore had to be imputed in order for the experts to calculate that election's efficiency gap.²⁰³ Applying the simplified formula to this election, but using different imputation methods, the *Whitford* plaintiffs' experts reached different results: Mayer's efficiency gap would have been 9.85% under the simplified formula, whereas Jackman's gap was 13%.²⁰⁴ The difference between these two estimates is greater than the difference between Mayer's estimate and the numeric

201. The proportion of contested districts is $1 - \bar{S}_U$. Therefore, $\bar{V}_1 = \bar{S}_U \bar{V}_{1U} + (1 - \bar{S}_U) \bar{V}_C$, and $\bar{V}_2 = \bar{S}_U \bar{V}_{2U} + (1 - \bar{S}_U) \bar{V}_C$.

Thus: $\bar{V}_2 - \bar{V}_1 = [\bar{S}_U \bar{V}_{2U} + (1 - \bar{S}_U) \bar{V}_C] - [\bar{S}_U \bar{V}_{1U} + (1 - \bar{S}_U) \bar{V}_C] = \bar{S}_U (\bar{V}_{2U} - \bar{V}_{1U})$, and $\Delta W_{1,2} = \left[S^* - 2 \left(\bar{V}_1 - \frac{1}{2} \right) \right] - \left[S^* - 2 \left(\bar{V}_2 - \frac{1}{2} \right) \right] = 2(\bar{V}_2 - \bar{V}_1) = 2\bar{S}_U (\bar{V}_{2U} - \bar{V}_{1U})$.

202. $\bar{S}_U = \frac{1}{3}$, $\bar{V}_{1U} = 0.70$, and $\bar{V}_{2U} = 0.73$.

Thus: $\Delta W_{1,2} = 2\bar{S}_U (\bar{V}_{2U} - \bar{V}_{1U}) = 2 \frac{1}{3} (0.73 - 0.70) = 0.02$.

203. See Mayer Report, *supra* note 146, at 39 (noting that 23 Democrats but only 4 Republicans ran unopposed); *id.* at 44-45 (discussing imputation); see also Jackman Report, *supra* note 102, at 22-31 (discussing imputation).

204. Compare Mayer Report, *supra* note 146, at 46 tbl.10, with Jackman Report, *supra* note 102, at 16, 36. For more on the experts' use of the simplified formula, see notes 147-49 and accompanying text above.

threshold for presumptive invalidity,²⁰⁵ suggesting that selecting an imputation method could have a nontrivial impact on efficiency gap analysis.

There may be real-world elections for which the existence of an above-threshold efficiency gap depends on the imputation method selected to assess uncontested races.²⁰⁶ If the efficiency gap were adopted as the definitive legal test for unconstitutional partisan gerrymandering, plaintiffs alleging such a gerrymander might strategically employ imputation methods that would produce larger gaps; defendants might strategically employ competing methods to produce lower gaps; and the resulting battle of the experts over the imputation method could be outcome-determinative.

Given that analysts calculating the efficiency gap often must impute hypothetical election results, one of the purported advantages of the measure over other measures of partisan symmetry is actually quite limited. The efficiency gap may be particularly appealing—especially to Justice Kennedy—because it relies upon directly observed election data rather than hypothetical results.²⁰⁷ But if calculating the gap requires imputing hypothetical results, and if the size of the gap depends in substantial part on which method an analyst selects, the gap is less of a straightforward measure of real-world data. It is possible that the Court might restrict efficiency gap analysis to circumstances in which a plan produces a durable above-threshold gap under *any* plausible imputation method. This would address the concern with hypotheticals and avoid a battle of the experts—but at the cost of limiting the circumstances in which plaintiffs can deploy the measure and, potentially, when those plaintiffs can prevail on their partisan gerrymandering claims at all.

It is particularly troubling that the efficiency gap calculation is more sensitive to imputation method when a plan has more uncontested races because an uncontested race is a signal that a district is highly uncompetitive.²⁰⁸ Thus, a highly uncompetitive plan may produce a relatively larger number of uncontested races, rendering efficiency gap analysis more sensitive to imputation method. That efficiency gap analysis may be more difficult for uncompetitive plans could perversely create a (further) incentive for mapmakers to draw uncompetitive plans. In Part III below, I argue that adopting the efficiency gap as the definitive measure of partisan gerrymandering may also unintentionally incentivize uncompetitive plans because

205. The academic proponents suggested a threshold of 8% based on historical trends. See Stephanopoulos & McGhee, *supra* note 7, at 884, 888–89. Jackman suggested a threshold of 7% based on durability analysis. See Jackman Report, *supra* note 102, at 66–69.

206. Cf. Jackman Report, *supra* note 102, at 22 (“Uncontested races are common in state legislative elections, and are even the norm in some states.”).

207. Cf. *League of United Latin Am. Citizens v. Perry (LULAC)*, 548 U.S. 399, 419–20 (2006) (opinion of Kennedy, J.) (“[W]e are wary of adopting a constitutional standard that invalidates a map based on unfair results that would occur in a hypothetical state of affairs.”).

208. See Mayer Report, *supra* note 146, at 39 (“[U]ncontested races occur largely when one party sees zero probability of winning because the majority party has such overwhelming majorities in the district.”).

the sensitivity of the efficiency gap to vote swings is a function of a plan's responsiveness. Combined, these considerations suggest that overreliance on the efficiency gap may present the risk of a vicious cycle: Endorsing the efficiency gap as the definitive measure may unintentionally encourage mapmakers to draw uncompetitive plans that produce a high number of uncontested races, and courts may struggle to evaluate the resulting plans because the high proportion of uncontested races renders efficiency gap analysis more sensitive to imputation methods.

2. Voter suppression

Because the efficiency gap is a measure of the relative number of wasted votes for each party, the only data points it requires are ballots cast and seats won. In the measure's most direct application, an analyst computes the efficiency gap produced by a given plan in a given election by inputting into the long-form equation (or simplified formula) the actual votes cast in that election by district and party (or the vote margin and seat margin produced by that election) after imputations for uncontested races. In a more advanced application, an analyst estimates the results of a hypothetical election by running regression techniques on historical and contemporaneous data correlated with election outcomes and then inputs those results into the relevant equation to compute the efficiency gap a given plan would likely produce in that hypothetical election.

The efficiency gap's focus on ballots cast means that it cannot detect any obstacles voters face in casting ballots and is blind to rules of electoral administration that disproportionately affect supporters of one party. If one such rule—for example, a stringent photo identification requirement for in-person voting—thwarts a voter's effort to cast a ballot for a party *y* candidate, the measure detects no problem; it simply assumes that party *y* has one ballot less support from the electorate.

This dynamic may unintentionally reward, and thereby further incentivize, voter suppression efforts because suppression may make a gerrymander seem less like a gerrymander—that is, suppression can hide gerrymanders from the efficiency gap. Suppressing one party's statewide vote total can have the effect of reducing the overall gap. If, for instance, the Democrats enact a partisan gerrymander and then adopt electoral reforms that disproportionately burden Republican voters, any resulting decrease in Republican turnout would operate to reduce the efficiency gap's measure of the pro-Democratic advantage conferred by the districting plan.²⁰⁹

209. To be clear, I do not claim that this lack of sensitivity to voter suppression is a problem unique to the efficiency gap. It is a feature of any measure of partisan gerrymandering that quantifies an ideal or acceptable relationship between votes and seats won. Specifically, it is a feature of any measure that satisfies the efficiency principle. Any definition that uses popular support as demonstrated by ballots cast to justify the number of seats won may incentivize voter suppression because political actors seeking to demonstrate greater relative support can inflate the appearance of that support by making it harder for their competitors' supporters to vote.

Unfortunately, the Court is considering the efficiency gap at a moment in U.S. politics when electoral administration is a partisan issue, with Democrats more likely to support “ballot access” measures ostensibly designed to reduce the perceived risk that an eligible voter will encounter difficulty casting a ballot and Republicans more likely to support “ballot integrity” measures ostensibly designed to reduce the perceived risk that an ineligible voter will cast a ballot.²¹⁰ In the years since the Court invalidated the Voting Rights Act’s coverage formula and thereby rendered inoperative its preclearance regime,²¹¹ twenty states have adopted “ballot integrity” laws²¹²—including notably the two states currently defending against partisan gerrymandering claims based on efficiency gap approaches, Wisconsin²¹³ and

210. See Stephen Ansolabehere, *Access Versus Integrity in Voter Identification Requirements*, 63 N.Y.U. ANN. SURV. AM. L. 613, 613-16 (2008).

211. See *Shelby County v. Holder*, 133 S. Ct. 2612, 2631 (2013); see also *supra* note 59.

212. See Brennan Ctr. for Justice, *New Voting Restrictions in America 2-11* (2017), <https://perma.cc/8BDU-GKAC> (detailing as of May 2017 that Alabama, Arizona, Arkansas, Georgia, Indiana, Iowa, Kansas, Mississippi, Missouri, Nebraska, New Hampshire, North Carolina, North Dakota, Ohio, Rhode Island, South Carolina, Tennessee, Texas, Virginia, and Wisconsin had enacted at least one type of voting restriction).

213. See *id.* at 8 (discussing Wisconsin’s restrictions on the right to vote). In 2011, two years before the Court decided *Shelby County*, Wisconsin adopted a stringent voter identification law. See Act of May 25, 2011, No. 23, 2011 Wis. Sess. Laws 104 (codified as amended in scattered sections of the Wisconsin Statutes). A federal district court concluded that the law violated the Fourteenth Amendment and the Voting Rights Act. See *Frank v. Walker*, 17 F. Supp. 3d 837, 863, 879 (E.D. Wis.), *rev’d*, 768 F.3d 744 (7th Cir. 2014). Before the Seventh Circuit heard argument in the appeal, the state supreme court softened the effects of the law by requiring the state to issue photo identifications free of charge. See *Frank v. Walker*, 768 F.3d 744, 747 (7th Cir. 2014); Milwaukee Branch of the NAACP v. Walker, 851 N.W.2d 262, 281 (Wis. 2014); see also Richard L. Hasen, *Softening Voter ID Laws Through Litigation: Is It Enough?*, 2016 WIS. L. REV. FORWARD 100, 110-11 (detailing much of the litigation challenging Wisconsin’s voter identification law in federal and state courts). The state also places restrictions on individual voter registration and early voting. See Brennan Ctr. for Justice, *supra* note 212, at 8.

North Carolina.²¹⁴ It is difficult to confidently estimate the effect of electoral reforms on voter turnout,²¹⁵ and it is not the objective of this Article to advance the debate about whether, and to what extent, “ballot integrity” measures such as those adopted in Wisconsin and North Carolina differentially burden and thereby reduce participation among Democratic-leaning

214. See Brennan Ctr. for Justice, *supra* note 212, at 11 (discussing North Carolina’s restrictions on the right to vote). North Carolina was subject to preclearance prior to *Shelby County* because some of its counties fell under the coverage formula. See *Jurisdictions Previously Covered by Section 5*, U.S. DEP’T JUST., <https://perma.cc/LY5X-QX9V> (last updated Aug. 6, 2015). Once the Court struck down the coverage formula, the North Carolina legislature quickly enacted an omnibus electoral reform bill with five key provisions: (1) a voter identification requirement (for in-person but not mail-in voting) limited to forms of identification that “African Americans disproportionately lacked”; (2) a reduction in early voting days; (3) elimination of same-day registration; (4) elimination of out-of-precinct voting; and (5) elimination of preregistration. See *N.C. State Conference of the NAACP v. McCrory*, 831 F.3d 204, 216–18 (4th Cir. 2016), *cert. denied*, 137 S. Ct. 1399 (2017); see also Act of July 26, 2013, No. 381, 2013 N.C. Sess. Laws 1505 (codified in scattered sections of the North Carolina General Statutes). The Fourth Circuit concluded that the omnibus bill constituted intentional racial discrimination, “target[ing] African Americans with almost surgical precision,” see *N.C. State Conference*, 831 F.3d at 214–15, and “[u]sing race as a proxy for party . . . to win an election,” see *id.* at 222. See also *id.* at 225 (noting that North Carolina is a state where “African-American race is a better predictor for voting Democratic than party registration” (quoting one of the state’s experts in the litigation)).

215. Turnout may go down after an electoral reform, but this correlation does not tell us how much of the turnout effect was caused by the reform as opposed to other factors such as reduced enthusiasm for candidates. We may be able to estimate the number of people potentially affected by an electoral reform—for instance, the number of registered voters who lack required voter identification. But this does not tell us the number of voters actually affected—for instance, the number of registered voters turned away on election day for lack of proper identification. Another complicating factor is that voter suppression efforts may affect turnout through information or misinformation. For example, an individual may actually be permitted by law to cast a ballot but decline to try to vote based on the mistaken assumption that she is not.

voters.²¹⁶ My only claim is that if electoral administration reform had the differential turnout effect some fear, that in turn would affect the efficiency gap analysis.

Consider the following. In the *Whitford* litigation, the plaintiffs proposed a numeric threshold of 7%,²¹⁷ and their expert Kenneth Mayer computed an efficiency gap for the actual plan (using the simplified formula) of about 9.85%.²¹⁸ According to the simplified formula ($\Delta W = S^* - 2V^*$), holding seat share constant, every 0.5 percentage point decrease in Democratic vote share decreases by 1 percentage point the Republican advantage conferred by the electoral map. About 2.8 million ballots were cast in the 2012 Wisconsin State Assembly election;²¹⁹ 0.5% of this number is 14,000 ballots. Thus, if 14,000 fewer Democrats cast ballots, the pro-Republican efficiency gap would decrease by about 1 percentage point. If about 40,000 fewer Democrats cast ballots, the pro-Republican efficiency gap would decrease by about 2.86 percentage points—and would fall below the proposed 7% threshold. If 140,000 fewer Democrats cast ballots, the pro-Republican efficiency gap would be eliminated entirely.

216. For a sampling of this debate, see Jack Citrin et al., *The Effects of Voter ID Notification on Voter Turnout Results from a Large-Scale Field Experiment*, 13 ELECTION L.J. 228, 235 (2014) (finding “little support for the hypothesis that notification of ID requirements depresses turnout”); Shelley de Alth, Essay, *ID at the Polls: Assessing the Impact of Recent State Voter ID Laws on Voter Turnout*, 3 HARV. L. & POL’Y REV. 185, 186 (2009) (“[P]hoto and non-photo ID laws decreased turnout by between 1.6 and 2.2 percentage points . . . , [but] states that amended their ID laws more recently experienced increased voter turnout, whereas states that changed their voting laws prior to 2004 showed a decline in turnout.”); Robert S. Erikson & Lorraine C. Minnite, *Modeling Problems in the Voter Identification-Voter Turnout Debate*, 8 ELECTION L.J. 85, 98 (2009) (“[T]he existing science regarding vote suppression [is] incomplete and inconclusive . . . not because of any reason to doubt the suppression effect but rather because the data that have been analyzed to date do not allow a conclusive test.”); Samuel Issacharoff, *Ballot Bedlam*, 64 DUKE L.J. 1363, 1381 (2015) (“To date, empirical studies . . . have been unable to find any substantial decline either in overall turnout or in the turnout of racial minorities as a result of [voter identification] laws.”); *id.* at 1381 n.67 (collecting studies); Pamela S. Karlan, *Turnout, Tenuousness, and Getting Results in Section 2 Vote Denial Claims*, 77 OHIO ST. L.J. 763, 774 (2016) (“[A]s a practical matter, determining whether a challenged practice has depressed minority turnout can be extraordinarily complex.”); *id.* at 774 n.63 (collecting studies); Spencer Overton, *Voter Identification*, 105 MICH. L. REV. 631, 659–61 & nn.136–41, 143–44 & 150–53 (2007) (citing studies suggesting that senior citizens, young people, people of color, people with disabilities, low-income people, and transient people differentially lack driver’s licenses); Michael J. Pitts, *Empirically Assessing the Impact of Photo Identification at the Polls Through an Examination of Provisional Balloting*, 24 J.L. & POL. 475, 480 (2008) (concluding that a study of provisional ballots cast in Indiana’s 2008 primary election “likely provide[s] a little something for both proponents and opponents of photo identification”).

217. See Jackman Report, *supra* note 102, at 66–69.

218. See Mayer Report, *supra* note 146, at 46 tbl.10; see also *supra* notes 146–47 and accompanying text (explaining the calculation of the 9.85% gap).

219. See Mayer Report, *supra* note 146, at 46 tbl.10.

A real-life example puts this into perspective. A federal judge in the challenge to Wisconsin's voter identification law concluded that over 300,000, or roughly 9%, of all registered voters in Wisconsin lacked the necessary identification, that "[a] substantial number of the 300,000 plus eligible voters who lack a photo ID are low-income," and that "it is likely that a substantial number" of those voters without a qualifying identification "will be deterred from voting."²²⁰ These considerations do not establish the intent or effect of Wisconsin's photo ID law. But they do suggest the risk inherent in the efficiency gap's focus on ballots cast. A party eager to construct a partisan gerrymander capable of withstanding any legal challenge based on the efficiency gap would have a strong incentive to engage in voter suppression efforts that reduce the turnout of its opponents' supporters.

The only way to avoid this perverse relationship between the measure of partisan gerrymandering and the turnout effects of partisan voter suppression is to explicitly account for partisan turnout effects in the measure. This can be accomplished by applying the efficiency gap measure to a hypothetical election result estimated in a way that controls for turnout-reducing electoral reforms or practices. In essence, this technique would ask what efficiency gap a plan would produce without voter suppression rather than what efficiency gap a plan did produce with voter suppression. The drawback is that this accommodation of the voter suppression problem makes the measure further reliant on hypotheticals and more sensitive to modeling assumptions. But it would avoid the risk of rewarding and further incentivizing partisan voter suppression. At the very least, this approach warrants consideration under the circumstances present in Wisconsin and North Carolina, where the same lawmakers who enacted the challenged maps also enacted electoral reforms—subsequently called into question by federal courts²²¹—that risk partisan voter suppression and a corresponding underestimation of the extent of advantage those maps conferred on Republicans. More generally, if efficiency gap measures are to play a role in partisan gerrymandering claims, this problem warrants further consideration.

III. The Efficiency Gap's Relationship to Proportionality and Competitiveness

Part II above focused on the efficiency gap's conceptual design and its underlying methodological choices. This Part turns to the efficiency gap's relationship to seats-votes proportionality and competitiveness. The efficiency gap was explicitly designed to capture asymmetrical partisan efficiency, not proportionality or competitiveness. Yet the efficiency gap bears a relationship

220. *Frank v. Walker*, 17 F. Supp. 3d 837, 854, 862 (E.D. Wis.), *rev'd*, 768 F.3d 744 (7th Cir. 2014).

221. See *supra* notes 213-14.

to proportionality and competitiveness that warrants better understanding and consideration.

As vote margin increases, the efficiency gap approves increasing departures from strict seats-votes proportionality. A party with 75% vote share can win every seat and achieve an efficiency gap of zero. In fact, the efficiency gap will report a disadvantage for a party with more than 75% vote share even if that party wins every seat. More realistically, a party with 59% vote share can win a 75% supermajority in the legislature and still produce a below-threshold efficiency gap.²²²

The efficiency gap bears a more nuanced relationship to competitiveness. The efficiency gap is a measure of differential, not overall, competitiveness, but the gap's sensitivity to vote swings is a function of seats-votes responsiveness, which depends on the proportion of relatively competitive districts. This relationship reveals that scholars and jurists focusing on competitiveness may be referring to the efficiency gap without realizing it. And it suggests that adopting the efficiency gap as the exclusive measure of partisan gerrymandering would permit and perhaps encourage mapmakers to draw uncompetitive plans.

The analysis proceeds as follows. Subpart A below explains why proportionality and competitiveness matter. Partisan symmetry may be a necessary but insufficient condition for a well-functioning democracy. We should be concerned by extreme departures from proportionality and competitiveness, especially when they occur simultaneously.

Subpart B then adopts the analytic technique introduced in Part II.C.1 above to offer a more intuitive understanding of the efficiency gap's operation. At the level of an individual district, the wasted vote disparity is a function of the district's competitiveness: It is zero when one party wins 75-25, such that the victory margin is precisely half of turnout, and it jumps discontinuously when the district flips from one party to the other. From this district-level analysis, we can derive key features of the measure's plan-level operation.

Subpart C examines the efficiency gap's relationship with proportionality. The proponents recognize an extreme vote share problem, one in which a party with 75% vote share can win all the seats and still achieve an efficiency gap of zero. I argue that the problem is not confined to this point. For example, a majority with 59% vote share can win a veto-proof 75% supermajority in the legislature with a below-threshold gap.²²³

Subpart D examines the efficiency gap's relationship with competitiveness. I define the competitiveness gap as the seat-share-weighted difference in average

222. See *infra* Part III.C.

223. Such a supermajority would also ward off filibusters in states with that procedure. See Paige Scobee, *Ahoy! The Future of the Filibuster*, NAT'L CONF. ST. LEGISLATURES: NCSL BLOG (June 29, 2016), <https://perma.cc/BU52-VXUK> ("Approximately 10 states have a cloture rule that requires more than a simple majority." (citing Meghan Reilly, *States Limiting Legislative Debate*, CONN. GEN. ASSEMBLY (July 8, 2009), <https://perma.cc/K3DQ-YDPA>)).

competitiveness between x -won and y -won districts and show that under the equal voter turnout assumption, the efficiency gap is equal to the competitiveness gap. This means that a mapmaker can reduce the efficiency gap by unpacking, and thereby increasing overall competitiveness, or by decracking, and thereby decreasing overall competitiveness. While a zero gap can be achieved at any level of competitiveness, a zero gap can only be *maintained* at a relatively low level of competitiveness because the gap's sensitivity to vote swings is a function of seats-votes responsiveness.

Subpart E offers a normative assessment of the efficiency gap's relationship with proportionality and competitiveness. I suggest that this relationship presents two problems: a false positive problem, in which the measure flags as suspect normatively desirable plans that reflect efforts to promote proportionality or competitiveness; and a false negative problem, in which it fails to detect normatively problematic plans such as skewed bipartisan gerrymanders with excessive seat bonuses and insufficient representation for minority parties. I suggest that the doctrinal tools of intent, justification, and sensitivity analysis only partially address the false positive problem and fail to address the false negative problem for want of a mechanism to overcome the presumption of validity triggered by a below-threshold gap.

Finally, Subpart F demonstrates how the efficiency gap's relationship with proportionality and competitiveness depends on the definition and weight of surplus votes.

A. The Relevance of Proportionality and Competitiveness

The efficiency gap's proponents, like the *LULAC* amici,²²⁴ have good reason to focus on partisan symmetry rather than other democratic norms like seats-votes proportionality and competitiveness. For one thing, proportionality and competitiveness are in tension with each other. A maximally competitive system, one in which each district race is a razor-thin nail-biter, may depart radically from proportionality because a small uniform swing across all districts in favor of one party would result in that party's winning every seat with just over half the votes. Conversely, a strictly proportional system exhibits relatively low competitiveness because any incremental change in vote share must translate into an equal change in seat share, and so the number of competitive districts is necessarily limited.

Moreover, the Supreme Court has demonstrated deep skepticism about the constitutional significance of proportionality and competitiveness. The Court has repeatedly insisted that the Constitution does not require strict seats-votes

224. See *supra* Part I.B.

proportionality.²²⁵ And in *Gaffney v. Cummings*, a majority of the Court was untroubled by the uncompetitiveness of a bipartisan gerrymander that carved the state up into safe Democratic districts and safe Republican districts so as to achieve seats-votes proportionality.²²⁶ Thus, the Court might reject as foreclosed by precedent any legal test that essentially requires strict proportionality or maximal competitiveness. In contrast, five Justices have expressed interest in partisan symmetry, a normatively appealing standard that closely tracks non-discrimination principles familiar to both equal protection and First Amendment law.²²⁷

But even if strict proportionality is not required, a significant departure from proportionality may be relevant to assessing a districting plan. So too with competitiveness. A plan need not maximize competitiveness, but we may be rightly concerned if a plan needlessly and intentionally minimizes it. And we ought to be particularly concerned by a plan that simultaneously departs significantly from both proportionality and competitiveness. Note that *Gaffney* approved a bipartisan gerrymander that reduced competitiveness to achieve seats-votes proportionality, not a plan that departed from both competitiveness and proportionality.²²⁸

If proportionality and competitiveness matter, the principle of partisan symmetry proposed by the *LULAC* amici and invoked by the efficiency gap proponents may constitute a necessary but insufficient condition of a well-functioning electoral system. Take some extreme examples. In a winner-take-all system, whichever party earns more votes gets all the seats. This system satisfies the principle of partisan symmetry and maximizes competitiveness, but it permits an extreme departure from seats-votes proportionality and thus denies any representation to the minority party. In contrast, consider a system in

225. See, e.g., *Vieth v. Jubelirer*, 541 U.S. 267, 288 (2004) (plurality opinion) (“[Appellants’] standard rests upon the principle that groups . . . have a right to proportional representation. But the Constitution contains no such principle. It guarantees equal protection of the law to persons, not equal representation in government to equivalently sized groups.”).

226. See 412 U.S. 735, 752-54 (1973) (“[J]udicial interest should be at its lowest ebb when a State purports fairly to allocate political power to the parties in accordance with their voting strength . . .”).

227. See *League of United Latin Am. Citizens v. Perry* (*LULAC*), 548 U.S. 399, 483-84 (2006) (Souter, J., concurring in part and dissenting in part) (“[N]or do I rule out the utility of a criterion of symmetry as a test. Interest in exploring this notion is evident. Perhaps further attention could be devoted to the administrability of such a criterion at all levels of redistricting and its review.” (citations omitted) (citing *id.* at 419-20 (opinion of Kennedy, J.); *id.* at 465-68 (Stevens, J., concurring in part and dissenting in part); and *id.* at 491-92 (Breyer, J., concurring in part and dissenting in part))); see also *supra* note 114 (discussing the equal protection and First Amendment conceptions of the nondiscrimination principle embodied in the notion that partisan gerrymandering is unconstitutional).

228. See 412 U.S. at 752, 754.

which each party gets half the seats no matter how many votes it earns. This system satisfies the principle of partisan symmetry and ensures minority representation, but it eliminates competitiveness. Whereas competitive races promote accountability,²²⁹ safe districts shift the action from the general election to the primary, which pushes legislators to ideological extremes, promotes polarization and gridlock,²³⁰ and reduces the responsiveness of legislators to the general electorate.²³¹ The proliferation of safe districts²³² may also discourage high-quality challengers, reduce party mobilization, and depress voter participation,²³³ giving incumbents an advantage unrelated to their prior performance

229. See Stephanopoulos, *supra* note 38, at 676-77.

230. See, e.g., Lillian V. Smith, Note, *Recreating the "Ritual Carving": Why Congress Should Fund Independent Redistricting Commissions and End Partisan Gerrymandering*, 80 BROOK. L. REV. 1641, 1644 (2015) ("[P]artisan gerrymandering plays a significant role [in creating congressional stagnation] by reducing the competitiveness of elections, contributing to the systemic entrenchment of partisan interests, and perpetuating congressional deadlock . . ."). Recent Congresses have been notoriously unproductive. See Drew DeSilver, *In Late Spurt of Activity, Congress Avoids "Least Productive" Title*, PEW RES. CTR.: FACT TANK (Dec. 29, 2014), <https://perma.cc/AA34-QSLR>. But see Drew DeSilver, *Congressional Productivity Is Up—But Many New Laws Overturn Obama-Era Rules*, PEW RES. CTR.: FACT TANK (Aug. 29, 2017), <https://perma.cc/2B7Q-7N6E> (reporting that the 115th Congress "is among the most productive in recent years"). When legislators fear primary challengers but take general elections for granted, the predictable result is increased partisanship and gridlock. See MARK TUSHNET, *THE NEW CONSTITUTIONAL ORDER* 13-16, 19 (2003).

231. See Josh Chafetz, Essay, *The Phenomenology of Gridlock*, 88 NOTRE DAME L. REV. 2065, 2086 (2013) (considering the possibility that "the combination of partisan primaries and bipartisan gerrymandering are resulting in a legislature that cannot be said to be broadly responsive to the American people"); Samuel Issacharoff & Pamela S. Karlan, *Where to Draw the Line?: Judicial Review of Political Gerrymanders*, 153 U. PA. L. REV. 541, 574 (2004) ("The perverse consequence of the incumbent gerrymander is that it skews the distribution politically by driving the center out of elected office at the legislative level.").

232. By one count, the number of swing congressional districts plunged from 103 in 1992 to 35 in 2012. Nate Silver, *As Swing Districts Dwindle, Can a Divided House Stand?*, N.Y. TIMES: FIFTYTHREE (Dec. 27, 2012, 9:46 AM), <https://perma.cc/5WJ9-BEET>; see also John Nichols, *Why Redistricting Threatens Democracy*, NATION (Feb. 1, 2012), <https://perma.cc/9NQN-EQQ5> (arguing that redistricting "explains why the vast majority of races are not competitive").

233. See, e.g., *League of United Latin Am. Citizens v. Perry (LULAC)*, 548 U.S. 399, 471 n.10 (2006) (Stevens, J., concurring in part and dissenting in part) (explaining multiple ways in which "[s]afe seats may harm the democratic process," including by decreasing voter turnout); Richard H. Pildes, *The Constitution and Political Competition*, 30 NOVA L. REV. 253, 260 (2006) ("[I]t is well documented that competitive elections encourage the appearance of strong challengers to incumbents and increase voter turnout and party mobilization."). Note that some scholars and jurists emphasize voter participation as a primary democratic value in itself. See Stephanopoulos, *supra* note 37, at 297 (describing Justice Breyer's, Christopher Elmendorf's, and Spencer Overton's views on the "primacy of voter participation").

or present popularity. For these reasons, it may be problematic if the Court defines unconstitutional partisan gerrymandering in a way that entirely neglects norms of competitiveness and seats-votes proportionality.

One approach would be to account for different norms with different causes of action: a partisan gerrymandering claim for excessive departures from partisan symmetry, a bipartisan gerrymandering claim for excessive departures from competitiveness, and a minority protection claim for excessive departures from seats-votes proportionality. This approach warrants consideration going forward. But at this moment, when no other political gerrymandering claim exists, it is prudent to consider the implications for other democratic norms of a partisan gerrymandering test designed to vindicate only the principle of partisan symmetry. If mapmakers can avoid excessive partisan asymmetry by sacrificing competitiveness, minority representation, or both, and if the Court prohibits excessive partisan asymmetry without offering any legal protection or incentive for competitiveness or minority representation, the result may be skewed incentives to produce problematic electoral plans. This is not a criticism of the efficiency gap measure itself, which was sensibly designed with the exclusive goal of quantifying partisan asymmetry. Rather, it is a call for careful study and use of the measure with the understanding that partisan symmetry is a necessary but insufficient condition of a well-functioning democracy.

B. The Simple Zero-Gap Plan

To develop a deeper understanding of the efficiency gap's relationship to proportionality and competitiveness, this Subpart examines the characteristics of districts and plans in which the same number of votes are wasted.

1. Measuring competitiveness

I begin the analysis by defining and deriving some properties of a district election's victory margin. Recall that V_{1i} and V_{2i} respectively denote the number of ballots cast for the first- and second-place candidates in a given district. The sum is the district's total voter turnout: $T_i = V_{1i} + V_{2i} = V_{yi} + V_{xi}$. The difference is the victory margin (M_i).

$$M_i = V_{1i} - V_{2i} = |V_{xi} - V_{yi}| = \begin{cases} V_{xi} - V_{yi} & i \in D_x \\ V_{yi} - V_{xi} & i \in D_y \end{cases}$$

The proportional victory margin (m_i) is the district-level victory margin (M_i) expressed as a proportion of district-level voter turnout (T_i): $m_i = \frac{M_i}{T_i}$. Rearranging, we can express V_{1i} and V_{2i} in terms of victory margin (M_i) and turnout (T_i).²³⁴

$$V_{1i} = \frac{T_i + M_i}{2} = \frac{T_i}{2} (1 + m_i)$$

$$V_{2i} = \frac{T_i - M_i}{2} = \frac{T_i}{2} (1 - m_i)$$

Let $m_p = \frac{\sum_{i \in D_p} m_i}{s_p}$ denote the average proportional victory margin in p -won districts and let $m = \frac{\sum_{i \in D} m_i}{s}$ denote the average proportional victory margin over all districts. Note that the seat-share-weighted sum of m_x and m_y is m and the seat-share-weighted difference of m_x and m_y is twice the vote margin (V^*).²³⁵

$$m = \bar{s}_x m_x + \bar{s}_y m_y$$

$$2V^* = \bar{s}_x m_x - \bar{s}_y m_y$$

This means we can express m_x and m_y in terms of m , V^* , and S^* .²³⁶

$$m_x = \frac{m + 2V^*}{1 + 2S^*}$$

$$m_y = \frac{m - 2V^*}{1 - 2S^*}$$

234. These expressions can be derived as follows:

$$V_{1i} = \frac{(2V_{1i}+0)}{2} = \frac{(V_{1i}+V_{1i})+(V_{2i}-V_{2i})}{2} = \frac{(V_{1i}+V_{2i})+(V_{1i}-V_{2i})}{2} = \frac{T_i+M_i}{2} = \frac{T_i}{2} (1 + m_i);$$

$$V_{2i} = \frac{(2V_{2i}+0)}{2} = \frac{(V_{2i}+V_{2i})+(V_{1i}-V_{1i})}{2} = \frac{(V_{1i}+V_{2i})-(V_{1i}-V_{2i})}{2} = \frac{T_i-M_i}{2} = \frac{T_i}{2} (1 - m_i).$$

235. Using the equal voter turnout assumption, the second of these expressions can be derived as follows:

$$2V^* = \bar{V}_x - \bar{V}_y = \frac{V_x - V_y}{V} = \frac{\sum_{i \in D} V_{xi} - \sum_{i \in D} V_{yi}}{ST^*} = \frac{\sum_{i \in D} (V_{xi} - V_{yi})}{ST^*};$$

$$= \frac{\sum_{i \in D_x} (V_{xi} - V_{yi}) - \sum_{i \in D_y} (V_{yi} - V_{xi})}{ST^*} = \frac{\sum_{i \in D_x} T_i m_i - \sum_{i \in D_y} T_i m_i}{ST^*};$$

$$= \frac{\sum_{i \in D_x} T^* m_i - \sum_{i \in D_y} T^* m_i}{ST^*} = \frac{\sum_{i \in D_x} m_i - \sum_{i \in D_y} m_i}{s} = \frac{s_x m_x - s_y m_y}{s};$$

$$= \bar{s}_x m_x - \bar{s}_y m_y.$$

236. These results can be derived as follows:

$$m_x = \frac{2\bar{s}_x m_x + 0}{2\bar{s}_x} = \frac{(\bar{s}_x m_x + \bar{s}_y m_y) + (\bar{s}_x m_x - \bar{s}_y m_y)}{2(\frac{1}{2} + S^*)} = \frac{m + 2V^*}{1 + 2S^*};$$

$$m_y = \frac{2\bar{s}_y m_y + 0}{2\bar{s}_y} = \frac{(\bar{s}_x m_x + \bar{s}_y m_y) - (\bar{s}_x m_x - \bar{s}_y m_y)}{2(\frac{1}{2} - S^*)} = \frac{m - 2V^*}{1 - 2S^*}.$$

These equations will prove critical in Part IV below.

2. A district's wasted vote disparity is a discontinuous linear function of its margin of victory

At the level of an individual district, the wasted vote disparity (Δw_i) is a discontinuous linear function of the victory margin m_i .

$$\Delta w_i = \frac{W_{yi} - W_{xi}}{T_i} = \begin{cases} \frac{1}{2} - m_i & i \in D_x \\ m_i - \frac{1}{2} & i \in D_y \end{cases}$$

With ties excluded, the disparity (Δw_i) is zero if and only if the victor prevails by half of turnout²³⁷—regardless which party wins. With this victory margin, three-quarters of ballots are cast for the winner and one-quarter for the loser. For example, suppose 100 ballots are cast, 75 for party x and 25 for party y . Party x wins 75% vote share and prevails by a margin of victory of 50 votes, precisely half of turnout. Under the proponents' formulation, party y has wasted 25 lost votes while party x has wasted 25 surplus votes ($75 - 50$), so each party wastes the same number of votes and the district's wasted vote disparity is zero.²³⁸

Call one-half the minimizing victory margin m^0 and define the competitiveness score (c_i) as the difference between the minimizing (m^0) and the actual (m_i) victory margin: $c_i = m^0 - m_i$. The wasted vote disparity is then:

$$\Delta w_i = \begin{cases} c_i & i \in D_x \\ -c_i & i \in D_y \end{cases}, \text{ and}$$

$$\Delta w_i = c_i = 0 \text{ if and only if } m_i = m^0 = 0.5.$$

A district's wasted vote disparity, then, is simply its competitiveness score—the difference between the minimizing and actual victory margins—with a sign convention such that a relatively competitive district favors the winning party while a relatively uncompetitive district favors the losing party. When m_i exceeds m^0 , c_i is negative because the district is less competitive than the minimizing level. This favors the losing party because surplus votes exceed lost votes. When m_i is less than m^0 , c_i is positive because the district is more competitive than the minimizing level. This favors the winning party because lost

237. This makes intuitive sense. The proponents note that in single-member district elections featuring two candidates, precisely half of all votes are wasted. See Stephanopoulos & McGhee, *supra* note 7, at 851 & n.107. Thus, equal wasted votes occur when each party wastes one-quarter of ballots cast. See Mira Bernstein & Moon Duchin, Opinion, *A Formula Goes to Court: Partisan Gerrymandering and the Efficiency Gap*, 64 NOTICES AM. MATHEMATICAL SOC'Y 1020, 1022 (2017).

238. For this reason, Bernstein and Duchin accuse the measure of "[f]etishiz[ing] three-to-one landslide districts." See Bernstein & Duchin, *supra* note 237, at 1022.

votes exceed surplus votes. The district-level disparity is positive (favors party x) when x wins a relatively competitive district or y wins a relatively uncompetitive district; the disparity is negative (favors party y) when x wins a relatively uncompetitive district or y wins a relatively competitive district.

The simple zero-gap plan has a competitiveness gap of zero because each district has a competitiveness score of zero. Given that each district is won by half of turnout ($m_i = m^0 = \frac{1}{2}$), party x earns $\frac{T^*}{2}$ more votes in x -won districts, and party x earns $\frac{T^*}{2}$ fewer votes in y -won districts. Because the plan must assign each voter to one district, the seat margin must be double the vote margin.

$$2V^* = \frac{V_x - V_y}{V} = \frac{T^*(\sum_{i \in D_x} m^0 - \sum_{i \in D_y} m^0)}{ST^*} = \frac{S_x - S_y}{2S} = S^*$$

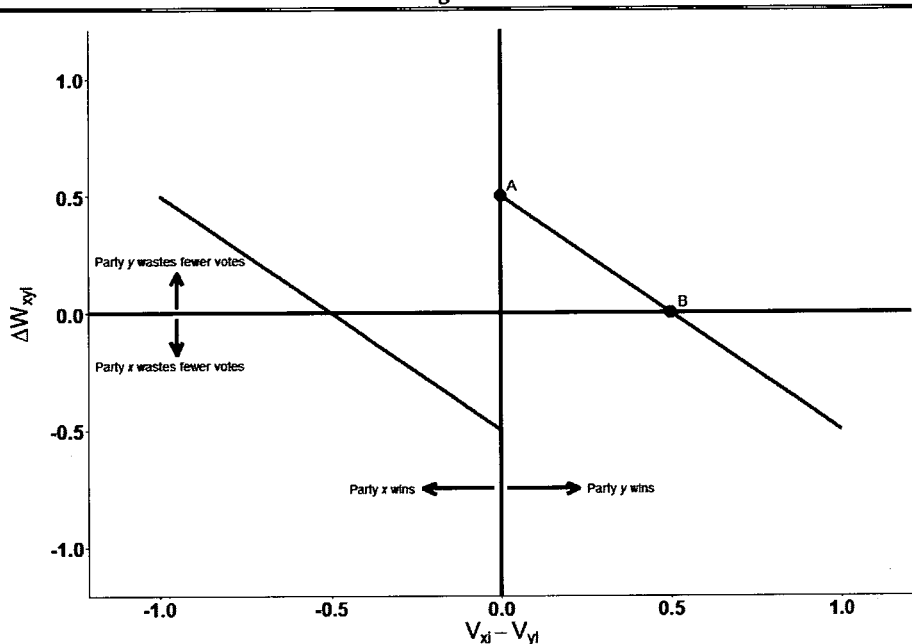
Recall that the simple zero-gap plan can be transformed to or from any zero-gap plan with the appropriate series of voter swaps. By design, a voter swap preserves equal voter turnout²³⁹ and statewide vote share. Because no swap alters a district's election outcome, the series preserves seat share. And because the series preserves vote share, seat share, and equal turnout, it must necessarily preserve the efficiency gap. Thus, the equality $S^* = 2V^*$ is a feature of any zero-gap plan, not just the simple zero-gap plan. This offers another way to understand the double proportionality and double responsiveness that emerges from the proponents' approach²⁴⁰: It is the seats-votes relationship exhibited by a simple minimizing plan composed exclusively of minimizing districts. When each party earns half the votes, the simple minimizing plan accords each party half the seats. When one party earns 75% of the votes, the simple minimizing plan accords that party all the seats because that party wins each district 75 to 25.

Figure 8 below illustrates the relationship between a district's disparity and the vote share difference between parties x and y . It looks like a double backslash—two downward-sloping lines with a discontinuous jump at the 50% mark when the seat flips from one party to the other. Much of the measure's operation at the plan level can be intuited from this district-level relationship. Given that a party can win a single seat with 75% of the votes in a zero-disparity district, a party can win all the seats with 75% of votes and an efficiency gap of zero. Because a district's disparity is its competitiveness score, the efficiency gap is the seat-share-weighted difference in average competitiveness. And because a district's disparity jumps discontinuously when a seat flips, the sensitivity of a plan's efficiency gap to vote swings is a function of how many of its districts are competitive.

239. Each participating district exchanges one voter for another and thus maintains the same voter turnout.

240. See McGhee, *supra* note 10, at 68 (presenting equation 5); *id.* app. B at 79-82 (deriving equation 5); Stephanopoulos & McGhee, *supra* note 7, at 853 & n.114.

Figure 8



At point A, there is maximal competitiveness and maximal wasted vote disparity. At point B, there is zero wasted vote disparity and a low level of competitiveness, with a minimizing proportional margin of victory of 50%. There is a discontinuous jump when the winning party flips.

C. The Efficiency Gap and Proportionality

The efficiency gap measures undeserved seat share relative to a baseline of double proportionality, not strict proportionality.²⁴¹ That the efficiency gap is distinct from a requirement of strict proportionality may be a doctrinal virtue because it avoids the argument that its use is foreclosed by precedent holding that the Constitution does not require strict proportionality.²⁴² But the efficiency gap's departure from strict proportionality is normatively problematic, particularly as one party's vote share increases. A party with 75% vote share can win every seat and still achieve an efficiency gap of zero. In fact, the efficiency

241. See *supra* notes 93-96 and accompanying text.

242. See, e.g., *Vieth v. Jubelirer*, 541 U.S. 267, 288 (2004) (plurality opinion); see also *supra* note 225 and accompanying text.

gap will report a *disadvantage* for a party with more than 75% vote share—even if that party wins every seat.²⁴³

The academic proponents recognize that the efficiency gap measure “fails to capture the idea of fairness at stake in redistricting” in such a case, but they conclude that “this is not a problem that is especially relevant to real-world redistricting” because “results this lopsided are extremely rare.”²⁴⁴ In other words, the proponents concede the normative correspondence problem but address that problem by limiting the scope of the measure’s operation: “All an analyst must do is flag elections in which a party received at least 75 percent of the statewide vote and 100 percent of the seats.”²⁴⁵ The proponents, writing in 2015, noted that “[n]o party has received more than 75 percent of the aggregate vote in state legislative elections since 1982, and there are only 18 such cases out of 800 in congressional elections (all of them either in the South or in states with fewer than four House districts).”²⁴⁶

The proponents provide strong evidence that it is historically rare for a majority party to enjoy a vote share above 75%. But this fact does not eliminate the concerns highlighted by this scenario. Eighteen cases out of 800 (about 2%) is a tiny proportion, but it is a significant absolute number of cases. Five of the eighteen cases involved a state exhibiting high vote share for a single congressional election—Wyoming in 1984, Mississippi in 1990, South Dakota and West Virginia in 1998, and Louisiana in 2000.²⁴⁷ The other thirteen cases involved a state exhibiting high vote share in multiple congressional elections—twice for North Dakota (1984 and 1986); thrice for Alaska (2000, 2002, and 2004) and Hawaii (1984, 1992, and 2008); and five times for Vermont (1982, 1984, 1990, 1992, and 1996).²⁴⁸ For this latter group of states, the efficiency gap would repeatedly “fail[] to capture the idea of fairness at stake” in the election results.²⁴⁹

Moreover, the proponents presented only those cases where majority vote share exceeded 75%, the vote share needed to win every seat with an efficiency gap of zero. The problem may reach its apex at this point, but it is not confined

243. For example, if party *x* wins every seat ($S^* = 0.5$) with 80% vote share ($V^* = 0.3$), the simplified formula computes a gap of negative 10%.

$$\Delta W = S^* - 2V^* = (0.5) - 2(0.3) = -0.1$$

The negative sign indicates that the gap disadvantages party *x* and favors party *y*, even though party *x* won every seat with only 80% of the vote.

244. See Stephanopoulos & McGhee, *supra* note 7, at 863–64.

245. *Id.* at 863.

246. *Id.*

247. See *id.* at 863 & n.148.

248. See *id.*

249. See *id.* at 863.

to this point.²⁵⁰ The proponents' standard does not require a gap of zero; it requires only a gap below the numeric threshold above which election plans are presumptively invalid. Let ΔW^* denote this threshold. The efficiency gap will fall below this threshold whenever the following inequality obtains:²⁵¹

$$\bar{V} > \frac{(\bar{S} - \Delta W^* + \frac{1}{2})}{2}.$$

With the proponents' suggested threshold of $\Delta W^* = 0.08$,²⁵² this reduces to:

$$\bar{V} > \frac{\bar{S}}{2} + 0.21.$$

Thus, a party with vote share exceeding 71% can win every seat with a below-threshold gap.²⁵³ If the presumption of validity were irrebuttable, a majority with this vote share could capture every seat with a plan impervious to judicial scrutiny.

Thus far, we have assumed that the majority insists on every last seat. But if the majority is willing to throw the minority a bone, it can enjoy less-than-total domination and a below-threshold gap at lower vote share. For example, with a vote share above 59%, the majority can achieve 75% seat share with a below-threshold gap.²⁵⁴ Presumably, 71% vote share is more common than 75% vote share, and 59% vote share is more common still.

Finally, historical patterns do not necessarily predict future trends. In the coming decades, more states may consistently exhibit high vote share. And if the Court were to adopt a legal standard for partisan gerrymandering that gave free rein to majorities with sufficiently high vote share, majorities would have an ever-stronger incentive to achieve it. This could encourage desirable behavior (like voter persuasion) or undesirable behavior (like voter suppression). Even if a majority party earned a sufficiently high vote share through legitimate means,

250. Compare *id.* (arguing that the problem "is easily identified" whenever a party receives 75% of the vote share and 100% of the seats), with *id.* (recognizing, presumably for elections even where the vote share is under 75%, "the unexpected results that begin to emerge when one party receives an extraordinarily high vote share").

251. Because $S^* = \bar{S} - \frac{1}{2}$, $V^* = \bar{V} - \frac{1}{2}$, and $\Delta W = S^* - 2V^*$ (under the simplified formula), this inequality ensures a below-threshold gap:

If $\bar{V} > \frac{\bar{S} - \Delta W^* + \frac{1}{2}}{2}$, then $\bar{V} > \frac{(\bar{S} - \frac{1}{2}) - \Delta W^*}{2} + \frac{1}{2}$, and so $2(\bar{V} - \frac{1}{2}) > (\bar{S} - \frac{1}{2}) - \Delta W^*$.

Thus: $S^* - 2V^* < \Delta W^*$, and so $\Delta W < \Delta W^*$.

252. See Stephanopoulos & McGhee, *supra* note 7, at 884, 888-89.

253. $\bar{S} = 1$ if the party wins every seat, so $\bar{V} > 0.5 + 0.21 = 0.71$.

254. This is because $0.59 > \frac{0.75}{2} + 0.21 = 0.585$.

courts might be unable to regulate that majority's subsequent redistricting decisions.

For these reasons, the proponents may be too quick to conclude that the operation of the measure in cases of relatively high vote share "is not a problem that is especially relevant to real-world redistricting."²⁵⁵

D. The Efficiency Gap and Competitiveness

The efficiency gap bears a more nuanced relationship to competitiveness. The value of the efficiency gap at a specified level of vote share is a function of differential average competitiveness between *x*-won and *y*-won districts—a function I call the *competitiveness gap*. The sensitivity of the efficiency gap to changes in vote share is a function of responsiveness, which is closely related to a plan's overall competitiveness. Subpart D.1 below explains the competitiveness gap, while Subpart D.2 explains the relationship between responsiveness and the efficiency gap's sensitivity to changes in vote share.

1. The competitiveness gap

This Subpart explains the relationship between the efficiency gap and the competitiveness gap. I first define the competitiveness gap and show that it equals the efficiency gap under the equal voter turnout assumption. I then explore two implications of this equality. First, scholars and jurists may invoke the competitiveness gap (or something close to it) without recognizing its relationship to the efficiency gap. Second, at a given level of vote share, mapmakers can achieve a gap of zero with either high or low overall competitiveness. Specifically, mapmakers can eliminate or reduce an efficiency gap in one of two ways: *unpacking*, which increases a plan's overall competitiveness; or *decracking*, which decreases a plan's overall competitiveness.

a. Definition and equality

Let $c_p = \frac{\sum_{i \in D_p} c_i}{s_p}$ denote the average competitiveness score in *p*-won districts, and let $c = \frac{\sum_{i \in D} c_i}{s}$ denote the average competitiveness score over all districts. Note that average competitiveness is the difference between the minimizing and the average victory margin ($c = m^0 - m$ and $c_p = m^0 - m_p$),²⁵⁶ and c is the seat-

255. Stephanopoulos & McGhee, *supra* note 7, at 864.

256. Proof: $c = \frac{\sum_{i \in D} c_i}{s} = \frac{\sum_{i \in D} (m^0 - m_i)}{s} = \frac{\sum_{i \in D} m^0 - \sum_{i \in D} m_i}{s} = \frac{sm^0 - sm}{s} = m^0 - m$;
 $c_p = \frac{\sum_{i \in D_p} c_i}{s_p} = \frac{\sum_{i \in D_p} (m^0 - m_i)}{s_p} = \frac{\sum_{i \in D_p} m^0 - \sum_{i \in D_p} m_i}{s_p} = \frac{s_p m^0 - s_p m_p}{s_p} = m^0 - m_p$.

share-weighted sum of c_x and c_y ($c = \bar{s}_x c_x + \bar{s}_y c_y$). Define the competitiveness gap (ΔC) as the seat-share-weighted difference of c_x and c_y .

$$\Delta C = \bar{s}_x c_x - \bar{s}_y c_y$$

While c measures a plan's overall average competitiveness, the competitiveness gap ΔC measures a plan's differential average competitiveness, comparing the average competitiveness of x - and y -won districts.

Under equal voter turnout, then, the efficiency gap is the competitiveness gap.

$$\Delta W = \frac{\sum_{i \in D} \Delta w_i}{S} = \frac{\sum_{i \in D_x} c_i + \sum_{i \in D_y} -c_i}{S} = \frac{S_x c_x - S_y c_y}{S} = \bar{s}_x c_x - \bar{s}_y c_y = \Delta C$$

And the competitiveness gap reduces to the simplified seats-votes formula.

$$\begin{aligned} \Delta C &= \bar{s}_x c_x - \bar{s}_y c_y = \bar{s}_x (m^0 - m_x) - \bar{s}_y (m^0 - m_y) \\ &= \frac{1}{2} (\bar{s}_x - \bar{s}_y) - (\bar{s}_x m_x - \bar{s}_y m_y) = S^* - 2V^* \end{aligned}$$

Thus, under the equal voter turnout assumption, one can frame the measure as relative wasted votes, undeserved seat share, or differential average competitiveness.

$$\Delta W = S^* - 2V^* = \Delta C$$

An efficiency gap of zero does not necessarily entail equal average competitiveness in x - and y -won districts. This equality obtains only in the special case in which each party wins half the seats. In the more general case, the relative competitiveness of x - and y -won districts needed to achieve an efficiency gap of zero will depend on the relative seat shares. Formally, let $\tilde{c} = \frac{c_x}{c_y}$ denote the competitiveness ratio and $\tilde{s} = \frac{s_x}{s_y}$ denote the seat ratio. The sign of the efficiency gap (ΔW) then depends on whether the product of the seat ratio ($\tilde{s}$) and the competitiveness ratio ($\tilde{c}$) is greater than, equal to, or less than one.²⁵⁷

$$\begin{cases} \Delta W > 0 & \tilde{s}\tilde{c} > 1 \\ \Delta W = 0 & \tilde{s}\tilde{c} = 1 \\ \Delta W < 0 & \tilde{s}\tilde{c} < 1 \end{cases}$$

When the product of the seat ratio and the competitiveness ratio is greater than one, the competitiveness gap and thus the efficiency gap are positive, indicating an advantage for party x . When the product is less than one, the

257. The equality can be derived as follows: $\Delta W = 0$ if and only if $\bar{s}_x c_x - \bar{s}_y c_y = 0$ if and only if $\bar{s}_x c_x = \bar{s}_y c_y$ if and only if $\frac{\bar{s}_x c_x}{\bar{s}_y c_y} = 1$ if and only if $\tilde{s}\tilde{c} = 1$. The inequalities can be derived analogously.

competitiveness gap and thus the efficiency gap are negative, indicating an advantage for party y . And when the product is equal to one, the competitiveness gap and thus the efficiency gap are zero, indicating partisan fairness in the sense of equal wasted votes. Thus, when each party wins half the seats ($\bar{s} = 1$), an efficiency gap of zero requires that x - and y -won districts exhibit equal average competitiveness ($\bar{c} = 1$). When party x wins a majority of seats ($\bar{s} > 1$), an efficiency gap of zero requires that y -won districts are more competitive than x -won districts on average ($\bar{c} < 1$). Specifically, if party x wins two-thirds of the seats ($\bar{s} = 2$), an efficiency gap of zero requires that y -won districts be twice as competitive as x -won districts on average ($\bar{c} = 0.5$).

b. Unacknowledged measure convergence

The equivalence of the efficiency gap and the competitiveness gap suggests that scholars and jurists may in fact be referring to the efficiency gap measure (or something quite like it) without realizing it. For example, Samuel Wang recently proposed three tests for partisan gerrymandering, including the following *lopsided outcomes test*:

Compare the difference between the share of Democratic votes in the districts that Democrats win, and the share of Republican votes in the districts that Republicans win. This test works because in a partisan gerrymander, the targeted party wins lopsided victories in a small number of districts, while the gerrymandering party's wins are engineered to be relatively narrow. To compare the winning vote shares for the two parties, I use a grouped t -test, an extremely common statistical test.²⁵⁸

Just like the efficiency gap, the lopsided outcomes test measures differential average competitiveness. There are three differences between this test and the efficiency gap, but only one is substantive. First, the efficiency gap focuses on the competitiveness score (c_i), whereas the lopsided outcomes test focuses on the victor's vote share (v_i^*)—"the share of Democratic votes in the districts that Democrats win, and the share of Republican votes in the districts that Republicans win."²⁵⁹ These two variables are linearly related: $v_i^* = \frac{3-2c_i}{4}$.²⁶⁰ Second, the lopsided outcomes test uses a statistical test—"a grouped t -test"²⁶¹—to estimate the likelihood that the average competitiveness in x -won districts is different from the average competitiveness in y -won districts, whereas the efficiency gap simply computes the difference.

258. Samuel S.-H. Wang, *Three Tests for Practical Evaluation of Partisan Gerrymandering*, 68 STAN. L. REV. 1263, 1306 (2016) (footnote omitted).

259. *Id.*

260. This relationship can be derived as follows:

$$v_i^* = \frac{V_{1i}}{T_i} = \frac{T_i + M_i}{2T_i} = \frac{1}{2} + \frac{m_i}{2} = \frac{1}{2} + \frac{\frac{1}{2} - c_i}{2} = \frac{1}{2} + \frac{1}{4} - \frac{c_i}{2} = \frac{3-2c_i}{4}.$$

261. Wang, *supra* note 258, at 1306.

The third difference is the substantive one: The lopsided outcomes test does not account for seat share. The test measures unweighted differential average competitiveness, assuming that the competitiveness ratio should be 1, whereas the efficiency gap measures seat-share-weighted differential average competitiveness, assuming that the competitiveness ratio should be equal to the seat ratio. As the seat ratio approaches 1, this last difference drops out; but when the seat ratio departs significantly from 1, the two tests will diverge. When one party wins most seats, the efficiency gap—but not the lopsided outcomes test—permits a competitiveness bonus: A party with a positive seat margin can win relatively more competitive districts while maintaining a competitiveness gap of zero, so long as the competitiveness ratio equals the seat ratio.

The *Whitford* litigation provides another example of this unacknowledged measure convergence. The majority of the three-judge district court in that case separated its discriminatory effect analysis into two steps. First, it discussed evidence of discriminatory effect other than the efficiency gap.²⁶² Only after concluding that the other evidence “ma[de] a firm case on the question of discriminatory effect” did the majority proceed to discuss how “that evidence [was] further bolstered by the plaintiffs’ use of the ‘efficiency gap.’”²⁶³ But as part of its initial discussion, the majority focused on competitiveness, noting that Democrat-won districts were far less competitive on average than Republican-won districts—that is, Democrats were packed into “safe” districts.²⁶⁴ The majority did not seem to realize that in talking about differential competitiveness, it was actually talking about the efficiency gap.

c. Unpacking versus decracking

The equivalence of the efficiency gap and the competitiveness gap also clarifies that plans with low efficiency gaps can, but need not, exhibit competitiveness. The efficiency gap is a function of differential competitiveness, not overall competitiveness. To achieve an efficiency gap of zero, the competitiveness ratio must equal the seat ratio, but overall competitiveness can take on any value. Mapmakers can design minimizing plans with districts as competitive or uncompetitive as they please while still achieving an efficiency gap of zero. They can even make some districts more competitive than others, provided the relative competitiveness of the average *x*-won district and the average *y*-won district is proportional to the relative number of seats won by parties *x* and *y*.

What mapmakers cannot do (assuming they want a zero or low efficiency gap) is systematically vary competitiveness by party so that the competitiveness

262. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 898-903 (W.D. Wis. 2016), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

263. See *id.* at 903.

264. See *id.* at 898-99.

ratio departs from the seat ratio. For example, the mapmaker cannot achieve a zero or low efficiency gap by drawing a plan that awards each party half the seats but has party *x* generally winning competitive races and party *y* generally winning by landslides. That competitiveness gap would entail an efficiency gap in favor of party *x*, systematically packing and cracking party *y* supporters and producing an efficiency gap (party *y* wastes more votes) as well as, equivalently, a competitiveness gap (*x*-won districts would be excessively competitive) and an undeserved seat share relative to the ideal 2:1 seats-votes ratio.

To eliminate (or limit) this undeserved seat share, mapmakers must flip one or more seats from the favored party to the disfavored party. But just as there are two fundamental gerrymandering strategies—packing and cracking—so too are there two analogous strategies for flipping the requisite seats to eliminate (or reduce) a large efficiency or competitiveness gap: *unpacking* and *decracking*. Unpacking flips the requisite seats by transferring supporters of the disfavored party from relatively uncompetitive districts won by the disfavored party. This unpacking strategy makes the districts won by the disfavored party, and thus the plan in general, more competitive. Decracking flips the requisite seats by transferring supporters of the disfavored party from relatively competitive districts won by the favored party. This decracking strategy makes the districts won by the favored party, and thus the plan in general, less competitive. Table 3 below demonstrates these two strategies in a hypothetical ten-district election.

Table 3
Hypothetical Ten-District Election Demonstrating the
Unpacking and Decracking Strategies

District	Original Plan		Unpacking Plan		Decracking Plan	
	<i>x</i>	<i>y</i>	<i>x</i>	<i>y</i>	<i>x</i>	<i>y</i>
1	700	300	700	300	800	200
2	700	300	700	300	800	200
3	700	300	700	300	700	300
4	540	460	440	560	440	560
5	540	460	440	560	440	560
6	540	460	540	460	540	460
7	540	460	540	460	540	460
8	540	460	540	460	540	460
9	350	650	450	550	350	650
10	350	650	450	550	350	650
Total Won	8	2	6	4	6	4

In both the unpacking plan and the decracking plan, districts 4 and 5 swap enough *x* voters for *y* voters to flip those seats from *x*-won to *y*-won, thereby producing a seat margin twice the vote margin and a zero efficiency gap. But the unpacking plan swaps voters out of *y*-won districts 9 and 10, making those districts and the plan overall more competitive. In contrast, the decracking plan swaps voters out of *x*-won districts 1 and 2, making those districts and the plan overall less competitive.

2. Sensitivity and responsiveness

Subpart D.1 above illustrated how both competitive and uncompetitive plans can produce an efficiency gap of zero. But while a plan of any competitiveness level can achieve a zero (or low) gap at one level of vote share, only a plan with a specified level of competitiveness can *maintain* a zero (or low) gap over a range of vote share. This is so because the sensitivity of the efficiency gap is a function of overall competitiveness.

a. Sensitivity as a function of responsiveness

Ignoring turnout effects, the efficiency gap (ΔW) is a function of two variables: the statewide vote margin (V^*) and the statewide seat margin (S^*): $\Delta W = S^* - 2V^*$. But the statewide seat margin is itself a function of the statewide vote margin: $S^* = S^*(V^*)$. This latter function is the real-world—rather than the ideal—seats-

votes curve. The responsiveness (r) of this curve tells us how much the seat margin changes for a given incremental change in vote margin. Mathematically, responsiveness is the derivative of the function $S^*(V^*)$ with respect to V^* : $r = \frac{dS^*(V^*)}{dV^*}$. Graphically, it is the slope of the tangent to the seats-votes curve at a specified point. Responsiveness is a measure of competitiveness because the more districts feature small victory margins, the greater the change in seat margin for a given change in vote margin will be.

The sensitivity of the efficiency gap to a change in statewide vote margin is captured by the derivative with respect to V^* of $\Delta W = S^*(V^*) - 2V^*$.

$$\frac{d\Delta W}{dV^*} = r - 2$$

According to this equation, the sensitivity of the efficiency gap to a change in vote margin depends on the responsiveness of the actual seats-votes curve. If responsiveness is precisely equal to two, the increase in seat margin perfectly offsets the increase in vote margin, and the efficiency gap remains constant. If responsiveness is greater than two, the increase in vote margin triggers an over-compensatory increase in seat margin, and the efficiency gap increases. If responsiveness is less than two, the increase in vote margin triggers an under-compensatory increase in seat margin, and the efficiency gap decreases.

This relationship means that both highly competitive ($r > 2$) and highly uncompetitive ($r < 2$) plans entail a risk: They may produce a zero gap at expected vote share but an above-threshold gap if vote share departs sufficiently from the expectation. However, highly competitive plans may present a greater risk because they require smaller shifts in vote share to produce above-threshold gaps. With a highly competitive plan, a small change in vote share can flip many seats, and that large change in seat share can produce an above-threshold gap. With a highly uncompetitive plan, conversely, a small change in vote share will flip no seats, so a 4% vote swing is needed to produce an 8% gap.²⁶⁵

b. The robust, minimizing plan

Now consider a mapmaker who knows (or can estimate well) the current statewide vote margin but who recognizes that the vote margin may vary over

265. Assume that the plan produces a zero gap ($S^* - 2V^* = 0$) at expected vote margin V^* and a 4% uniform vote swing fails to flip a single seat. Thus, the seat margin is still S^* , but the vote margin is now $V^* + 0.04$, so the efficiency gap is $S^* - 2(V^* + 0.04) = S^* - 2V^* - 0.08 = -0.08$. For both the highly competitive and highly uncompetitive plans, the above-threshold gap may not be durable under sensitivity analysis. But as we await a ruling from the Court in *Whitford* and possible elaboration in future cases, there is uncertainty about whether and how sensitivity analysis will play a role doctrinally. Will mapmakers prefer to avoid generating unstable above-threshold gaps? If so, they may prefer less competitive plans.

time with changing electoral circumstances. Suppose this mapmaker wishes to design a plan that is both *minimizing*, in the sense that it produces an efficiency gap of zero under the current statewide vote margin, and *robust*, in the sense that it maintains a zero (or low) efficiency gap if the vote margin varies. How can the mapmaker design such a robust, minimizing plan? The trick is to find a plan that produces a particular seats-votes relationship—one that exhibits the right ratio at the current vote margin (double proportionality) and that flips seats at the right rate as vote margin varies (double responsiveness).

In the example provided by Table 4 below, the statewide vote margin is 5%.²⁶⁶ To produce an efficiency gap of zero, the statewide seat margin must therefore be 10%.²⁶⁷ Thus, party *x* must win 6 of the 10 seats.²⁶⁸ And for every 5% change in vote margin, one seat must flip.²⁶⁹ Assuming uniform swing, this means that the proportional vote margin in each district must be a distinct multiple of 5%. The result is to spread out the competitiveness of each district so that seats flip at the right rate.

Table 4
The Robust, Minimizing Plan

District	Total Votes by Party			Winner
	<i>x</i>	<i>y</i>	V^*_i	
1	775	225	.275	<i>x</i>
2	725	275	.225	<i>x</i>
3	675	325	.175	<i>x</i>
4	625	375	.125	<i>x</i>
5	575	425	.075	<i>x</i>
6	525	475	.025	<i>x</i>
7	475	525	-.025	<i>y</i>
8	425	575	-.075	<i>y</i>
9	375	625	-.125	<i>y</i>
10	325	675	-.175	<i>y</i>

Note that only 2 of 10 districts are relatively competitive: District 6 currently favors party *x* but would flip if party *y* earned a 5% uniform vote swing,

266. Party *x* wins 5500 votes, or 55%, and party *y* wins 4500 votes, or 45%.

267. $0 = S^* - 2(0.05)$, so $S^* = 0.1$.

268. $S^* = \bar{S}_x - \frac{1}{2}$, so $0.1 = \bar{S}_x - \frac{1}{2}$ and thus $\bar{S}_x = 0.6$.

269. For example, with a 5% increase, $V^* = 0.1$, so to maintain an efficiency gap of zero, we need $0 = S^* - 2(0.1) = S^* - 0.2$, so $S^* = 0.2$, and thus $\bar{S}_x = 0.7$, or 7 out of 10 seats.

and district 7 currently favors party *y* but would flip if party *x* earned a 5% uniform vote swing. Under a 5% uniform vote swing, all other districts would have the same winning party. As this example illustrates, a mapmaker eager to produce and maintain an efficiency gap of zero has an incentive to limit the competitiveness of the plan she draws. Most districts must be safe seats.

E. False Negatives and False Positives

The preceding examination reveals how the efficiency gap measure privileges a form of symmetric partisan efficiency over electoral competitiveness and strict seats-votes proportionality. If proportionality and competitiveness matter, the efficiency gap's relationship to them presents two problems of normative correspondence: The measure may favor normatively undesirable plans (the false negative problem) and may disfavor normatively desirable plans (the false positive problem). Consider each problem in turn.

First, a plan may be gerrymandered in a way the measure cannot detect, achieving the ideal of equal wasted votes at the expense of both competitiveness and seats-votes proportionality. In this sense, the efficiency gap measure has a significant false negative problem: It approves plans that exhibit one democratic ideal (equal wasted votes) even if those plans subvert competitiveness, seats-votes proportionality, or both.

Second, the measure may flag normatively desirable plans as suspect. A plan that produces rough proportionality even when one party enjoys a significant popular majority will necessarily produce a large efficiency gap in favor of the minority party. While both competitive and uncompetitive plans can produce low efficiency gaps, a more competitive plan may present a greater risk of an above-threshold gap in the face of vote swings. In this sense, the efficiency gap measure has a significant false positive problem: It flags as suspect plans that depart from the ideal of equal wasted votes even if that departure reflects an effort to promote democratic norms like competitiveness and seats-votes proportionality.

For both reasons, the efficiency gap measure may promote fairness in the sense of symmetric partisan efficiency but unintentionally encourage uncompetitive elections that accord one party a legislative majority disproportional to its popular support.

The efficiency gap proposal consists not only of the measure itself but also of sensitivity analysis, analysis of partisan intent on the part of the mapmakers,

and an opportunity for state justification.²⁷⁰ These other doctrinal tools may mitigate the false positive problem: If a highly competitive plan produces a large efficiency gap in one election, that gap will likely be unstable under sensitivity analysis. And a plan that promotes competitiveness or proportionality is more likely to be justified by legitimate districting principles and less likely to be branded the result of discriminatory intent. But these tools do not eliminate the problem altogether because mapmakers may fear not only liability but also litigation. An above-threshold gap may be unstable or justified, but it is likely to invite legal challenge, and the state may only prevail after the considerable delay, expense, and risk of a trial before a three-judge federal district court.

Perhaps more significantly, these doctrinal tools do not address the false negative problem because the proponents offer no mechanism to overcome the presumption of validity triggered by a below-threshold gap.²⁷¹ If the Court were

270. The academic proponents suggested that “sensitivity testing” be “incorporat[ed] into the thresholds,” see Stephanopoulos & McGhee, *supra* note 7, at 889-90, and that states ought to be able to offer nondiscriminatory justifications to rebut the presumptive invalidity of an above-threshold efficiency gap, see *id.* at 891, but they did not include any requirement of showing partisan intent, see *id.* at 832 n.2 (“Our conception of gerrymandering is strictly effects-based . . .”). The *Whitford* plaintiffs chose to offer evidence of partisan intent as a relevant portion of the test for an unconstitutional gerrymander, and the district court concluded that the Wisconsin State Assembly plan at issue was drawn in part “to entrench the Republican Party in power.” See *Whitford v. Gill*, 218 F. Supp. 3d 837, 890-96 (W.D. Wis. 2016), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

271. The proponents repeatedly refer to both the invalidity of an above-threshold gap and the validity of a below-threshold gap as presumptive rather than irrebuttable. See Stephanopoulos & McGhee, *supra* note 7, at 884 (“[C]ourts would need to choose an efficiency gap threshold above which district plans would be presumptively unlawful and below which they would be presumptively valid.”); *id.* at 886 (arguing that the Supreme Court could avoid the concerns raised in *Vieth* by “specifying an efficiency gap level above which plans would be presumptively unlawful and below which they would be presumptively legitimate”); *id.* at 890 (“[C]ourts may be reluctant in early cases to set particular levels above which plans are presumptively unlawful and below which they are presumptively legitimate.”); see also *id.* at 891 (“Throughout our discussion to this point, we have spoken of *presumptive* rather than *irrebuttable* validity and invalidity.”). While the proponents explain in detail how a state could rebut the presumption of invalidity, see *id.* at 891-95, they say nothing about how a plaintiff could rebut the presumption of validity triggered by a below-threshold efficiency gap. Moreover, the proponents suggest that their “approach would neatly slice *Vieth*’s Gordian knot, informing lower courts and political actors, in clear quantitative terms, exactly ‘[h]ow much political . . . effect is too much.’” *Id.* at 886 (alterations in original) (quoting *Vieth v. Jubelirer*, 541 U.S. 267, 297 (2004) (plurality opinion)). This characterization suggests that a plan producing a below-threshold gap must be valid because its partisan effect is not too much. Finally, the *Whitford* panel majority seems to have understood the plaintiffs’ proposal to include an irrebuttable presumption of validity for low-gap plans. See 218 F. Supp. at 908 (“[A] challenge to a map enacted with egregious partisan intent but demonstrating a low [efficiency gap] also will fail because the plaintiffs cannot demonstrate the required discriminatory effect.”).

to embrace the efficiency gap as the definitive measure of partisan gerrymandering, plans that consistently produced low efficiency gaps would be judicially bulletproof. This would be problematic in jurisdictions where one party enjoys a large majority, as the majority party could achieve a low gap by carving the state up into safe districts that accorded it most or all of the seats.

F. Definition and Weight of Surplus Votes

This Subpart demonstrates how the efficiency gap's relationships with proportionality and competitiveness vary once we modify the definition of wasted votes. Specifically, I alter the definition and weight of surplus votes using parameters α (alpha) and β (beta) and derive a more generalized simplified formula as a function of these parameters.²⁷²

The parameter α captures weighting of surplus votes. The efficiency gap, as proposed, measures the number of wasted votes through the equation.

$$W_{pi} = L_{pi} + E_{pi}$$

To weight lost (L_{pi}) and surplus (E_{pi}) votes differently, we can simply multiply the number of surplus votes by the parameter α , which we can adjust to capture how heavily we wish to weight surplus votes as compared to lost votes. The wasted votes equation becomes:

$$W_{pi} = L_{pi} + \alpha E_{pi}.$$

Under the proponents' definition, α is set to 1.²⁷³ If, alternatively, we set α to zero, then we would ignore surplus votes entirely. If we set α to one-half, then we would weight surplus votes half as heavily as lost votes. If we set α greater than 1, we would weight surplus votes more heavily than lost votes.

The parameter β captures the definition of surplus votes. The efficiency gap as proposed defines the number of surplus votes as one-half the victory margin.

$$E_{pi} = \frac{1}{2} M_i$$

But surplus votes could also plausibly be defined as the entire victory margin.²⁷⁴ Thus, we can replace the number one-half with the parameter β to allow an analyst to vary the definition of a surplus vote. If we do so, we get the following generalized equation defining surplus votes:

$$E_{pi} = \beta M_i.$$

272. I use the term *parameter* to denote a variable for which an analyst selects a real number rather than a variable an analyst observes.

273. Cf. Stephanopoulos & McGhee, *supra* note 7, at 851.

274. See, e.g., Nagle, *supra* note 20, at 199 & n.16, 203 & n.24.

We can set β to one-half (using proponents' definition) or to 1 (using the alternative definition)—or at some intermediate value.²⁷⁵

The parameter γ (gamma) captures the combined effect of weight and definition. It is simply the product of α and β .

$$\gamma = \alpha\beta$$

We can now repeat all the relevant steps of our prior analysis, this time in terms of these parameters, to quantify the precise effect of how we define and weight surplus votes. The long-form definition-generalized efficiency gap is:

$$\Delta W(\gamma) = \frac{W_y - W_x}{V_y + V_x}$$

where

$$W_p = \sum_{i \in D} W_{pi}$$

and

$$W_{pi} = \begin{cases} E_{pi} = \gamma M_i & i \in D_p \\ L_{pi} = V_{pi} & i \notin D_p \end{cases}$$

As we will soon see, it will prove useful to define the measure's generalized ideal responsiveness as the following function of parameter γ : $r(\gamma) = 1 + 2\gamma$. Note that when $\alpha = 1$ and $\beta = \frac{1}{2}$, as under the proponents' traditional definition, $\gamma = \alpha\beta = \frac{1}{2}$, and $r = 1 + 2\gamma = 2$: the responsiveness of the ideal seats-votes curve associated with the simplified formula.

At the level of an individual district, the wasted vote disparity (Δw_i) remains a simple linear discontinuous function of the proportional victory margin (m_i), but that function now depends on ideal responsiveness (r).²⁷⁶

$$\Delta w_i = \begin{cases} \frac{1}{2}(1 - rm_i) & i \in D_x \\ -\frac{1}{2}(1 - rm_i) & i \in D_y \end{cases}$$

275. An analyst might use an intermediate value if she thought that both definitions had some merit and some flaws and that the best approach was somewhere in between.

276. Because $V_{2i} = \frac{T_i}{2}(1 - m_i)$, both surplus and lost votes can be expressed in terms of m_i :

$$E_{pi} = \gamma M_i = \gamma m_i T_i \text{ and so } \frac{E_{pi}}{T_i} = \gamma m_i;$$

$$L_{pi} = V_{pi} = V_{2i} = \frac{T_i}{2}(1 - m_i) \text{ and so } \frac{L_{pi}}{T_i} = \frac{1}{2}(1 - m_i).$$

By definition $\Delta w_i = \frac{w_{yi} - w_{xi}}{T_i}$ and $r = 1 + 2\gamma$, so Δw_i can be expressed in terms of m_i .

Excluding ties, this function is zero if and only if $m_i = \frac{1}{r}$. Call this the generalized minimizing proportional victory margin, denoted $m^0(\gamma)$. For example, when $\alpha = 1$ and $\beta = \frac{1}{2}$, as under the proponents' traditional definition, $\gamma = \alpha\beta = \frac{1}{2}$, $r = 1 + 2\gamma = 2$, and $m^0 = \frac{1}{r} = \frac{1}{2}$. This is the minimizing victory margin, derived above, in which the victor earns 75% vote share.²⁷⁷

Define the generalized competitiveness score $c_i(\gamma)$ as the difference between this new minimizing victory margin ($m^0(\gamma)$) and the actual victory margin (m_i): $c_i(\gamma) = m^0(\gamma) - m_i$. At the level of an individual district, the

$$\text{For } i \in D_x, \Delta w_i = \frac{w_{yi} - w_{xi}}{T_i} = \frac{L_{yi}}{T_i} - \frac{E_{xi}}{T_i} = \frac{1}{2}(1 - m_i) - \gamma m_i = \frac{1}{2}(1 - rm_i).$$

$$\text{For } i \in D_y, \Delta w_i = \frac{w_{yi} - w_{xi}}{T_i} = \frac{E_{yi}}{T_i} - \frac{L_{xi}}{T_i} = \gamma m_i - \frac{1}{2}(1 - m_i) = -\frac{1}{2}(1 - rm_i).$$

277. See *supra* text accompanying notes 237-38.

wasted vote disparity Δw_i is a simple function of $c_i(\gamma)$ and $r(\gamma)$ ²⁷⁸:

$$\Delta w_i = \begin{cases} \frac{r}{2} c_i & i \in D_x \\ -\frac{r}{2} c_i & i \in D_y \end{cases}$$

With this new competitiveness score, define average competitiveness in p -won districts as $c_p(\gamma) = \frac{\sum_{i \in D_p} c_i(\gamma)}{s_p}$. Note that $c_p(\gamma) = m^0(\gamma) - m_p$ where m_p is the average victory margin in p -won districts. Finally, define the competitiveness gap as the seat-share-weighted difference between average competitiveness in x - and y -won districts.

$$\Delta C(\gamma) = \bar{s}_x c_x(\gamma) - \bar{s}_y c_y(\gamma)$$

With these new definitions, the efficiency gap and the competitiveness gap are related by the factor $\frac{r}{2}$.²⁷⁹

$$\Delta W(\gamma) = \frac{r}{2} \Delta C(\gamma)$$

The definition-generalized simplified formula is:

$$\Delta W(\gamma) = S^* - r(\gamma)V^*.$$

This is the traditional simplified formula with the constant 2 replaced by function $r(\gamma)$. Of course, under the proponents' traditional definition, $\gamma = \frac{1}{2}$ and $r\left(\frac{1}{2}\right) = 2$, so the definition-generalized simplified formula reduces to the traditional formula. More generally, the definition-generalized simplified formula

278. By definition, $c_i = m^0 - m_i$ and $m^0 = \frac{1}{r}$. Rearranging: $m_i = \frac{1}{r} - c_i$. Thus:

$$1 - rm_i = 1 - r\left(\frac{1}{r} - c_i\right) = 1 - 1 + rc_i = rc_i. \text{ And so:}$$

$$\begin{aligned} \Delta w_i &= \begin{cases} \frac{1}{2}(1 - rm_i) & i \in D_x \\ -\frac{1}{2}(1 - rm_i) & i \in D_y \end{cases} \\ &= \begin{cases} \frac{r}{2} c_i & i \in D_x \\ -\frac{r}{2} c_i & i \in D_y \end{cases} \end{aligned}$$

279. Using the equal voter turnout assumption, this result is readily derived by expressing the efficiency gap as the average wasted vote disparity:

$$\Delta W = \frac{\sum_{i \in D} \Delta w_i}{s} = \frac{\frac{r}{2} [\sum_{i \in D_x} c_i + \sum_{i \in D_y} -c_i]}{s} = \frac{\frac{r}{2} [s_x c_x - s_y c_y]}{s} = \frac{r}{2} [\bar{s}_x c_x - \bar{s}_y c_y] = \frac{r}{2} \Delta C.$$

and the definition-generalized competitiveness gap are also related by the same factor $\frac{r}{2}$.

$$\frac{r}{2}\Delta C(\gamma) = S^* - rV^*$$

Combining these results with the new, generalized definitions—and under equal voter turnout—the efficiency gap, competitiveness gap, and simplified formula are related as follows.

$$\Delta W(\gamma) = \frac{r}{2}\Delta C(\gamma) = S^* - rV^*$$

Because r is 2 under the proponents’ definition, the long-form equation and simplified formula equal the competitiveness gap. In the general case, the long-form equation and simplified formula equal the competitiveness gap multiplied by the factor $\frac{r}{2}$.

As the preceding examination reveals, even when we alter the definition and weight of surplus votes, the measure still captures relative wasted votes, undeserved vote share, and differential competitiveness. The measure still generally privileges equal wasted votes over competitiveness and seats-votes proportionality. But the precise tradeoffs between these competing norms are determined by the precise values the analyst selects for parameters α and β . Table 5 below summarizes the effects of varying these two parameters.

Table 5
Effects of Varying the Definition and Weight of Surplus Votes

	Weight of surplus vote relative to lost vote	Relationship between surplus votes and vote margin	Product of α & β	Ideal responsiveness	Generalized simplified formula	Minimizing victory margin	Minimizing winning vote share	Statewide vote share for total capture with zero gap	Statewide vote share for supermajority with sub-threshold gap
Parameter	α	β	γ	r	ΔW	m^0	v^*	$V_{100\%,0\%}$	$V_{100\%,8\%}$ $V_{75\%,9\%}$
Equation			$\alpha\beta$	$1 + 2\gamma$	$S^* - rV^*$	$\frac{1}{r}$	$\frac{1 + m^0}{2}$	v^*	$\frac{(\frac{5}{2} - \frac{1}{2}) - \Delta W^*}{r} + \frac{1}{2}$
Alternative definition	1	1	1	3	$S^* - 3V^*$	1/3	2/3	67%	64%, 56%
Traditional interpretation	1	1/2	1/2	2	$S^* - 2V^*$	1/2	3/4	75%	71%, 59%
Alternative interpretation	1/2	1	1/2	2	$S^* - 2V^*$	1/2	3/4	75%	71%, 59%
Alternative definition & weighing	1/2	1/2	1/4	3/2	$S^* - \frac{3}{2}V^*$	2/3	5/6	83%	78%, 61%
Alternative weighing	0	1	0	1	$S^* - V^*$	1	1/1	100%	92%, 67%

A simple function of γ , $r = 1 + 2\gamma$, determines the responsiveness of the ideal seats-votes curve. The generalized simplified formula is just like the traditional one, except vote margin (V^*) is multiplied by responsiveness (r), which is a function of γ . The minimizing victory margin (m^0) is the multiplicative inverse of r : $m^0 = \frac{1}{r}$. The winner's minimizing vote share (v^*) is a simple function of m^0 : $v^* = \frac{1+m^0}{2}$. And the vote share a party must exceed to maintain a gap below threshold ΔW^* with a seat share of $\bar{S}$ is $\bar{V}_{\bar{S}, \Delta W^*} = \frac{(\bar{S} - \frac{1}{2}) - \Delta W^*}{r} + \frac{1}{2}$.²⁸⁰ Note that the winner's minimizing vote share is the vote share a party needs in order to win every seat ($\bar{S} = 1$) with an efficiency gap of zero ($\Delta W^* = 0$): $\bar{V}_{1,0} = v^*$.²⁸¹ This makes sense because a simple plan composed exclusively of x -won districts will produce a gap of zero, a seat share of one, and a statewide vote share equal to the winner's minimizing vote share.

As Table 5 above shows, the parameter γ is like a dial the analyst can turn to calibrate the relationship among the efficiency gap, competitiveness, and seats-votes proportionality. When γ is one, the minimizing district is won 67% to 33%, mitigating the competitiveness problem, but the minimizing plan exhibits triple seats-votes proportionality, exacerbating the proportionality problem. When γ is zero, the minimizing district is won 100% to 0%, exacerbating the competitiveness problem, but the minimizing plan exhibits strict seats-votes proportionality, eliminating the proportionality problem.

The proponents eschew either of these “pure” approaches.²⁸² Rather than turning the dial all the way in one direction or the other, they adjust the dial to an intermediate position, setting γ equal to one-half. This intermediate calibration avoids the most extreme tensions with proportionality and competitiveness, opting instead for more moderate tension with both norms.

There are two mathematically equivalent ways to set γ equal to one-half: (1) defining a surplus vote as half the vote margin while equally weighting lost and surplus votes, and (2) defining a surplus vote as the entire vote margin while weighting a surplus vote half as heavily as a lost vote. In the latter approach, the definition is more intuitive, but the weight is more arbitrary and therefore

280. Because $S^* = \bar{S} - \frac{1}{2}$, $V^* = \bar{V} - \frac{1}{2}$, and $\Delta W = S^* - rV^*$ (under the generalized simplified formula), this inequality ensures a below-threshold gap.

If $\bar{V} > \frac{(\bar{S} - \frac{1}{2}) - \Delta W^*}{r} + \frac{1}{2}$, then $r(\bar{V} - \frac{1}{2}) > (\bar{S} - \frac{1}{2}) - \Delta W^*$, and so $S^* - rV^* < \Delta W^*$, and $\Delta W < \Delta W^*$.

281. $\bar{V}_{1,0} = \frac{(\frac{1}{2} - \frac{1}{2}) - 0}{r} + \frac{1}{2} = \frac{1}{2}(1 + m^0) = v^*$.

282. See Stephanopoulos & McGhee, *supra* note 7, at 851.

harder to discern and justify. The proponents opt instead for the former approach, adopting a more intuitive weight but a less intuitive definition.

IV. A New Wasted Vote Measure

In this Part, I suggest the plausibility and superiority of a measure that compares shares rather than absolute numbers of wasted votes, with a surplus vote defined as the full vote margin and lost and surplus votes weighted equally. This measure better accords with the prevailing individual rights framework because it is voter- rather than party-centric in both scale and its definition of surplus votes. And the measure is more structurally resonant because it coheres more closely with electoral reality and with the democratic values of seats-votes proportionality and competitiveness.

Consider a modified efficiency gap measure (ΔW^V) with a voter-centric scale and definition of surplus vote.

$$\Delta W^V = \frac{W_y}{V_y} - \frac{W_x}{V_x}$$

where

$$W_p = \sum_{i \in D} W_{pi}$$

and

$$W_{pi} = \begin{cases} \gamma M_i & i \in D_p \\ V_{pi} & i \notin D_p \end{cases}$$

Recall that we can express vote totals, and thus wasted votes, in terms of victory margin and turnout.²⁸³

$$W_{pi} = \begin{cases} \gamma T_i m_i & i \in D_p \\ \frac{T_i}{2} (1 - m_i) & i \notin D_p \end{cases}$$

$$W_p = \sum_{i \in D} W_{pi} = \sum_{i \notin D_p} \frac{T_i}{2} (1 - m_i) + \sum_{i \in D_p} \gamma T_i m_i$$

If we apply equal voter turnout ($T_i = T^*$), then $\frac{S_x}{S} = \bar{S}$, $\frac{S_y}{S} = 1 - \bar{S}$, $V_x = ST^*\bar{V}$, and $V_y = ST^*(1 - \bar{V})$. We can therefore express each party's wasted vote share

283. See *supra* note 234 and accompanying text.

as a function of these variables, the parameter γ , and the average victory margin in x - and y -won districts (m_x and m_y).²⁸⁴

$$\frac{w_y}{v_y} = \frac{2\bar{S} - 2\bar{S}m_x + 4\gamma(1-\bar{S})m_y}{4(1-\bar{V})} \quad (EQ1)$$

$$\frac{w_x}{v_x} = \frac{4\gamma\bar{S}m_x + 2(1-\bar{S}) - 2(1-\bar{S})m_y}{4\bar{V}} \quad (EQ2)$$

284. By definition, the following equalities hold:

$$\bar{S} = \bar{S}_x = \frac{S_x}{S}, \text{ so } 1 - \bar{S} = \bar{S}_y = \frac{S_y}{S},$$

$$\bar{V} = \bar{V}_x = \frac{V_x}{V}, \text{ so } 1 - \bar{V} = \bar{V}_y = \frac{V_y}{V},$$

$$T^* = \frac{V}{S}, \text{ so } V = ST^*, V_x = V\bar{V}_x = ST^*\bar{V}, \text{ and } V_y = V\bar{V}_y = ST^*(1 - \bar{V});$$

$$\sum_{i \in D_x} 1 = S_x = S\bar{S} \text{ and } \sum_{i \in D_y} 1 = S_y = S(1 - \bar{S});$$

$$m_x = \frac{\sum_{i \in D_x} m_i}{S_x}, \text{ so } \sum_{i \in D_x} m_i = S_x m_x = S\bar{S}m_x;$$

$$m_y = \frac{\sum_{i \in D_y} m_i}{S_y}, \text{ so } \sum_{i \in D_y} m_i = S_y m_y = S(1 - \bar{S})m_y.$$

Using these equalities:

$$W_y = \sum_{i \in D_x} \frac{T^*}{2} (1 - m_i) + \sum_{i \in D_y} \gamma T^* m_i = \frac{T^*}{2} \sum_{i \in D_x} 1 - \frac{T^*}{2} \sum_{i \in D_x} m_i + \gamma T^* \sum_{i \in D_y} m_i;$$

$$W_y = \frac{T^*}{2} S\bar{S} - \frac{T^*}{2} S\bar{S}m_x + \gamma T^* S(1 - \bar{S})m_y = \frac{ST^*}{4} (2\bar{S} - 2\bar{S}m_x + 4\gamma(1 - \bar{S})m_y);$$

$$\frac{w_y}{v_y} = \frac{\frac{ST^*}{4} (2\bar{S} - 2\bar{S}m_x + 4\gamma(1 - \bar{S})m_y)}{ST^*(1 - \bar{V})} = \frac{2\bar{S} - 2\bar{S}m_x + 4\gamma(1 - \bar{S})m_y}{4(1 - \bar{V})} \quad (EQ1).$$

And:

$$W_x = \sum_{i \in D_x} \gamma T^* m_i + \sum_{i \in D_y} \frac{T^*}{2} (1 - m_i) = \gamma T^* \sum_{i \in D_x} m_i + \frac{T^*}{2} \sum_{i \in D_y} 1 - \frac{T^*}{2} \sum_{i \in D_y} m_i;$$

$$W_x = \gamma T^* S\bar{S}m_x + \frac{T^*}{2} S(1 - \bar{S}) - \frac{T^*}{2} S(1 - \bar{S})m_y = \frac{ST^*}{4} (4\gamma\bar{S}m_x + 2(1 - \bar{S}) - 2(1 - \bar{S})m_y);$$

$$\frac{w_x}{v_x} = \frac{\frac{ST^*}{4} (4\gamma\bar{S}m_x + 2(1 - \bar{S}) - 2(1 - \bar{S})m_y)}{ST^*\bar{V}} = \frac{4\gamma\bar{S}m_x + 2(1 - \bar{S}) - 2(1 - \bar{S})m_y}{4\bar{V}} \quad (EQ2).$$

Nagle proceeded as though m_x and m_y were independent variables.²⁸⁵ But recall that the seat-share-weighted sum of m_x and m_y is the average victory margin over all districts (m), and the seat-share-weighted difference of m_x and m_y is twice the vote margin.²⁸⁶

$$\begin{aligned} m &= \bar{S}_x m_x + \bar{S}_y m_y \\ 2V^* &= \bar{S}_x m_x - \bar{S}_y m_y \end{aligned}$$

Let $\bar{S}$ and $\bar{V}$ denote seat and vote share for party x . Then $\bar{S}_x = \bar{S}$, $\bar{S}_y = 1 - \bar{S}$, and $V^* = \bar{V} - \frac{1}{2}$. Thus, we can express these relationships in terms of $\bar{S}$ and $\bar{V}$.

$$m = \bar{S} m_x + (1 - \bar{S}) m_y \quad (EQ3)$$

$$2\bar{V} - 1 = \bar{S} m_x - (1 - \bar{S}) m_y \quad (EQ4)$$

Rearranging, we can express m_x and m_y in terms of m , $\bar{V}$, and $\bar{S}$.

$$2\bar{S} m_x = m + 2\bar{V} - 1 \quad (EQ5)$$

$$2(1 - \bar{S}) m_y = m - 2\bar{V} + 1 \quad (EQ6)$$

Substituting *EQ5* and *EQ6* into *EQ1* and *EQ2* and simplifying, we can express each party's wasted vote share as a function of $\bar{S}$, $\bar{V}$, γ , and m .

$$\frac{w_y}{v_y} = \frac{2\bar{S} - m(1 - 2\gamma) - 2\bar{V}(1 + 2\gamma) + 1(1 + 2\gamma)}{4(1 - \bar{V})} \quad (EQ7)$$

$$\frac{w_x}{v_x} = \frac{-2\bar{S} - m(1 - 2\gamma) + 2\bar{V}(1 + 2\gamma) + 1(1 - 2\gamma)}{4\bar{V}} \quad (EQ8)$$

The new measure is the difference between each party's wasted vote share. Subtracting *EQ8* from *EQ7* and simplifying, we get *EQ9*.

$$\Delta W^V = \frac{2\bar{S} - 2\bar{V}(1 + 2\gamma) - (1 - m(1 - 2\gamma))(1 - 2\bar{V}) + 2\gamma}{4\bar{V}(1 - \bar{V})} \quad (EQ9)$$

Setting ΔW^V to zero yields the following ideal seats-votes formula.

$$\bar{S} = (1 + 2\gamma)\bar{V} + \frac{1}{2}(1 - m(1 - 2\gamma))(1 - 2\bar{V}) - \gamma$$

285. Nagle called his parties *A* and *B* instead of *x* and *y* and focused on “the average *A* vote for those districts won by *A* and the average *A* vote for those districts won by *B*.” See Nagle, *supra* note 20, app. B at 208. These two values are related to the parties’ respective average victory margins. Only after setting equal the parties’ respective wasted vote shares did Nagle conclude that these two values “are not independent; [one] can be determined from [the other] through a quadratic formula.” See *id.* at 201; *id.* app. B at 209. However, even when respective wasted vote shares are not equal, the two values are related to one another, as demonstrated in *EQ3* and *EQ4* below.

286. See *supra* note 235 and accompanying text.

We can now express the ideal seats-votes formula in terms of seat margin ($S^* = \bar{S} - \frac{1}{2}$) and vote margin ($V^* = \bar{V} - \frac{1}{2}$).

$$S^* = (2\gamma - m(2\gamma - 1))V^*$$

With equal weighting of lost and surplus votes ($\alpha = 1$) and the proponents' definition of surplus votes based on half the vote margin ($\beta = \frac{1}{2}$) such that $\gamma = \frac{1}{2}$, the victory margin term (m) drops out entirely, and the ideal seats-votes relationship reduces to strict proportionality.

$$S^* = \left(2\left(\frac{1}{2}\right) - m\left(2\left(\frac{1}{2}\right) - 1\right)\right)V^* = (1 - m(0))V^* = V^*$$

This result accords with Nagle's.²⁸⁷ However, if we maintain equal weighting of lost and surplus votes ($\alpha = 1$) but alternatively define surplus votes based on the entire vote margin ($\beta = 1$) so that $\gamma = 1$, then the victory margin term m remains and the ideal seats-votes relationship reduces to the following:

$$S^* = (2(1) - m(2(1) - 1))V^* = (2 - m)V^*.$$

Under maximal uncompetitiveness, when each prevailing candidate earns all the votes (that is, as m approaches 1), this reduces to strict proportionality. Under maximal competitiveness, when each prevailing candidate wins by a single vote (that is, as m approaches 0), this reduces to the same double proportionality that emerges from the proponents' party-centric approach based on equal wasted vote totals and surplus votes defined as one-half of the victory margin. But unlike the proponents' approach, this seats-votes curve depends on the victory margin term m . As m varies from 0 to 1, the curve's slope varies from 2 to 1. The seat bonus is still capped at 2, but now the majority party can achieve that maximal seat bonus only if it maximizes competitiveness. If the system is less than maximally competitive, the seat bonus must be less than 2.

Nagle recognized that his voter-centric approach under an alternative definition of surplus votes would produce an ideal seats-votes curve that depended on competitiveness.²⁸⁸ But Nagle, following McGhee, views this dependence on

287. See Nagle, *supra* note 20, at 202 (equation 6).

288. See *id.* at 203 ("Making districts more competitive allows a [majority] party ... to increase its [seat share] with no change in this measure"). Nagle used the proponents' definition of surplus votes as half the victory margin but noted that the alternative definition was equivalent to doubling the relative weight of surplus votes. See *id.* at 199 & n.16, 203 & n.24.

competitiveness as an undesirable—indeed, a fatal—feature of the measure under the alternative definition of surplus votes ($\gamma \neq \frac{1}{2}$):

However, as Eric McGhee has kindly pointed out, the possibility that different values of [seat share] for the same vote [share] may give the same value of bias violates [the efficiency principle] . . . Making districts more competitive allows a gerrymandering party that has [a majority vote share] to increase its [seat share] with no change in this measure of bias when $[\gamma > \frac{1}{2}]$.²⁸⁹

Nagle and McGhee are right that when $\gamma = 1$, or more generally when $\gamma > \frac{1}{2}$, the majority party can increase its seat share while maintaining constant vote share and equal wasted vote shares by increasing average district competitiveness. And this property violates McGhee's strict efficiency principle, which defines efficiency as increasing seat share at constant vote share.²⁹⁰ But this property does not necessarily violate the modified efficiency principle that defines efficiency as increasing expected seat share at constant vote share.²⁹¹ This is because the majority party can only increase its seat share at current vote share by increasing the competitiveness of the system, which makes the outcome less robust to vote swings.²⁹²

But there is another way to address the measure's dependence on competitiveness when $\gamma > \frac{1}{2}$. Rather than using the system's actual average victory margin (m), the analyst can compare the difference in wasted vote shares that would have obtained if the system exhibited an ideal baseline level of average victory margin ($\hat{m}$). Political scientists estimate that real-world seats-votes curves tend to exhibit competitiveness that varies with vote share, exhibiting high competitiveness when the majority enjoys a modest vote margin and low competitiveness when the majority enjoys a significant vote margin.²⁹³ This produces an S-shaped curve that is relatively flat far away from the (0.5,0.5) point where each party earns half the votes but relatively steep near the (0.5,0.5) point.²⁹⁴ This accommodation between competitiveness and proportionality favors the majority when the minority is large but protects the minority when it

289. See Nagle, *supra* note 20, at 203.

290. See McGhee, *Measuring Efficiency*, *supra* note 86, at 418.

291. See *supra* Part II.A.2.

292. See *supra* Part III.D.2.

293. See Browning & King, *supra* note 53, at 312 fig.1, 313.

294. See, e.g., *id.* at 312 fig.1.

is small. Because the seat bonus decreases with majority vote share, the majority must earn almost all the votes in order to win all the seats.

This suggests an alternative wasted vote measure. First, construct an average victory margin measure $\hat{m}(V^*)$ that aligns with these features of electoral reality. Second, compute the difference in the parties' wasted vote shares (rather than wasted vote totals)—using the full (not one-half) definition of surplus votes ($\gamma = 1$)—that would obtain under $\hat{m}(V^*)$.²⁹⁵ This measure may be more intuitive given that it adopts more voter-centric approaches to scale and definition. And it may avoid the extreme vote share problem in which a majority can completely (or effectively) shut out a minority and capture all (or most) of the seats while maintaining a gap of zero. Moreover, this measure satisfies McGhee's strict efficiency principle. The drawback, of course, is that it relies on a baseline competitiveness measure selected by the analyst, which introduces conjecture. However, the analyst could derive this relationship from real election data.²⁹⁶ Thus the baseline could be what the relationship has generally been in real-world elections, not the analyst's subjective judgment about what the relationship ought to be. A large gap would indicate that the majority has won significantly more seats than the majority would have won had the plan equalized the parties' respective wasted vote shares under a level of competitiveness consistent with the generally prevailing electoral relationship between competitiveness and vote share.

V. Doctrinal Implications and Conclusions

This Article has analyzed the methodological choices underlying the efficiency gap measure as well as the tensions between the measure and democratic

295. To make this idea more concrete, I offer an example for illustrative purposes. Suppose the analyst selected the following competitiveness measure: $\hat{m}(V^*) = 4V^{*2}$. This is the simplest function that satisfies three sensible properties: maximal competitiveness at minimal vote margin, $\hat{m}(0) = 4 * 0 = 0$; minimal competitiveness at maximal vote margin, $\hat{m}(\frac{1}{2}) = 4(\frac{1}{2})^2 = 4(\frac{1}{4}) = 1$; and symmetric treatment of parties, $\hat{m}(-V^*) = 4(-V^*)^2 = 4V^{*2} = \hat{m}(V^*)$. The analyst would then substitute this competitiveness measure into the seats-votes formula $S^* = (2 - m)V^*$.

$$S^*(\gamma = 1) = (2 - \hat{m}(V^*))V^* = (2 - (4V^{*2}))V^* = 2V^* - 4V^{*3}$$

Expressed in terms of seat share and vote share, the relationship is:

$$\bar{S} = 2\left(\bar{V} - \frac{1}{2}\right) - 4\left(\bar{V} - \frac{1}{2}\right)^3 + \frac{1}{2}.$$

This equation represents a nonlinear seats-votes curve exhibiting higher responsiveness when vote margin is small and lower responsiveness when vote margin is large.

296. Such a statistical exercise lies beyond the scope of this Article but warrants future consideration.

norms of proportionality, competitiveness, and voter participation. This analysis has implications for the Court as it decides *Whitford*. I suggest that the Court should reaffirm the justiciability of partisan gerrymandering claims, adopt the conclusion of the majority of the three-judge district court that the Wisconsin State Assembly plan constitutes an unlawful partisan gerrymander, and acknowledge that efficiency gap analysis provides evidence of that plan's discriminatory effect. As its architects intended, that plan packed and cracked Democratic voters such that many more Democratic voters than Republican voters cast votes that had no effect on electoral outcomes. The efficiency gap analyses performed by the plaintiffs' experts Mayer and Jackman capture this effect by computing efficiency gap values that indicate a significant advantage for Republicans, one that constitutes a historical outlier compared to modern elections and that is likely to persist in future elections.

While the Court should view efficiency gap analysis as helpful evidence supporting the panel's conclusion, it should not adopt the efficiency gap as the exclusive definitional measure of partisan gerrymandering in such a way that a plan would be an unconstitutional partisan gerrymander if and only if it produced a sufficiently large, durable, and unjustified efficiency gap. Instead, the Court should recognize the efficiency gap as one indicative measure of partisan gerrymandering while affording some flexibility for courts, litigants, and scholars to refine measurement approaches over time. Both methodological and normative considerations support this approach.

A. Methodological Considerations

Were the Court to proclaim the efficiency gap the definitive measure of partisan gerrymandering, it would place a heavy burden on the measure: the need to be able to assess the validity of all future plans in a way that yields a single correct answer. Yet the answer efficiency gap analysis yields depends on a series of methodological choices analysts are still exploring: what imputation methods to use for uncontested races, how to account for variation in voter turnout, how to define and weight surplus votes, and whether to compare wasted vote totals or shares.²⁹⁷ Each choice has the potential to change the calculated gap. If, as proposed, a challenged plan's efficiency gap is compared to some numeric threshold, both the challenged plan's gap and the numeric threshold itself may depend heavily on methodological choices.

At the level of a single plan, methodological choices may (1) change the sign of the gap and thereby toggle the assessment of which party is favored; (2) drive the gap above or below the numeric threshold; or (3) change the magnitude of the gap enough to influence the justification analysis (for example, a proffered

297. Another question is whether to look to statewide or district-level races when calculating the efficiency gap. See, e.g., Krasno et al., *supra* note 20, at 5 (arguing in favor of the use of statewide rather than district-level races).

excuse may suffice to justify a gap of 10% but not a gap of 20%). And while used to assess an individual plan's efficiency gap, the numeric threshold is itself based on a claim about the distribution of gaps associated with modern elections—and is therefore dependent on methodological choices.

Consider how the proponents arrived at an 8% threshold for state legislative maps. The proponents' 2015 article used only the simplified formula to compute the efficiency gaps for a large number of congressional and state legislative elections between 1972 and 2012.²⁹⁸ Based on the distributions, the proponents advocated for an 8% numeric threshold for state legislative maps.²⁹⁹ They reasoned, "A gap of at least eight points placed a [state legislative] plan in the worst 12 percent of all plans in this period, . . . about 1.5 standard deviations from the mean."³⁰⁰

However, this 8% threshold depends on all the methodological choices discussed above. Variation in imputation method, as described in Part II.E.1 above, can sometimes produce significant changes in the gap. Because the gaps were calculated using the simplified formula rather than the long-form calculation, all differential turnout effects were ignored. And of course, all gaps were calculated with the proponents' definition and weighting of surplus votes and by comparing wasted vote totals rather than shares. The precise impact of all of these methodological choices can only be quantified by recalculating all the historical gaps with alternative methodological approaches—an intensive project that is certainly worth undertaking but lies beyond the scope of this Article. Thus, there is substantial uncertainty as to whether and to what degree the 8% threshold would change were the underlying methodological choices to change.

It should be noted that the proponents set a different threshold for congressional plans—two undeserved seats—than the 8% threshold for state legislative

298. See Stephanopoulos & McGhee, *supra* note 7, at 867–68.

299. See *id.* at 887–89. In the *Whitford* litigation, Jackman, one of the plaintiffs' experts, used only the simplified formula to compute the efficiency gaps for a similarly large number of elections. Jackman Report, *supra* note 102, at 16. Based on these distributions, he proposed a 7% threshold. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 860 (W.D. Wis. 2016), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

300. See Stephanopoulos & McGhee, *supra* note 7, at 888–89.

plans.³⁰¹ In arriving at this threshold, the proponents added a new methodological choice to the mix: When calculating the historical undeserved seat share in congressional races, they omitted all election results from states with fewer than eight districts.³⁰² They justified this choice on the ground that “redistricting in smaller states has only a minor influence on the national balance of power.”³⁰³ Yet twenty-two states (whose representatives make up more than one-fifth of the House of Representatives) have more than one but fewer than eight congressional districts.³⁰⁴ Removing these states from the full data set eliminates those data points from consideration and may significantly change the historical analysis.

Special problems do arise in trying to apply the efficiency gap measure to plans with low district numerosity, such as congressional maps for small states or electoral maps for small local governing bodies. If the threshold is set at two undeserved congressional seats, then voters in many small states could never make political gerrymandering claims because there could never be two undeserved seats. Conversely, if the threshold is set at 8%, maps with low district numerosity may be particularly likely to exhibit above-threshold gaps. For example, if a state has only two congressional districts, there are only two possible values for seat margin: 0% (each party wins one district) or 50% (one party wins both districts).³⁰⁵ Unless this state has the right vote share, it will necessarily have a large efficiency gap. There is, then, a problem of scope. The efficiency gap either works poorly or does not work at all in capturing gerrymandering dynamics in plans with low district numerosity.

301. See *id.* at 837 (“To take into account both the severity and durability of gerrymanders, we recommend setting the bar at two seats for congressional plans . . .” (emphasis omitted)). Note that the League of Women Voters in the *Rucho* litigation has focused not on the two-seat threshold but rather on a numeric efficiency gap threshold. It appears that in calculating historical gaps across congressional plans and evaluating the extent of North Carolina’s deviation from historical norms, the *Rucho* plaintiffs excluded states with low district numerosity from the historical baseline. See *Common Cause v. Rucho*, 240 F. Supp. 3d 376, 380-81 (M.D.N.C. 2017) (per curiam) (“According to the League Amended Complaint, the Plan produced an efficiency gap of 19 percent in the 2016 election, which is ‘in approximately the worst 4 percent of the historical distribution, and the single worst score of all relevant congressional plans in the country in 2016.’” (quoting Amended Complaint ¶¶ 61-62, *Rucho*, 240 F. Supp. 3d 376 (No. 1:16-cv-01164-WO-JEP), 2017 WL 6887476)).

302. See Stephanopoulos & McGhee, *supra* note 7, at 868.

303. *Id.*

304. See U.S. CENSUS BUREAU, U.S. DEP’T OF COMMERCE, CONGRESSIONAL APPORTIONMENT: 2010 CENSUS BRIEFS 2 tbl.1 (2011), <https://perma.cc/7WPB-BTFH>. In total, these states were allocated ninety-one representatives—approximately 21% of the total number of representatives nationwide. See *id.*

305. See *supra* text accompanying notes 132-33.

In short, efficiency gap analysis is not robust to methodological choices and electoral circumstances. And those methodological choices are currently the subject of dynamic debate. For example, the *Whitford* panel majority viewed the simplified formula as a “shortcut” and the long-form equation as “preferable.”³⁰⁶ But McGhee now suggests that the long-form equation violates his efficiency principle and that the simplified formula represents the proper definition of the efficiency gap.³⁰⁷ Nagle has suggested comparing wasted vote shares rather than totals.³⁰⁸ And I have proposed a new wasted vote measure that compares wasted vote shares rather than totals, as Nagle suggests, while defining surplus votes as the full victory margin.³⁰⁹ My proposal could allow responsiveness to vary with vote share in a way better aligned with democratic values and electoral reality, but like all the methodological choices implicated by efficiency gap analysis, it warrants further exploration.

If the efficiency gap is *the* measure of partisan gerrymandering, courts will struggle when a plan’s validity depends on methodological approach, and litigants will have a powerful incentive to advance the methodological approach that best supports their cause. If conversely the efficiency gap is only an indicative measure of partisan gerrymandering, courts will have more flexibility to give it more or less weight depending on whether the measure’s assessment of the plan at issue is more or less robust to methodological choices.

B. Normative Considerations

The other reason the Court should recognize the efficiency gap as an indicative but nondefinitive measure is that it exhibits tensions with democratic norms of electoral competitiveness, seats-votes proportionality, and voter participation.

The efficiency gap measure is at best agnostic and at worst antagonistic toward the goal of electoral competitiveness. Because the measure privileges the perspective of mapmakers serving party interests, it does not recognize harm to voters when elections are uncompetitive. The measure may fail to recognize even extreme bipartisan gerrymanders. And more competitive plans pose a greater risk of above-threshold gaps in the face of vote swings.³¹⁰

306. See *Whitford v. Gill*, 218 F. Supp. 3d 837, 907-08 (W.D. Wis. 2016), *stay granted*, 137 S. Ct. 2289, and *jurisdiction postponed*, 137 S. Ct. 2268 (2017).

307. See *supra* notes 160-62 and accompanying text.

308. See *supra* notes 171-77 and accompanying text.

309. See *supra* Part IV.

310. See *supra* Part III.D.

With a simplifying assumption of equal voter turnout across districts, the efficiency gap is equivalent to double seats-votes proportionality. It idealizes districting plans in which a party supported by 75% of the electorate wins all the seats. To the extent that normative intuitions support a system in which vote share and seat share should be roughly equal, the efficiency gap undermines that norm—and not only in extreme scenarios.³¹¹

Finally, the efficiency gap measure, like other measures justifying seats won in terms of ballots cast, registers voter suppression simply as reduced support for the targeted party. In some circumstances, suppressing the disfavored party's voters will actually serve to lower the gap. Thus, the efficiency gap does not condemn—and may in fact encourage—voter suppression.³¹²

The efficiency gap also gives rise to both false positive and false negative concerns.³¹³ The proposed test's other doctrinal elements—sensitivity analysis, intent, and state justification—only partially address the false positive problem. In pursuit of competitiveness or proportionality, a mapmaker may devise a districting plan that produces an above-threshold gap but still avoid a finding of invalidity unless that gap is durable, unjustified, and intentional. But these doctrinal tools are imperfect; they may not always work, and even when they do, they help avoid ultimate invalidation, not litigation. Moreover, because below-threshold efficiency gaps trigger an apparently irrebuttable presumption of validity, the other doctrinal elements do nothing to address the false negative problem.

Were the Court to embrace the efficiency gap as the definitive measure of partisan gerrymandering, any plan that wastes (roughly) equal votes would enjoy absolute immunity from judicial scrutiny under the partisan gerrymandering doctrine—regardless how severely it subverted other democratic norms like competitiveness or seats-votes proportionality. In this way, the efficiency gap proposal would provide mapmakers a powerful incentive to draw uncompetitive plans that give the majority a double proportionality seat bonus. Such plans can sensibly be called gerrymanders if that term implicates norms of competitiveness and seats-votes proportionality. Yet those are precisely the plans the efficiency gap would be most likely to approve and thus the plans mapmakers would be encouraged to design.

Because electoral districting implicates—and gerrymandering threatens—multiple democratic norms, it is unsurprising that a single measure would fail to adequately address them all. Indeed, it may be that no one measure can satisfactorily reduce to a single number the multiple democratic norms at stake. I do not fault the efficiency gap measure for failing to perform an impossible task. And I recognize that the efficiency gap does measure one significant democratic

311. See *supra* Part III.C.

312. See *supra* Part II.E.2.

313. See *supra* Part III.E.

norm, which may powerfully capture, and thereby help curb, certain partisan gerrymanders. For this reason, the efficiency gap is a helpful indicative measure courts can and should consider when analyzing claims of partisan gerrymandering—so long as courts recognize its technical and normative limits.